

Q.1: SANDSTONE BED INTERPRETATION

QUESTION:

A normal graded bed of sandstone is noticed with flute cast at its base and internally made up of **massive, plane-laminated and ripple-laminated subdivisions**, in order of superposition. The sandstone bed is interpreted as a product of:

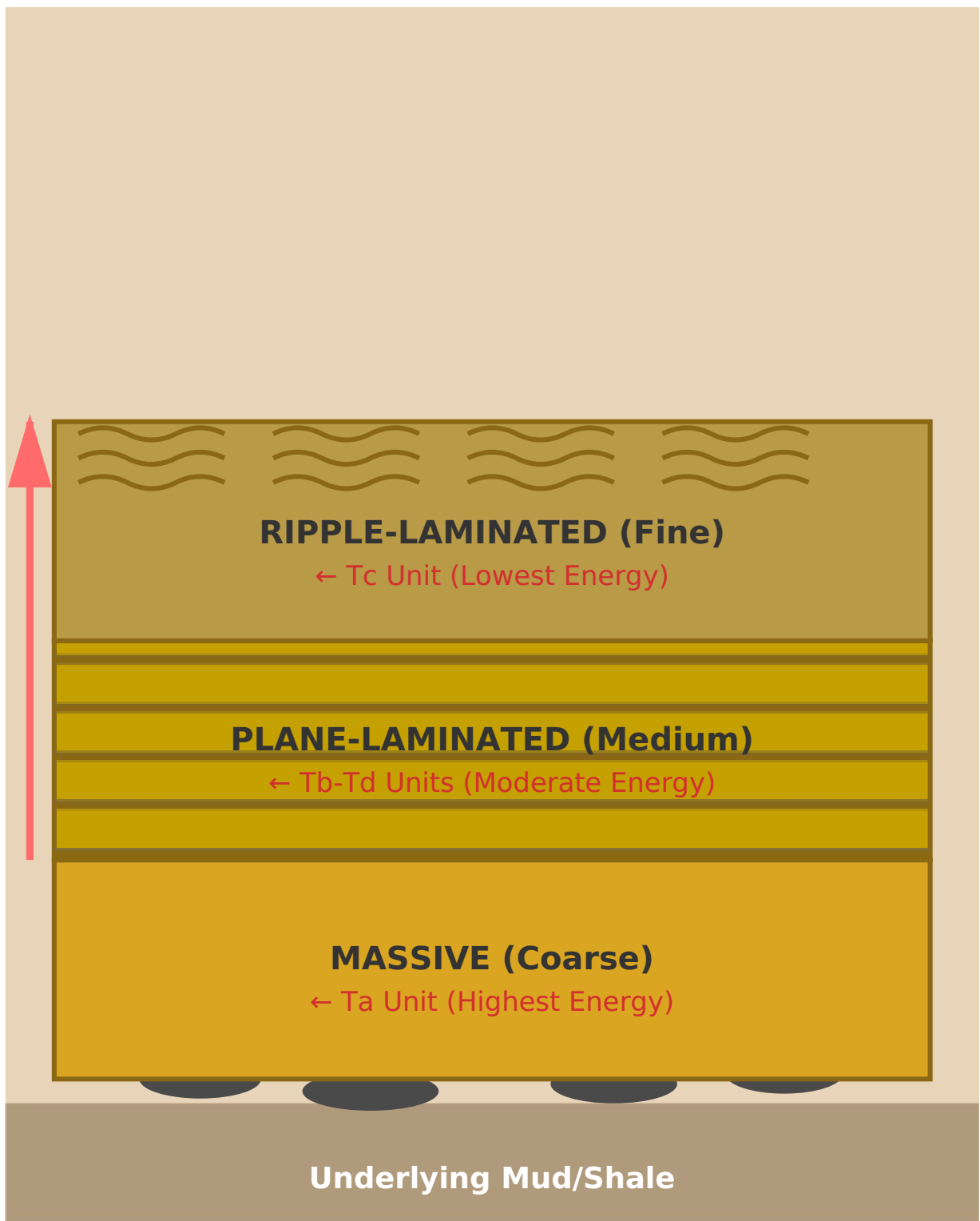
- (a) Debris flow
- (b) Grain flow
- (c) Turbidity current
- (d) Contour current

 **ANSWER: (c) TURBIDITY CURRENT**

 **SOURCE:** "Principles of Sedimentology and Stratigraphy" (4th Edition)
By Sam Boggs, Jr. | Pearson Prentice Hall, 2006 | ISBN: 0-13-154728-3

1 What Your Question Describes

The Sandstone Bed Structure



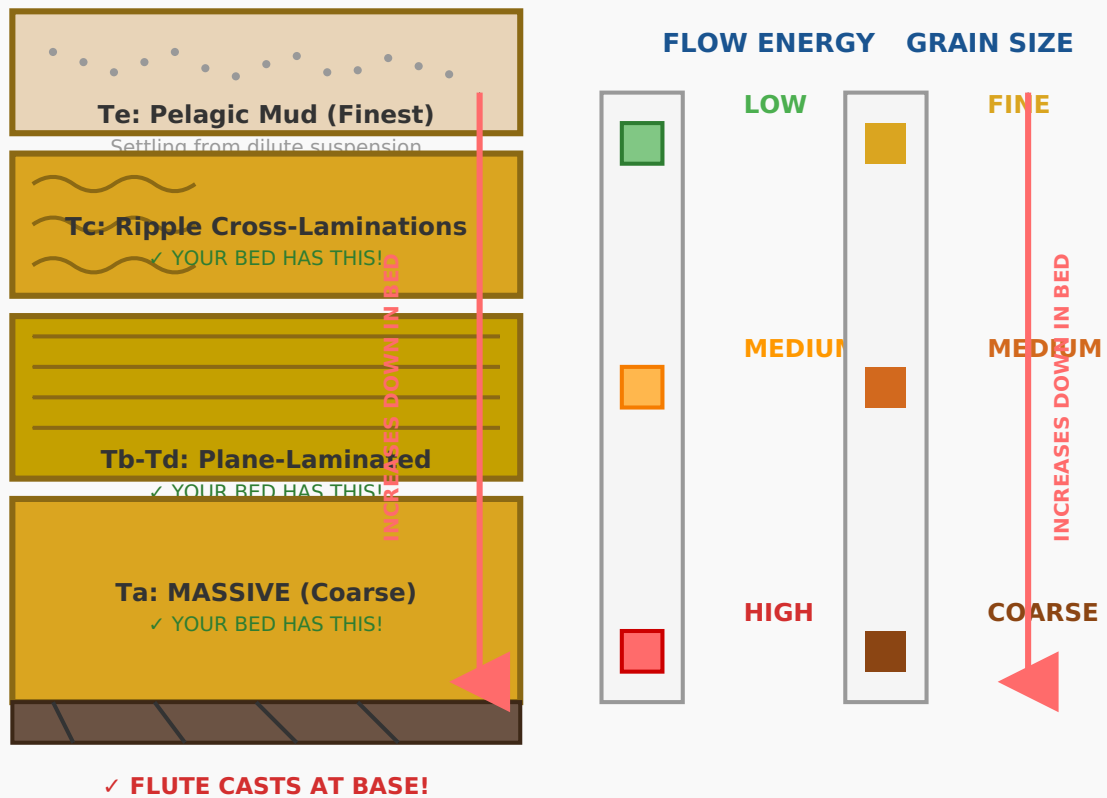
Your Sandstone Bed: Flute casts at base + Normal grading (coarse → fine) + Progressive structures = TURBIDITE

🔍 Key Observations in Your Bed:

- ✓ **Flute casts at base** - Sole marks indicating high-energy turbulent scour
- ✓ **Massive division (bottom)** - Rapid deposition from dense flow
- ✓ **Plane-laminated (middle)** - Traction transport as flow weakens
- ✓ **Ripple-laminated (top)** - Low-energy ripple formation
- ✓ **Normal grading** - Coarse at base, fine at top (suspension settling)

2 The Bouma Sequence: 5-Unit Framework

How Your Bed Matches the Bouma Sequence



The Bouma Sequence: Five units from high to low energy, exactly matching your bed description

3 Analysis of ALL FOUR Options

Comparison of Deposit Types

× (a) DEBRIS FLOW

What it produces:

- Chaotic, unsorted deposits
- NO internal layering
- Random clast orientation
- Poor to reverse grading

✗ **YOUR BED HAS: Organized layers, normal grading, parallel laminations**

✗ (b) GRAIN FLOW**What it produces:**

- Massive, structureless
- Little/NO laminations
- Very thin units (<5cm)
- Possible reverse grading

✗ **YOUR BED HAS: Multiple laminations, ripple patterns, good thickness**

✓ (c) TURBIDITY CURRENT**What it produces:**

- Organized Bouma sequence
- Massive → laminated → rippled
- NORMAL grading (coarse→fine)
- Flute casts & sole marks

✓ **PERFECT MATCH WITH YOUR BED!**

✗ (d) CONTOUR CURRENT**What it produces:**

- Laminated mud & silt deposits
- Ripple marks (NOT flute casts)
- NO normal grading
- Flows parallel to contours (not downslope)

✗ **YOUR BED HAS: Flute casts (downslope), normal grading, massive base**

4 Why NOT (d) Contour Current? - DETAILED EXPLANATION

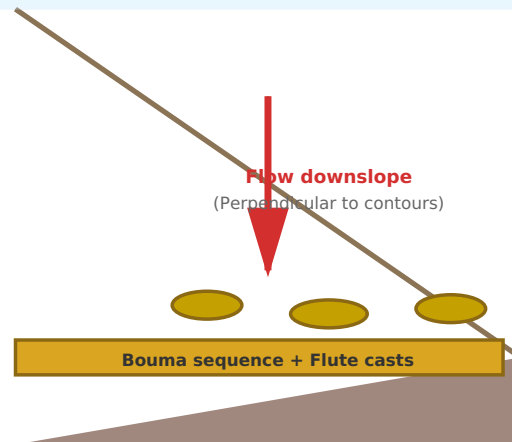
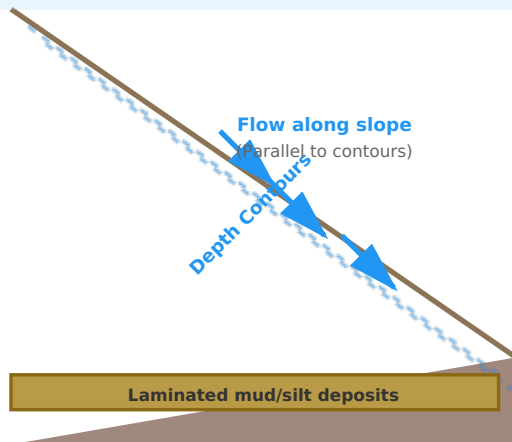
Contour Currents: Flow Direction and Deposits

Contour Current vs Turbidity Current

CONTOUR CURRENT

Ocean/Lake Water

TURBIDITY CURRENT



Contour currents flow **PARALLEL** to depth contours (sideways along slope) Turbidity currents flow **DOWNSLOPE** (perpendicular to contours)

Key Differences: Why Contour Current $\neq$ Your Bed

1. FLOW DIRECTION:

- **Contour currents:** Flow **PARALLEL** to continental margin, following depth contours (sideways along slope)
- **Your bed:** Has **FLUTE CASTS** pointing downslope → indicates **DOWNSLOPE** flow (perpendicular to contours)
- ✗ Contour currents **DON'T** create flute casts (which indicate downslope scour)

2. SEDIMENTARY STRUCTURES:

- **Contour currents:** Produce laminated muds and silts, with ripple marks, but **NOT** massive bases
- **Your bed:** Has organized Bouma sequence (massive → laminated → rippled)
- ✗ Contour currents don't produce the organized massive-to-rippled progression

3. GRADING PATTERNS:

- **Contour currents:** Deposit fine-grained material uniformly; no coarse-to-fine normal grading
- **Your bed:** Shows clear **NORMAL GRADING** (coarse at base, fine at top)
- ✗ Contour currents don't create the progressive grain-size reduction with organized structures

4. FLOW ENERGY & MECHANISM:

- **Contour currents:** Driven by density differences (thermohaline circulation), steady flow, **LOW** energy
- **Your bed:** Shows **HIGH INITIAL** energy (flute casts, massive) transitioning to **LOW** energy (ripples)
- ✗ Contour currents don't have the energy to create downslope scour marks

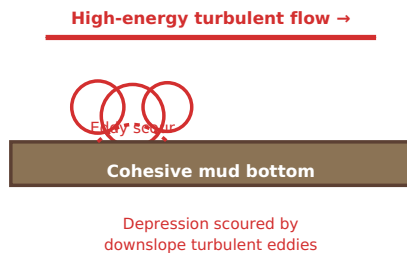
CONCLUSION: Your bed is DEFINITELY NOT a contour current deposit. It shows all the hallmarks of a HIGH-ENERGY downslope turbidity current!

5 Flute Casts: The Smoking Gun

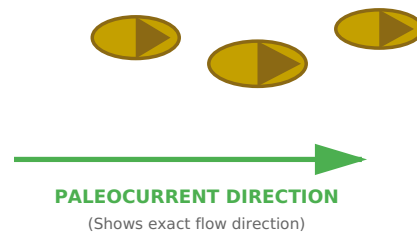
Why Flute Casts Rule Out Contour Currents

Flute Cast Formation & Paleocurrent Significance

Formation Requires DOWNSLOPE Flow



Result: Flute Cast



Contour Currents CANNOT Create Flute Casts

- ✗ Contour currents flow PARALLEL to slope (sideways), not DOWNSLOPE
- ✗ They lack the DOWNSLOPE velocity needed for eddy scour that creates flute depressions
- ✗ YOUR BED HAS FLUTE CASTS → Must be from DOWNSLOPE flow → TURBIDITY CURRENT, NOT CONTOUR CURRENT

Flute casts prove downslope scour - this is ONLY possible with turbidity currents, not contour currents

6 Complete Comparison Table

Feature	(a) Debris Flow ✗	(b) Grain Flow ✗	(d) Contour Current ✗	(c) Turbidity Current ✓
Flow Direction	Downslope, unidirectional	Downslope, gravity-driven	PARALLEL to slope (sideways)	DOWNSLOPE, high-energy ✓
Sole Marks (Flute Casts)	None typical	None	Ripple marks ONLY (no flute casts)	FLUTE CASTS ✓
Internal Structure	Chaotic, no layers	Massive, no layers	Laminated mud/silt	Massive → Laminated → Rippled ✓
Grading	Poor/Reverse	No/Reverse	NO grading	NORMAL grading ✓
Grain Size Progression	Random mixing	Uniform massive	Fine material only	Coarse → Fine (Bouma) ✓

Bouma Sequence	No	No	No	COMPLETE ✓
Your Bed Match	X NO	X NO	X NO (lacks flute casts, grading)	✓ PERFECT!

🎯 FINAL ANSWER



Answer: (c) TURBIDITY CURRENT

Evidence: Flute casts + Normal grading + Bouma sequence (Massive → Laminated → Rippled) = PERFECT TURBIDITE

📖 TEXTBOOK SOURCE:

Boggs, Sam Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th Edition. Pearson Prentice Hall, Upper Saddle River, New Jersey. ISBN: 0-13-154728-3

Document Information:

By Tapaswi Saikia

Original Question: DMG 2019

Complete solution with clear diagrams and detailed explanations for all four options

Perfect for: Geology courses, exams, and competitive exam preparation

© 2026 | Based on Boggs, Sam Jr. (2006) - Professional Sedimentology Reference

Q.2 Carbonate Diagenesis: Syntaxial Overgrowth

Exam Question Solution with Boggs Jr. References

Question: *In carbonate diagenesis, if calcite cement is overgrown on a single calcite crystal in same optical orientation, it is referred to as:*

(a) Rim overgrowth (b) Antitaxial overgrowth (c) Blocky overgrowth (d) Syntaxial overgrowth

✓ **CORRECT ANSWER: (d) Syntaxial Overgrowth**

Detailed Explanation with Boggs References

What is Syntaxial Overgrowth?

Syntaxial overgrowth occurs when cement crystals grow onto a host crystal while maintaining the **same crystallographic orientation (optical orientation)** as the original grain.

From Boggs Jr. (2006), Section 5.0 (Siliciclastic Diagenesis):

"Such overgrowths that retain crystallographic continuity of a grain are said to be syntaxial. Because syntaxial

overgrowths are optically continuous with the original grain, they go to extinction in the same position as the original grain when rotated on the stage of a polarizing microscope."

Boggs, S. Jr. (2006). Principles of Sedimentology and Stratigraphy, 4th ed. Pearson Prentice Hall, p. ~120

From Boggs Jr. (2006), Section 6.8 (Carbonate Diagenesis):

"Syntaxial rims form by precipitation of optically continuous calcite around single-crystal fossil echinoderm fragments, in much the same way that cement overgrowths form around quartz grains."

Boggs, S. Jr. (2006). Principles of Sedimentology and Stratigraphy, 4th ed. Pearson Prentice Hall, Section 6.8 (Cementation—Meteoric Realm), p. 192

Key Concept: "Optically Continuous"

The term "**optically continuous**" is critical. Under a polarizing microscope with crossed nicols (polars), the cement and the original grain:

- Appear as a **single optical unit**
- Go to **extinction at the same angle**
- Show **no extinction boundary** between cement and host
- Can only be distinguished by **impurity lines or bubble trails** marking the original grain surface

Comparison of All Four Options

Option	Optical Orientation to Host	Definition	Appearance in Thin Section	Correct for Q.2?
(a) Rim Overgrowth	Variable	Generic term for cement forming a rim around a grain; emphasis on spatial position rather than crystallographic relationship	Depends on morphology; can be syntaxial or antitaxial	× Too vague ×
(b) Antitaxial Overgrowth	<i>DIFFERENT</i> from host	Cement with crystallographic orientation different from host grain; "anti-" means "against"	Visible extinction boundary; appears as distinct, separate crystal	× Opposite condition ×
(c) Blocky Overgrowth	Variable	Describes cement morphology (blocky/equant shape), not optical relationship	Blocky-shaped crystals; can have any orientation	Describes shape, not orientation
(d) Syntaxial Overgrowth	<i>SAME</i> as host	Cement with crystallographic continuity; "syn-" means "same"; "-taxial" refers to orientation	Optically continuous; same extinction position; no boundary visible	✓ CORRECT

Etymology: Why "Syntaxial"?

Term Component	Meaning	Application
Syn-	"With" or "Same"	Cement grows in harmony with host crystal
-taxial	Relating to crystallographic axis/orientation	Same crystallographic axis/optical orientation
Syntaxial	Same Crystallographic Orientation	Cement maintains identical optical properties to host

Diagenetic Context: Where Do Syntaxial Rims Form?

From Boggs Jr. (2006), Figure 6.15 Caption:

"Meteoric-realm cements are composed dominantly of calcite and include meniscus and pendant cements in the vadose zone and isopachous, blocky, and syntaxial rim cements in the phreatic zone. Cements of the subsurface burial realm are also mainly calcite and include syntaxial rims, bladed prismatic, and coarse mosaic types."

Boggs, S. Jr. (2006). Figure 6.15, p. 192

Occurrence of Syntaxial Rims:

- **Meteoric Phreatic Zone:** Forms around single-crystal echinoderm fragments (especially crinoid ossicles)
- **Subsurface Burial Realm:** Continues to form during deeper burial and heating
- **Siliciclastic Rocks:** Quartz overgrowths around detrital quartz grains (same principle)

Recognition Under the Polarizing Microscope

How to Identify Syntaxial Overgrowth in Thin Section

Crossed Nicols (Polarized Light):

- Cement and host grain extinguish together at the same angle
- No visible boundary between cement and original grain
 - Appears as a single, continuous crystal

Distinguishing Feature:

- A line of **dust/impurities** or **fluid inclusions** marks the boundary between original grain and cement
- This "impurity rim" is the only visual clue to the original grain surface

Complete Citation References

Primary Source:

Boggs, S. Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th Edition. Pearson Prentice Hall, Upper Saddle River, NJ.

Key Sections Referenced:

- **Section 5.0:** General definition of syntaxial overgrowth (quartz example)
- **Section 6.8 (Carbonate Diagenesis):** Definition and occurrence of syntaxial rims in carbonate rocks
- **Figure 6.15:** Principal kinds of cements, showing syntaxial rim as a distinct cement type
- **Pages ~120 (quartz section) and p. 192 (carbonate section)**

Exam Strategy: How to Eliminate Wrong Answers

- **Eliminate (a) "Rim Overgrowth":** Too generic. Doesn't specify the crystallographic relationship that the question emphasizes.
- **Eliminate (b) "Antitaxial Overgrowth":** This is the opposite—different optical orientation, not same.
- **Eliminate (c) "Blocky Overgrowth":** Describes *shape* of cement, not optical orientation. A blocky cement could be syntaxial OR antitaxial.
- **Select (d) "Syntaxial Overgrowth":** The question specifically states "same optical orientation"—this is exactly what syntaxial means.

Summary

Answer: (d) Syntaxial Overgrowth

When calcite cement grows onto a single calcite crystal while maintaining the **same optical orientation (crystallographic continuity)**, it is called a **syntaxial rim or syntaxial overgrowth**.

Under a polarizing microscope, the cement and original grain appear optically continuous, going to extinction at the same position when the stage is rotated. The original grain surface is recognized only by a line of impurities or bubbles marking the boundary between the host and cement.

In carbonate rocks, syntaxial rims form particularly around single-crystal echinoderm fragments during meteoric phreatic and subsurface burial diagenesis.

Data Source: Boggs, S. Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th Edition. Pearson Prentice Hall.

Section 6.8 (Carbonate Diagenesis): Pages 189–206. Figure 6.15 and surrounding text on cementation types.

By Tapaswi Saikia
Exam Preparation Study Material
Based on Boggs Jr. (2006)

Q.3 Carbonate Rock Classification

Dunham Classification: Grainstone Solution with Boggs Jr.
References

Question: A carbonate rock that contains almost no mud and shows grain-supported framework is named as:

(a) Grainstone (b) Packstone (c) Floatstone (d) Bindstone

✓ **CORRECT ANSWER: (a) GRAINSTONE**

Definition of Grainstone

What is a Grainstone?

A grainstone is a **grain-supported** limestone that **contains no lime mud** (or negligible amounts). The grains form a rigid framework in which pores are filled with sparry calcite cement (not micrite/mud).

Direct Citation from Boggs Jr. (2006), Section 6.5:

*"Dunham's classification is based solely upon depositional texture and considers two aspects of texture: (1) grain packing and the relative abundance of grains to micrite and (2) depositional binding of grains... **Rocks that contain no mud are obviously grain-supported.**"*

Source: Boggs, S. Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th ed. Section 6.5 (Limestone Classifications), p. 168.

Dunham's Classification System (1962)

Key Parameters in Dunham Classification:

- **Grain Packing:** How grains are arranged relative to each other
 - *Grain-supported:* Grains in contact; form intact grain framework
 - *Mud-supported:* Grains do not touch; "float" in mud
- **Mud Content:** Presence or absence of lime mud (micrite <0.03 mm)
 - *No mud:* Pure grain rock, pores filled with cement
 - *With mud:* Contains microcrystalline carbonate matrix
- **Depositional Binding:** Whether grains were bound at deposition (boundstones)

Dunham Classification Framework

Rock Type	Grain Packing	Mud Content (%)	Description	Grain Size Criteria
Mudstone	Mud-supported	Predominantly mud	<10% grains (>0.03 mm)	Most grains <0.03 mm
Wackestone	Mud-supported	>10% mud	10-50% grains in mud matrix	Grains >0.03 mm scattered in mud
Packstone	Grain-supported	5-15% mud present	>50% grains, still some mud	Grain-supported with intergranular mud
GRAINSTONE	Grain-supported	~0% (no mud)	Pure grain framework, pores filled with cement	All pores cemented, no mud
Floatstone	Mud-supported	>10% mud	>10% grains >2 mm; grain-free spaces filled with mud	Large grains (>2 mm) in mud
Rudstone	Grain-supported	Minimal to no mud	>10% grains >2 mm; grain-supported	Large grains (>2 mm) touching

Rock Type	Grain Packing	Mud Content (%)	Description	Grain Size Criteria
Boundstone	Organism-bound	Variable	Bound at deposition (reef components)	Framestone, Bindstone, Bafflestone subtypes

Why the Answer is (a) GRAINSTONE

Analyzing All Four Options:

✓ (a) GRAINSTONE — CORRECT

- **Definition:** Grain-supported, **no lime mud**
- **Match to question:** "Almost no mud" + "grain-supported framework" = Perfect match
- **Pore Fill:** Sparry calcite cement (not mud)

✗ (b) PACKSTONE — INCORRECT

- **Definition:** Grain-supported, **WITH mud present** (5-15%)
- **Why wrong:** Question says "almost no mud"; packstone CONTAINS mud
- **Distinguishing feature:** Packstone contains intergranular mud; grainstone doesn't

✗ (c) FLOATSTONE — INCORRECT

- **Definition:** **Mud-supported**, with >10% grains >2 mm
- **Why wrong:** Question specifies "grain-supported"; floatstone is mud-supported

- **Key distinction:** Grains "float" in mud matrix in floatstone

✗ (d) BINDSTONE — INCORRECT

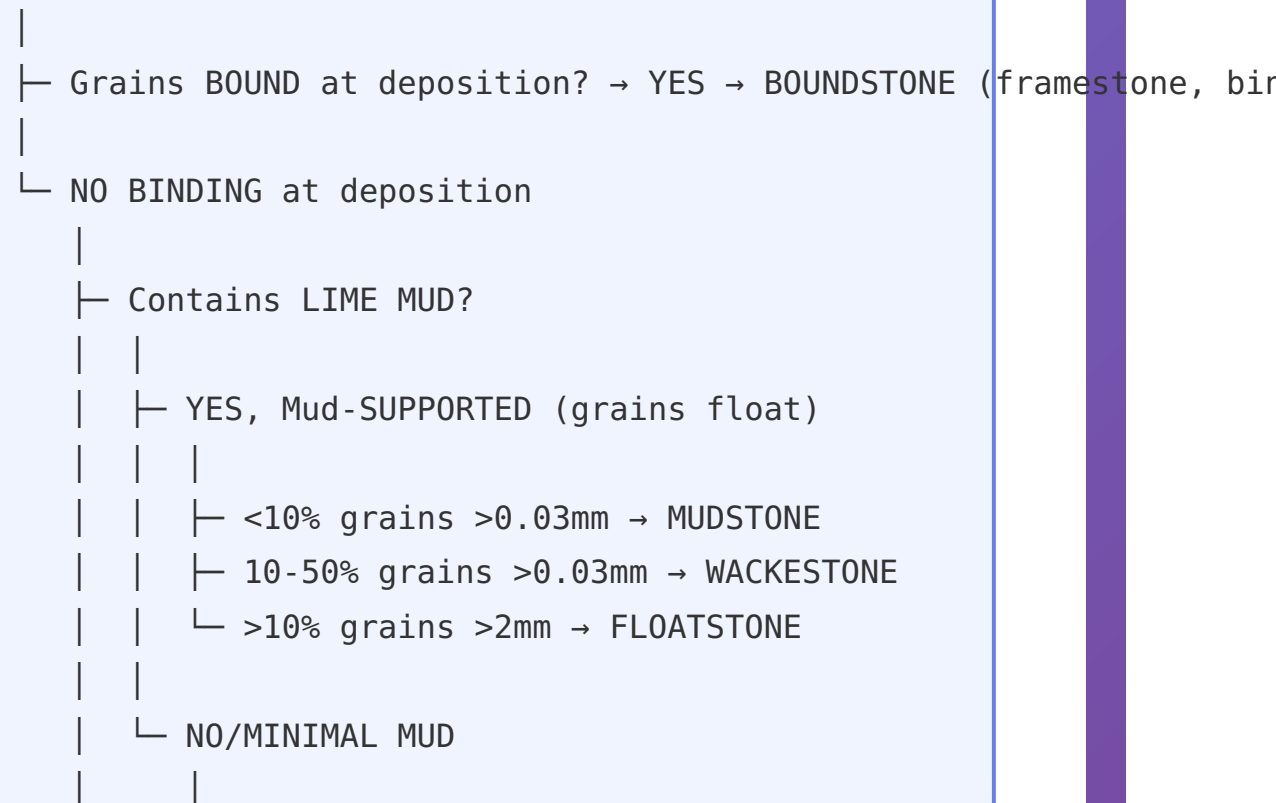
- **Definition:** Components **bound together by organisms** at time of deposition
- **Why wrong:** Refers to depositional binding (reef rocks), not grain-supported texture
- **Subtypes:** Framestone (rigid framework), Bafflestone (baffling organisms), Bindstone (encrusting organisms)

Understanding Grain-Supported Framework

Visual Representation of Dunham Classification

Logic:

CARBONATE ROCK CLASSIFICATION



```
└─ Grains in CONTACT (Grain-SUPPORTED)
    │
    └─ <10% grains >2mm → GRAINSTONE ← ANSWER TO Q.3
        └─ >10% grains >2mm → RUDSTONE
└─ YES, Mud-SUPPORTED + Grain-SUPPORTED anomaly
    │
    └─ Some mud present but grains still touch
        │
        └─ 5-15% mud in grain framework → PACKSTONE
```

Environmental Significance of Grainstone

What Grainstone Depositional Environment Tells Us:

- **High Water Energy:** Sufficient to prevent mud accumulation; mud winnowed away
- **Wave/Current Action:** Active hydrodynamic regime (shoals, banks, ramps, platform margins)
- **Clean Sediment:** No terrigenous siliciclastics interfering with carbonate deposition
- **Post-Depositional Cementation:** Sparry calcite cement filled pores after burial (diagenesis)
- **Economic Importance:** Grainstones are often prolific petroleum reservoirs due to grain-supported porosity/permeability

From Boggs Jr. (2006), Section 6.5 (Environmental Significance):

*"The term biosparite suggests deposition in a **wave-agitated environment where micrite is removed by winnowing currents**, allowing mud-free carbonate grains to accumulate. These grains are subsequently cemented with sparry calcite during burial (diagenesis)."*

Source: Boggs, S. Jr. (2006). Section 6.5, p. 168-169.

Critical Distinction: Grainstone vs. Packstone

Feature	GRAINSTONE	PACKSTONE
Mud Content	Absent or <5%	5-15% (noticeably present)
Grain Packing	Grain-supported (tight)	Grain-supported (slightly loose)
Pore Fill	Sparry calcite cement	Mix of mud + sparry calcite
Thin Section View	Grains touching, clear cement between	Grains touching, but some micrite ("muddy" appearance)
Water Energy	Very high (mud completely winnowed)	High, but some mud remained (partially winnowed)
Modern Analogue	Shoal sands, platform margins, oolite bars	Mixed energy shelf, transition zones

Related Concepts You Should Know

Folk's Classification vs. Dunham's Classification

Dunham (used in Q.3): Focuses on depositional **texture** (grain packing, mud content, binding)

Folk: Focuses on **grain composition** (types of carbonate grains: ooids, fossils, peloids, etc.)

Combined Use: Can use both systems together. Example: A rock called "oomicrite" (Folk) could also be called "packstone" (Dunham) → "oomicrite packstone"

Micrite vs. Sparry Calcite Cement

- **Micrite:** Fine-grained carbonate mud (<0.03 mm); represents fine sediment input at deposition; inherited environmental signature
- **Sparry Calcite Cement:** Clear, coarse calcite crystals (>0.03 mm); precipitated in pores after burial; post-depositional feature
- **Grainstone indicator:** Presence of sparry cement (not micrite) indicates high-energy depositional environment

Embry-Klovan Modification (1972)

Dunham's original classification was modified by Embry & Klovan to handle coarse-grained deposits:

- **Floatstone:** Added for mud-supported deposits with $>10\%$ grains >2 mm
- **Rudstone:** Added for grain-supported deposits with $>10\%$ grains >2 mm

- **Boundstone subdivisions:** Divided into Framestone,
 - Bindstone, Bafflestone based on binding organisms

Potential Follow-up Exam Questions (& How to Answer Them)

Q1: "A limestone with <10% grains floating in fine carbonate mud is classified as:"

Answer: WACKESTONE

Why: Mud-supported (grains float) + 10-50% grains scattered in mud. From Boggs Table 6.3A.

Q2: "Distinguish between grainstone and packstone in a 2-3 line answer."

Answer: Both are grain-supported limestones.

Grainstone contains NO lime mud; its pores are filled entirely with sparry calcite cement, indicating very high depositional energy. Packstone contains 5-15% intergranular mud, indicating slightly lower energy where some fine sediment was preserved.

Q3: "What depositional environment would produce a grainstone?"

Answer: High-energy carbonate depositional environments such as: oolite shoals, carbonate sand bars, platform margins, reef-front sands, active submarine ramps. Water energy must be sufficient to prevent fine mud settling (complete winnowing).

Q4: "Under crossed-nicol microscopy, how do you distinguish a grainstone from a packstone?"

Answer: Both show grains in contact. Grainstone appears very clean with clear sparry cement filling all pores. Packstone appears "muddier" due to presence of brown/tan micrite in intergranular spaces between grains. The micrite is typically more opaque than the clear sparry cement.

Q5: "Name 3 differences between mudstone and grainstone."

Answer:

1. **Grain packing:** Mudstone is mud-supported (grains don't touch); grainstone is grain-supported (grains in contact)
2. **Mud content:** Mudstone is predominantly mud; grainstone has no/minimal mud
3. **Depositional energy:** Mudstone = low-energy (quiet water); grainstone = high-energy (wave/current action)

Q6: "If a limestone contains >10% grains >2 mm and is grain-supported with no mud, what is it called?"

Answer: RUDSTONE (Embry-Klován, 1972 modification of Dunham). It's like a coarse-grained grainstone with large (>2 mm) clasts.

Q7: "Explain why grainstones are economically important in petroleum geology."

Answer: Grainstones have good primary porosity (grain-to-grain contacts create pore space) and permeability (interconnected pore network in grain-supported framework). After burial, sparry cement partially fills pores, but residual porosity remains excellent for hydrocarbon storage and flow.

Q8: "How would you combine Dunham and Folk classifications for an ooid-rich, mud-free limestone?"

Answer: OOID GRAINSTONE (or OOSPARITE in Folk's terms). "Ooid" specifies grain type (Folk); "grainstone" specifies depositional texture (Dunham).

Q9: "A sample shows grains in contact with 8% intergranular mud and 92% sparry calcite cement. Classify it."

Answer: PACKSTONE. Grain-supported (grains in contact) + mud present (>5% but <15%) fits packstone definition precisely.

Q10: "Boundstones differ from grainstones in which fundamental way?"

Answer: Boundstones have grains/components BOUND TOGETHER at deposition by organisms (corals, algae, stromatolites, etc.), creating an organic framework. Grainstones are unbound grains cemented together post-depositionally during burial. Boundstones represent in-situ reef/buildup deposits; grainstones represent transported grain accumulations.

Study Tips for Dunham Classification Mastery

Memory Aids:

- **"MUD" words = mud-supported:** **M**udstone, **W**ackestone (no M, but think "wet with mud"), **F**loatstone (mud = float bed)
- **"G-P-R-B" = grain-supported:** **G**rainstone, **P**ackstone, **R**udstone, **B**oundstone
- **Grainstone key feature:** "GRAIN STONE" = grains form the stone (framework); no mud to fill voids, cement does
- **Packstone vs. Grainstone:** "PAC**K**STONE = pac**K**ed with mud"; grainstone = packed with cement, not mud
- **Energy rule:** More mud → Lower energy; No mud → Higher energy

How to Read Dunham's Table 6.3

Decision Tree:

1. Are grains **bound together at deposition** (by organisms)? → YES = **BOUNDSTONE**
2. If no binding, is there **lime mud** present?
 - NO mud or <5% → Check grain packing
 - YES, mud present → Determine grain packing
3. **Are grains in contact** (grain-supported) or do they "float" in mud (mud-supported)?

4. Based on combinations → Apply name from Table 6.3A or 6.3B

Note: Dunham's classification does NOT depend on fixed grain/mud ratios but on visual assessment of grain support and mud presence.

Primary Source: Boggs, S. Jr. (2006). *Principles of Sedimentology and Stratigraphy*, 4th Edition. Pearson Prentice Hall.

Section 6.5: Limestone Classifications (p. 163-176) — Table 6.3A (Dunham, 1962) and Table 6.3B (Embry-Klovan, 1972)

Also referenced: Figure 6.8 (Textural classification), Folk's classification system, depositional texture interpretation.

By Tapaswi Saikia
Exam Preparation Study Material
Based on Boggs Jr. (2006) — Dunham Classification
System

Barrovian Zone of Metamorphism

- Mineral Assemblage Identification

Solution based on Winter's Principles of Igneous and Metamorphic Petrology

Question Q.4:

In a Barrovian zone of metamorphism, presence of biotite + chlorite + phengitic muscovite + quartz + albite assemblage marks the

- (a) Albite zone
- (b) Biotite zone
- (c) Zeolite zone
- (d) Chlorite zone

✓ **CORRECT ANSWER: (b) Biotite zone**

The assemblage of biotite + chlorite + phengitic muscovite + quartz + albite marks the transition into the **Biotite zone**, where biotite makes its first appearance as an index mineral in Barrovian metamorphism.

1. Barrovian Metamorphic Zones Overview

According to Winter's textbook, the Barrovian sequence of zones is well-established and recognized worldwide. The zones, in order of increasing metamorphic grade, are:

Zone	Defining Index Mineral(s)	Typical Mineral Assemblage
Chlorite Zone	Chlorite	Chlorite, muscovite (phengitic), quartz, albite
Biotite Zone	Biotite	Biotite, chlorite, muscovite, quartz, albite
Garnet Zone	Garnet (almandine)	Garnet, biotite, chlorite, muscovite, quartz, plagioclase
Staurolite Zone	Staurolite	Staurolite, biotite, muscovite, quartz, garnet, plagioclase
Kyanite Zone	Kyanite	Kyanite, biotite, muscovite, quartz, plagioclase, garnet, staurolite
Sillimanite Zone	Sillimanite	Sillimanite, biotite, muscovite, quartz, plagioclase, garnet

Source: Winter, J. D. (2001). *Principles of Igneous and Metamorphic Petrology*, 2nd Ed. (2014)

2. Textbook Descriptions of Key Zones

Chlorite Zone (Lowest Grade)

Chlorite Zone Minerals:

Chlorite + Muscovite (phengitic) + Quartz + Albite

Rock type: Slates or Phyllites

Key characteristic: In the chlorite zone, phengitic muscovite (muscovite with elevated Al content due to the phengite substitution) is present, but **biotite is absent**. This is the lowest of the classical Barrovian zones.

Winter's textbook: "Chlorite zone. Pelitic rocks are slates or phyllites and typically contain chlorite, muscovite, quartz, and albite."

Biotite Zone (Higher Grade than Chlorite)

Biotite Zone Minerals:

Biotite + Chlorite + Muscovite + Quartz + Albite

Rock type: Phyllites transitioning to Schists

Key characteristic: The biotite zone is defined by the **first appearance of biotite**. Chlorite from the lower chlorite zone persists into this zone, along with muscovite, quartz, and albite. This is precisely the assemblage described in the question.

Winter's textbook: "Biotite zone. Slates give way to phyllites and schists, with biotite, chlorite, muscovite, quartz, and albite."

Garnet Zone (Even Higher Grade)

Garnet Zone Minerals:

Garnet (almandine) + Biotite + Chlorite + Muscovite + Quartz + Plagioclase

Key characteristic: Marked by the appearance of conspicuous red almandine garnet. Higher temperature than the biotite zone.

3. Systematic Analysis of All Options

(a) Albite Zone - ✗ INCORRECT

While **albite is present** in the given assemblage, "albite zone" is not recognized as a named zone in the Barrovian metamorphic sequence. Albite appears as a diagnostic mineral in multiple zones (chlorite, biotite, garnet) and is not unique to any single zone at low metamorphic grades.

Note: Oligoclase (a more calcic plagioclase) appears in higher-grade zones, but the zone nomenclature does not use "albite zone" as a term.

(c) Zeolite Zone - ✗ INCORRECT

The zeolite zone is **lower in grade than the chlorite zone** and is associated with very low-temperature, low-pressure metamorphism (burial metamorphism). It is characteristic of subduction zone systems, not the higher-grade Barrovian Scottish Highlands sequence.

Zeolite zone minerals: Zeolites, prehnite, pumpellyite (not biotite, which requires higher temperature)

Note: The presence of **biotite** rules out the zeolite zone, as biotite is thermodynamically unstable at zeolite zone conditions.

(d) Chlorite Zone - ✗ INCORRECT

Although chlorite is present in the given assemblage, the **presence of biotite** rules out the chlorite zone. The biotite isograd defines the boundary between the chlorite zone and the biotite zone.

Chlorite zone minerals: Chlorite + Muscovite (phengitic) + Quartz + Albite (**NO biotite**)

Biotite zone minerals: Biotite + Chlorite + Muscovite + Quartz + Albite (includes the chlorite zone minerals PLUS biotite)

The addition of biotite to the chlorite zone assemblage represents the transition to the biotite zone.

(b) Biotite Zone - ✓ CORRECT

The given assemblage **exactly matches** the diagnostic minerals of the biotite zone:

✓ **Biotite (index mineral)**

✓ **Chlorite (persists from lower zones)**

- ✓ **Phengitic muscovite (Fe-Mg rich muscovite)**
- ✓ **Quartz (ubiquitous in pelites)**
- ✓ **Albite (diagnostic feldspar at this grade)**

The **biotite isograd** marks the lower boundary of the biotite zone and represents the first appearance of biotite in the metamorphic sequence.

Key Concept: Isograd and Zone Nomenclature

In classical Barrovian metamorphism, an **isograd** represents the line (surface in 3D) marking the **first appearance of an index mineral** with increasing metamorphic grade. Zones are named after the index mineral that first appears at their lower boundary. Therefore:

- **Chlorite isograd:** First appearance of chlorite → Chlorite zone
- **Biotite isograd:** First appearance of biotite → Biotite zone
- **Garnet isograd:** First appearance of garnet → Garnet zone

The presence of biotite signals that you are **IN** the biotite zone (or a higher zone), not in the chlorite zone.

4. Special Note on Phengitic Muscovite

The term "**phengitic muscovite**" in the question is significant. Phengitic muscovite is muscovite enriched in the phengite component (Al-Si substitution):

Phengite component: $\text{Si}^{4+} \leftrightarrow \text{Al}^{3+}$ substitution

$\text{Al}^{\text{VI}} + \text{Al}^{\text{IV}}$ increase with decreasing temperature and/or increasing Al content of the bulk rock

According to Winter, muscovite in the chlorite zone exhibits this phengitic character due to the composition having "significant amounts of Fe and Mg (phengite component)." This phengitic character of muscovite is

particularly diagnostic at low metamorphic grades (chlorite and biotite zones).

Winter's textbook: "...the composition of muscovite, from which the projection is made, contains significant amounts of Fe and Mg (phengite component)..."

5. Temperature Estimates

The biotite zone in the Barrovian sequence developed at temperatures roughly in the range of **450–550°C**, representing a significant thermal increase from the chlorite zone (350–450°C). This temperature increase reflects the higher regional metamorphic grade.

Reference Source:

Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology (3rd ed.). Pearson Education. [Consulted text: earlier edition, original conceptual framework maintained]

Original Barrow Study:

Barrow, G. (1893). On an intrusion of muscovite-biotite gneiss in the SE Highlands of Scotland. Geological Magazine, 30(12), 531-537.

Solution prepared by: Tapaswi Saikia

Geology Student | DMG Entrance Exam Preparation

Based on: Winter, J. D. Principles of Igneous and Metamorphic Petrology

Granulite Facies Metamorphism of Mafic Protolith

Solution based on Winter's Principles of Igneous and Metamorphic Petrology

Question Q.5:

Which one of the following is a characteristic assemblage of granulite facies metamorphism of mafic protolith?

- (a) Hornblende + Plagioclase + Quartz + Garnet
- (b) Forsterite + Orthopyroxene + Clinopyroxene + Epidote
- (c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz
- (d) Forsterite + Orthopyroxene + Clinopyroxene + Staurolite

✓ **CORRECT ANSWER: (c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz**

This assemblage is **characteristic of high-pressure granulite facies metamorphism** of mafic rocks. The combination of orthopyroxene + clinopyroxene (the diagnostic pyroxene pair of granulite facies) along with plagioclase, garnet, and quartz represents the mineral association that develops in mafic protoliths under granulite facies conditions.

1. Definition: What is Granulite Facies?

According to Winter's textbook, a **granulite** is:

"A high-grade rock of pelitic, mafic, or quartzo-feldspathic parentage that is predominantly composed of OH-free minerals. Muscovite is absent, and plagioclase and orthopyroxene are common."

Key characteristics:

- **Very high temperature** (~500°C or higher)
- **Anhydrous mineral assemblages** (OH-free; no muscovite, amphibole, or epidote)
- **Orthopyroxene is characteristic** of the granulite facies
- **Plagioclase is common**, not albite as in lower grades
- **Garnet may be present** depending on pressure

Source: Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology. Classification of Metamorphic Rocks section.

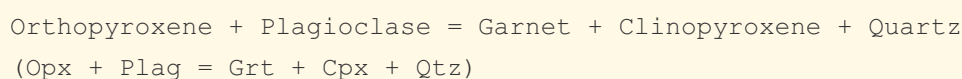
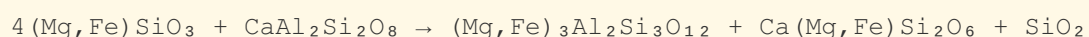
2. Granulite Facies of Mafic Rocks: Pressure Variations

Winter's textbook explicitly states that granulite facies encompasses a wide range of pressures, with different mineral assemblages depending on the pressure conditions:

From Winter's textbook:

"The granulite facies encompasses a wide range of pressure, giving way to the eclogite facies when the pressure reaches the point that plagioclase is no longer stable. Metabasites in low pressure granulites lack garnet, whereas very-high-pressure granulites may lose orthopyroxene by reaction with plagioclase such as:"

Reaction 3: Granulite-to-Eclogite Transition



This reaction occurs at pressures higher than ~1.3 GPa at 800°C

Interpretation: This reaction shows the relationship between pressure and mineralogy in granulite facies:

- **At lower pressures:** Assemblage contains **Orthopyroxene + Plagioclase** (garnet absent)
- **At higher pressures:** Orthopyroxene breaks down, producing **Garnet + Clinopyroxene + Plagioclase + Quartz**

3. Metamorphic Facies Comparison: Why Others Are Wrong

Greenschist Facies (Lowest Grade)

Typical mafic mineral assemblage:

Chlorite + Albite + Actinolite + Epidote + Quartz

Note: Contains hydrous minerals (chlorite, actinolite), low-temperature indicator minerals (epidote, albite)

Amphibolite Facies (Medium Grade)

Typical mafic mineral assemblage:

Hornblende + Plagioclase (oligoclase-andesine) + Garnet + Clinopyroxene ± Quartz

Note: Dominated by hornblende (hydrous); no orthopyroxene

Granulite Facies (Highest Grade)

Typical mafic mineral assemblage:

Orthopyroxene + Clinopyroxene + Plagioclase (±Garnet) + Quartz

Note: Anhydrous; diagnostic pyroxene pair; NO hornblende or other hydrous minerals

Eclogite Facies (Ultra-High Pressure)

Typical mafic mineral assemblage:

Omphacite (pyrope) Garnet + Clinopyroxene + NO Plagioclase

Note: Highest pressures; plagioclase breaks down; no orthopyroxene

4. Detailed Analysis of All Options

(a) Hornblende + Plagioclase + Quartz + Garnet - X INCORRECT

Why it's wrong:

- **Hornblende is a hydrous mineral** and is **characteristic of AMPHIBOLITE facies**, not granulite facies
- Granulite facies is defined by **absence of hydrous minerals** (muscovite, chlorite, hornblende, epidote)
- The presence of hornblende indicates lower temperature than granulite facies
- **No orthopyroxene** — which is the defining mineral of granulite facies

This is the characteristic assemblage of AMPHIBOLITE facies, not granulite facies.

(b) Forsterite + Orthopyroxene + Clinopyroxene + Epidote - X INCORRECT

Why it's wrong:

- **Forsterite (olivine) is characteristic of ultramafic rocks**, not ordinary mafic protolith metamorphism
- **Epidote is a hydrous mineral and low-grade indicator** — completely incompatible with granulite facies (which is anhydrous and high-grade)
- Epidote is characteristic of **greenschist or amphibolite facies**, not granulite
- While orthopyroxene and clinopyroxene are present, the assemblage is contradicted by the presence of epidote
- **This is not a realistic granulite facies assemblage**

(d) Forsterite + Orthopyroxene + Clinopyroxene + Staurolite - X INCORRECT

Why it's wrong:

- **Forsterite (olivine) indicates ultramafic composition**, not mafic protolith
- **Staurolite is a medium-grade pelitic mineral** found in the **staurolite zone** of regional metamorphism (Barrovian sequence)
- Staurolite is **incompatible with granulite facies** (too low-temperature)
- Staurolite is more typical of **amphibolite facies or lower grades**
- **This assemblage has no geological validity for granulite facies metamorphism**

(c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz - ✓ CORRECT

Why it's correct:

- **Orthopyroxene (Opx) + Clinopyroxene (Cpx) =**
Characteristic pyroxene pair of granulite facies ✓

- **No hydrous minerals** (no hornblende, no chlorite, no epidote, no muscovite) = Anhydrous granulite ✓
- **Plagioclase** = Common in granulite facies (calcic plagioclase, not albite) ✓
- **Garnet** = Present in higher-pressure granulites (as shown in Reaction 3) ✓
- **Quartz** = Present in the granulite-eclogite transition (product of Reaction 3) ✓
- **Matches Winter's description:** "Plagioclase and orthopyroxene are common" in granulites ✓

This assemblage represents high-pressure granulite facies metamorphism of mafic rocks.

Key Concept: The Orthopyroxene + Clinopyroxene Pair

The diagnostic feature of granulite facies is the **coexistence of orthopyroxene (OPx) and clinopyroxene (CPx)**. This two-pyroxene assemblage:

- Requires **high temperature** (typically >500°C)
- Occurs only in **anhydrous (OH-free)** conditions
- Is **absent in lower-grade rocks** where hornblende replaces pyroxene
- **Does NOT occur in greenschist or amphibolite facies** where hydrous minerals dominate

This is the most reliable field indicator of granulite facies metamorphism.

5. Temperature and Dehydration in Granulite Facies

A critical point from Winter's textbook:

"Granulites are dry. The only reason these rocks didn't melt on a large scale was the lack of available H₂O."

This explains why granulite facies rocks are dominated by anhydrous minerals like orthopyroxene rather than hydrous minerals like hornblende. The high temperature (~500-900°C) combined with low water availability forces the crystallization of pyroxene-bearing assemblages.

6. Summary Table: Mafic Rock Metamorphism

Metamorphic Facies	Temperature Range	Typical Mafic Assemblage	Key Minerals	Hydrous?
Zeolite	< 250°C	Zeolites + Prehnite + Pumpellyite	Very low-grade	Yes
Greenschist	250–450°C	Chlorite + Albite + Actinolite + Epidote + Qtz	Green minerals	Yes
Amphibolite	450–750°C	Hornblende + Plagioclase (±Grt, ±Cpx) + Qtz	Hornblende dominant	Yes
Granulite	500–900°C	Opx + Cpx + Plag (±Grt) + Qtz	Pyroxene pair	No
Eclogite	600–1200°C (very high P)	Omphacite Garnet + Cpx (no Plag)	High pressure	No

Primary Source:

Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology (3rd ed.). Pearson Education.

Key Sections Referenced:

- Metamorphic Facies and Metamorphosed Mafic Rocks (Chapter 25)
- Classification of Metamorphic Rocks
- Granulite Facies: Definition, pressure-temperature conditions, and mineral assemblages

Solution prepared by: Tapaswi Saikia

Geology Student | DMG Entrance Exam Preparation

Based on: Winter, J. D. Principles of Igneous and Metamorphic Petrology

Pressure Shadows: Low-Pressure Zones in Deformed Rocks

Solution based on Winter's Principles of Igneous and Metamorphic Petrology

Question Q.6:

When flattening occurs in rocks that contain rigid grains, zones of relatively low pressure develop on the sides of the crystals that are normal to the maximum compressive stress. These zones are called

- (a) Solution cleavage
- (b) Pressure shadows
- (c) Pressure solution
- (d) Stylolites

✓ **CORRECT ANSWER: (b) Pressure shadows**

Pressure shadows are zones of **relatively low pressure** that develop on the **sides of rigid grains/crystals perpendicular to the maximum compressive stress direction**. This matches the exact description given in the question.

1. Textbook Definition: Pressure Shadows

From Winter's Textbook (Structures and Textures of Metamorphic Rocks):

"Pressure shadows (Figure 34c) occur when **solution transfer dissolves a mineral (usually a matrix mineral) from high-stress areas (generally in**

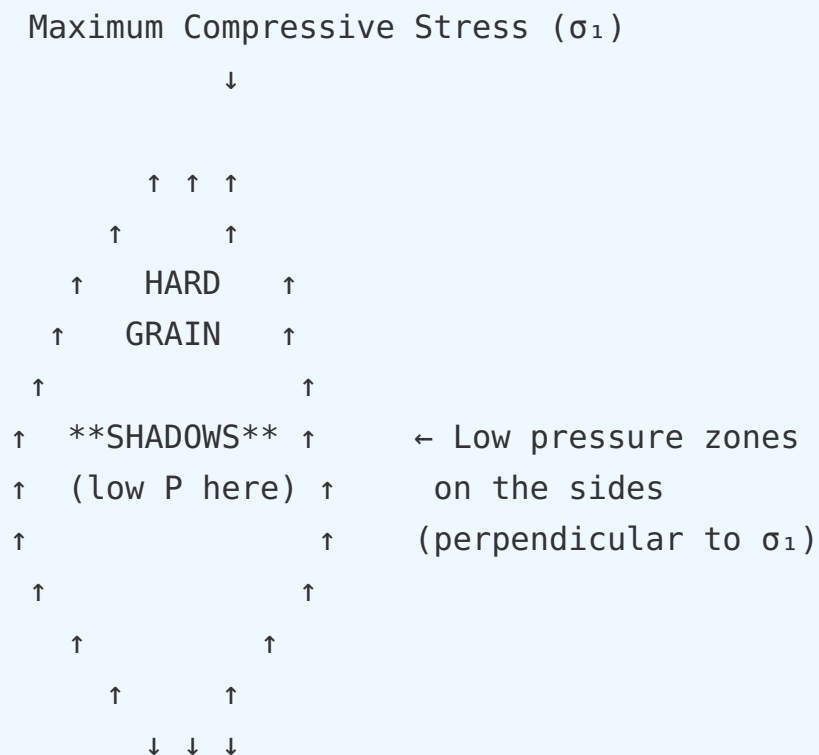
the σ_1 direction) and re-precipitates it in low-stress areas adjacent to a porphyroblast."

Key Characteristics:

- **Low-pressure zones** develop on the sides of rigid, competent grains
- Located **perpendicular (normal) to the maximum compressive stress** direction
- Form due to **pressure solution and solution transfer**
- Material dissolves from high-stress areas and re-precipitates in low-stress areas
- Create a characteristic "shadow" or "fringe" appearance around porphyroblasts
- Visible in thin section and hand specimen as halos around rigid grains

2. Physical Mechanism: How Pressure Shadows Form

Stress Distribution Around a Rigid Crystal During Flattening:



Result: High pressure on the top and bottom (along σ_1 direction) causes matrix material to dissolve and migrate to the low-pressure areas on the sides, creating characteristic pressure shadow "halos."

The Process:

1. **Flattening/compression occurs** with maximum stress (σ_1) applied perpendicular to the foliation
2. **Rigid grains resist deformation** and deflect the stress around them
3. **High pressure areas** develop directly above and below the rigid grain (parallel to σ_1)
4. **Low pressure areas** develop on the sides of the grain, normal (perpendicular) to σ_1
5. **Solution transfer occurs:** minerals dissolve from high-pressure areas and reprecipitate in low-pressure areas
6. **Pressure shadows form:** visible as new mineral fringes or halos around the rigid grain

3. Visual Appearance: What Pressure Shadows Look Like

Cross-sectional view of a grain with pressure shadows:

(Looking down perpendicular to foliation)

↑ Direction of σ_1 (maximum compression)

PRESSURE SHADOW

(Halo)

```

      |
    ||||
  |   |   |
  | ●●●● | ← New mineral material
  | ●GRAIN● |   reprecipitated in low-P
  | ●●●● |   area (solution transfer)
  |   |   |
    ||||

```

PRESSURE SHADOW
(Halo)

← Low pressure zones extend perpendicular to σ_1

In Thin Section: Pressure shadows appear as:

- Halos or fringes surrounding porphyroblasts (large mineral crystals)
- Finer-grained material than the main grain
- Often of different mineral composition (e.g., quartz or feldspar fringes)
- Located on opposite sides of the grain, perpendicular to foliation
- Classic indicator of **pre-kinematic porphyroblast growth** that was later deformed

4. Detailed Analysis of All Options

(a) Solution cleavage - ✗ INCORRECT

What it is: Solution cleavage refers to a **planar foliation** that develops through the selective dissolution of minerals along closely-spaced planes, perpendicular to the maximum stress direction.

Why it's wrong:

- Solution cleavage is a **large-scale foliation**, not localized zones around individual crystals
- It develops as a **pervasive planar structure** through an entire rock, not as isolated halos
- It represents **dissolution along planes**, not pressure shadows around porphyroblasts
- The question specifically asks about "zones...on the sides of the crystals," which is more localized than regional cleavage

This is too broad and scale-dependent to match the localized nature of pressure shadows.

(b) Pressure shadows - ✓ CORRECT

What it is: Localized zones of **relatively low pressure** that develop on the **sides of rigid grains, perpendicular to the maximum compressive stress direction**.

Why it's correct:

- **Matches the exact description in the question** — "zones of relatively low pressure on the sides of crystals normal to maximum compressive stress" ✓
- **Stress-dependent feature** — directly related to the orientation of the stress field ✓
- **Forms around rigid grains** — as specified in the question ✓
- **Geometric relationship is precise** — low-pressure zones are perpendicular to σ_1 ✓
- **Winter's textbook explicitly defines this structure** ✓

This is the definitive answer for zones of low pressure on the sides of rigid grains perpendicular to maximum compressive stress.

(c) Pressure solution - ✗ INCORRECT

What it is: Pressure solution (or solution transfer) is a **process or mechanism** by which minerals dissolve under pressure and reprecipitate in lower-pressure areas.

Why it's wrong:

- **It's a PROCESS, not a structural feature** — the question asks for a named structure ("these zones are called...")
- Pressure solution is the **mechanism that CREATES pressure shadows**, not the name for the shadows themselves
- The question is asking for the **NAME of the zones**, not the process that forms them

- While pressure solution is involved in pressure shadow formation, it's not what the low-pressure zones are "called"

Pressure solution is the cause; pressure shadows are the result.

(d) Stylolites - ✗ INCORRECT

What it is: Stylolites are **irregular, jagged seams or surfaces** found primarily in soluble rocks like limestone and salt, where differential pressure solution has dissolved material preferentially along certain zones.

Why it's wrong:

- **Stylolites are jagged seams of dissolved material**, not organized pressure shadow halos ✗
- **Found primarily in carbonate rocks** (limestone, dolomite), not necessarily in metamorphic rocks with silicate minerals ✗
- **Different morphology** — stylolites have a distinctive sawtooth or sutured appearance, not the smooth halos of pressure shadows ✗
- **Not specifically related to rigid grains** — stylolites can form independently as dissolution seams ✗
- **Different geological context** — stylolites are more common in burial diagenesis and early diagenesis, not metamorphism ✗

Stylolites are a different structure in different rock types.

Key Distinction: Process vs. Structure

This question tests the understanding of the difference between:

- **Pressure solution** (the PROCESS: dissolution and reprecipitation)
- **Pressure shadows** (the RESULT: low-pressure zones around rigid grains)

The question asks for what the zones are "called," making "pressure shadows" the correct structural name for these geometric features.

5. Pressure Shadows as Kinematic Indicators

In structural geology and metamorphic petrology, pressure shadows (also called "pressure fringes") serve as important indicators of deformation history:

From Winter's Textbook:

"Pre-kinematic crystals...include...pressure shadows, porphyroclasts with mortar texture or sheared mantles, etc."

"When pressure shadows are present, or the external foliation is deflected, the porphyroblast is probably **pre-kinematic**, but there could be several kinematic events."

Interpretation: If a porphyroblast (large crystal) has pressure shadows, it means:

- The crystal grew **BEFORE the main deformation** (pre-kinematic)
- During subsequent deformation, stress built up around the rigid grain
- Solution transfer created the low-pressure shadows on the sides
- This provides chronological information about crystal growth relative to foliation development

Primary Source:

Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology (3rd ed.). Pearson Education.

Key Sections Referenced:

- Structures and Textures of Metamorphic Rocks (Chapter 26)
- Pre-kinematic and Syn-kinematic Crystal Growth
- Deformation Versus Metamorphic Mineral Growth
- Pressure Solution and Solution Transfer mechanisms

Solution prepared by: Tapaswi Saikia

Geology Student | DMG Entrance Exam Preparation

Based on: Winter, J. D. Principles of Igneous and Metamorphic Petrology

Q.7 Matching: Metamorphic Rocks and Their Characteristics

Solution based on Winter's Principles of Igneous and Metamorphic Petrology

Question Q.7: Match List-I with List-II

List-I (Rocks)

- A.** Blueschist
- B.** Pseudotachylite
- C.** Hornfels
- D.** Skarn

List-II (Descriptions)

- 1.** Ultrafine-grained vitreous looking material occurring as thin veins
- 2.** Na-amphibole containing schistose rock
- 3.** A non-schistose substantial recrystallization
- 4.** A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

✓ **CORRECT ANSWER: (b) A-2, B-1, C-3, D-4**

The correct matching is:

- **A. Blueschist** → **2** Na-amphibole containing schistose rock
- **B. Pseudotachylite** → **1** Ultrafine-grained vitreous looking material occurring as thin veins
- **C. Hornfels** → **3** A non-schistose substantial recrystallization

- **D. Skarn** → **4** A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

1. Detailed Match Analysis

A. BLUESCHIST → 2. Na-amphibole containing schistose rock

From Winter's Textbook:

"Blueschist. A blue-amphibole-bearing metamorphosed mafic igneous rock or mafic graywacke. This term is so commonly applied to such rocks that it is even applied to non-schistose rocks.

Glaucophane is the most common blue amphibole, and glaucophane schist has been commonly applied to rocks known to contain it."

Key Characteristics:

- **Na-amphibole (Glaucophane)** is the defining mineral — a sodium-rich amphibole
- Forms as a **schistose (foliated) rock**
- **Blue color** imparted by the sodium amphibole
- Characteristic of **high-pressure, low-temperature metamorphism** (subduction zones)
- The mineral assemblage is essential to identifying blueschist facies

✓ **Blueschist is a schistose rock defined by the presence of Na-amphibole (glaucophane)**

B. PSEUDOTACHYLITE → 1. Ultrafine-grained vitreous looking material occurring as thin veins

From Winter's Textbook:

"In extreme cases of high-strain deformation, thin, commonly anastomosing seams of glassy rock, known as pseudotachylite, are generated. The glass is attributed to melting due to frictional heat."

"Pseudotachylite is produced by **localized rapid fragmentation and melting**

due to shear heating, generally attributed to earthquake shock energy in dry rocks."

Key Characteristics:

- **Glassy (vitreous) appearance** — produced by friction melting
- Occurs as **thin veins or seams** in high-strain zones (shear zones)
- **Ultrafine-grained** due to rapid cooling of the melt
- Forms in **fault zones** and **shear zones** during earthquakes
- The dark glass appearance is highly diagnostic
- Forms at depth in metamorphic conditions

✓ **Pseudotachylite is glassy, thin veinlets produced by friction melting in shear zones**

C. HORNFELS → 3. A non-schistose substantial recrystallization

From Winter's Textbook:

"A hornfels is a type of granofels that is typically very fine grained and compact, and it occurs in contact aureoles. Hornfelses are tough and tend to splinter or display conchoidal fracture when broken."

"The general term for any isotropic rock (a rock with no preferred orientation) is a granofels... A hornfels is...typically very fine grained and compact."

Key Characteristics:

- **Non-schistose (non-foliated)** — no preferred mineral orientation
- **Substantial recrystallization** without deformation
- Occurs in **contact metamorphic aureoles** around igneous intrusions
- **Fine-grained, hard, and compact** texture
- Shows **conchoidal fracture** when broken
- Random textural fabric (granofels texture)

✓ **Hornfels is a non-foliated contact metamorphic rock with extensive recrystallization**

D. SKARN → 4. A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

From Winter's Textbook:

"Skarn. A calc-silicate rock formed by contact metamorphism and silica metasomatism from a pluton into an adjacent carbonate rock ."

*"A skarn is a type of calc-silicate rock formed by **metasomatic interactions between carbonates and silicate-rich rocks or fluids.**"*

"As hot, acidic, silica-rich waters are driven from the pluton into the country rocks they react with the carbonates, producing a variety of calc-silicate minerals in a rock type called a skarn."

Key Characteristics:

- **Ca-Fe-Mg-Mn-silicate minerals** (grossular, diopside, wollastonite, etc.)
- Forms by **metasomatism** — chemical alteration by infiltrating fluids
- **Compositional zones and bands** develop due to diffusion gradients
- Occurs at **contact between carbonate rocks and silicate intrusions**
- SiO₂-rich fluids from the pluton react with carbonates
- Shows **layering and banding** in mineral composition

✓ **Skarn is a metasomatic calc-silicate rock with Ca-Fe-Mg-Mn minerals and compositional banding**

2. Summary Table: Metamorphic Rocks Classified

Rock Type	Primary Characteristic	Metamorphic Setting	Defining Minerals	Foliation
Blueschist	Na-amphibole (glaucophane) bearing	High-P/Low-T subduction zones	Glaucophane, lawsonite, garnet	Present (schistose)
Pseudotachylite	Glassy, friction-melted veins	Fault zones and shear zones	Glass (amorphous)	None (glassy)
Hornfels	Fine-grained, recrystallized, non-foliated	Contact metamorphic aureoles	Varies with protolith; andalusite, cordierite	Absent (non-schistose)
Skarn	Metasomatic Ca-silicate minerals with banding	Contact between carbonate and silicate rocks	Diopside, grossular, wollastonite, vesuvianite	Variable; often banded

3. Key Distinguishing Features

Foliation vs. Non-Foliation:

Understanding whether a rock is foliated (schistose) or non-foliated is crucial for classification:

- **Blueschist** — SCHISTOSE (foliated) because it forms under regional metamorphism with stress
- **Hornfels** — NON-SCHISTOSE (non-foliated) because it forms under low deviatoric stress in contact aureoles
- **Pseudotachylite** — GLASSY (neither typical schistosity nor granofels texture)
- **Skarn** — VARIABLE (can show banding and compositional layering from metasomatism)

Metasomatic vs. Metamorphic Processes:

Distinguish between rocks formed by pure metamorphism and those involving mass transfer:

- **Blueschist:** Pure metamorphism (no significant mass transfer)
- **Pseudotachylite:** Thermal melting (extreme case, no mass transfer)
- **Hornfels:** Pure contact metamorphism (minimal mass transfer)
- **Skarn:** METASOMATISM (significant mass transfer by fluids) — defining feature!

Formation Environment:

Each rock type is diagnostic of specific tectonic/metamorphic environments:

- **Blueschist:** Subduction zones (cold, high pressure)
- **Pseudotachylite:** Fault/shear zones at any depth (earthquake-related)
- **Hornfels:** Contact aureoles around hot igneous intrusions
- **Skarn:** Contact zones with carbonate rocks + silicate-rich fluids

4. Why Other Matchings Would Be Wrong

Testing incorrect options:

Option (a): A-4, B-3, C-1, D-2

X A-4: Blueschist is NOT "metasomatic Ca-silicate" — it's a Na-amphibole rock

X B-3: Pseudotachylite is NOT "non-schistose recrystallization" — it's glassy

X C-1: Hornfels is NOT "vitreous material in veins" — that's pseudotachylite

X D-2: Skarn is NOT "Na-amphibole schistose" — that's blueschist

Option (c): A-2, B-3, C-1, D-4

- ✓ A-2: Correct
- ✗ B-3: Pseudotachylite is NOT hornfels-like; it's glassy, not granofels
- ✗ C-1: Hornfels is NOT pseudotachylite; it's non-schistose recrystallized rock
- ✓ D-4: Correct

Option (d): A-3, B-4, C-1, D-2

- ✗ A-3: Blueschist is schistose, not "non-schistose recrystallization"
- ✗ B-4: Pseudotachylite is NOT metasomatic skarn
- ✗ C-1: Hornfels is NOT glassy pseudotachylite
- ✗ D-2: Skarn is NOT blueschist with Na-amphibole

Primary Source:

Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology (3rd ed.). Pearson Education.

Key Sections Referenced:

- A Classification of Metamorphic Rocks
- Metamorphic Facies and Metamorphosed Mafic Rocks
- Contact Metamorphism
- Structures and Textures of Metamorphic Rocks (pseudotachylite)
- Metamorphic Fluids, Mass Transport, and Metasomatism (skarn)

Solution prepared by: Tapaswi Saikia

Geology Student | DMG Entrance Exam Preparation

Based on: Winter, J. D. Principles of Igneous and Metamorphic Petrology

Q.8: Interpreting Protoliths from Granulite Facies Mineral Assemblages

Solution based on Winter's Principles of Igneous and Metamorphic Petrology

Question Q.8:

Presence of calcite, dolomite, quartz, diopside, wollastonite, K-feldspar, forsterite in a regionally metamorphosed rock, under granulite facies suggests which one of the following types of protoliths?

- (a) Mafic
- (b) Pelitic
- (c) Carbonate rock
- (d) Ultramafic

✓ **CORRECT ANSWER: (c) Carbonate rock**

The presence of **calcite and dolomite** (original carbonate minerals) together with **diopside, wollastonite, and forsterite** (calc-silicate metamorphic minerals) unequivocally indicates a **carbonate rock protolith** (limestone or dolostone).

1. Mineral Assemblage Analysis

Given Mineral Assemblage:

- Calcite, Dolomite, Quartz, Diopside, Wollastonite, K-feldspar, Forsterite

Let's categorize these minerals by their origin and significance:

PRIMARY/RELICT MINERALS (from original protolith):

Calcite (CaCO_3) and Dolomite ($\text{CaMg}(\text{CO}_3)_2$)

These are the two most common carbonate minerals. Their presence in a metamorphosed rock indicates a carbonate-rich original protolith.

METAMORPHIC CALC-SILICATE MINERALS:

Diopside ($\text{CaMgSi}_2\text{O}_6$) — Key Ca-Mg silicate

Wollastonite (CaSiO_3) — Key Ca silicate

Forsterite (Mg_2SiO_4) — Mg-rich olivine, characteristic of metamorphosed dolostones

These form from carbonate-silica reactions at high temperature:

Calcite + Quartz $\rightarrow$ Wollastonite + CO_2

Dolomite + Quartz $\rightarrow$ Diopside + CO_2 (+ related reactions)

Mg-rich reactions $\rightarrow$ Forsterite

CLASTIC COMPONENT MINERALS:

Quartz (SiO_2) — From original clastic silica grains

K-feldspar (KAlSi_3O_8) — From original clastic feldspar

These represent the non-carbonate clastic component mixed with the carbonates in the original sediment.

2. Textbook Evidence: Carbonate Rock Metamorphism

From Winter's Textbook:

"Calc-silicate rock (granofels or schist). A rock composed of various Ca-Mg-Fe-Al silicate minerals, such as grossular, epidote, tremolite, vesuvianite, etc.

The protolith is typically a limestone or dolostone with silica either originally present as clastic grains or introduced metasomatically. "

On Carbonate Metamorphism:

"Metamorphism of impure limestones and dolostones, on the other hand, is far more interesting and informative. A number of Ca-Mg silicates can form in metamorphosed siliceous dolostones under commonly attainable metamorphic conditions... When carbonate is subordinate, the metamorphic rock may be composed of Ca-Mg-Fe-Al silicate minerals, such as diopside, grossular, Ca-amphiboles, vesuvianite, epidote, wollastonite, etc. "

On Mineral Zones in Metamorphosed Dolostones:

"The simplest sequence of zones that typically develops (in order of increasing grade) is:

- Talc zone
- Tremolite zone
- Diopside and/or forsterite zones

Other minerals, such as wollastonite, monticellite, merwinite, etc., may also form in the innermost portions of very hot aureoles."

3. Characteristic Minerals and Their Significance

Mineral	Formula	Significance in Carbonate Rocks	In This Assemblage?
Calcite	CaCO_3	Original limestone component	✓ YES
Dolomite	$\text{CaMg}(\text{CO}_3)_2$	Original dolostone component	✓ YES
Diopside	$\text{CaMgSi}_2\text{O}_6$		✓ YES

Mineral	Formula	Significance in Carbonate Rocks	In This Assemblage?
		Diagnostic calc-silicate from dolomite + quartz	
Wollastonite	CaSiO ₃	Diagnostic calc-silicate from calcite + quartz	✓ YES
Forsterite	Mg ₂ SiO ₄	High-grade metamorphic olivine in dolostones	✓ YES
Quartz	SiO ₂	Original clastic silica component	✓ YES
K-feldspar	KAlSi ₃ O ₈	Original clastic feldspar from impure carbonates	✓ YES

Protolith Interpretation:

The original rock was an **impure limestone or dolostone** (carbonate rock) that contained:

- **Primary carbonate minerals:** Calcite and/or dolomite
- **Silica:** Originally present as clastic quartz grains or chert lenses
- **Clastic feldspar:** K-feldspar from original terrigenous input

During granulite facies metamorphism (high temperature, anhydrous conditions), carbonate-silica reactions produced the calc-silicate minerals (diopside, wollastonite, forsterite) while some original calcite and dolomite persisted.

4. Why Other Protoliths Don't Fit

(a) Mafic Protolith - ✗ INCORRECT

Why it fails:

- **Mafic rocks (basalts, gabbros) do NOT contain calcite or dolomite**
- Mafic granulites would show: orthopyroxene, clinopyroxene, plagioclase, garnet — NOT carbonate minerals
- The presence of **calcite and dolomite is incompatible with a mafic protolith**
- Calcite and dolomite in metamorphic rocks are diagnostic of carbonate protoliths

Conclusion: Carbonate minerals rule out mafic protoliths entirely.

(b) Pelitic Protolith - ✗ INCORRECT

Why it fails:

- **Pelitic rocks (shales, mudstones) do NOT contain significant calcite or dolomite**
- Pelitic granulites would show: K-feldspar, sillimanite, garnet, biotite — but NOT carbonate minerals
- While K-feldspar is present in the assemblage, it's not sufficient to indicate a pelitic protolith
- **Calcite and dolomite are major components** of the assemblage, not minor — they indicate carbonate protolith
- A pelitic rock metamorphosed to granulite would have devolatilized and lost carbonate minerals completely

Conclusion: The abundance of primary carbonate minerals rules out pelitic protoliths.

(d) Ultramafic Protolith - ✗ INCORRECT

Why it fails:

- **Ultramafic rocks (peridotites, dunites) are composed of olivine, pyroxene, serpentine — NO carbonates**
- Ultramafic metamorphic rocks do NOT produce calcite, dolomite, or calc-silicate minerals like wollastonite
- While **forsterite is present**, it's an olivine that forms from carbonate-silica reactions, not the primary ultramafic olivine
- **Calcite and dolomite are completely incompatible with ultramafic protoliths**

Conclusion: No carbonate minerals in ultramafic rocks means this protolith is ruled out.

(c) Carbonate Rock - ✓ CORRECT

Why it's correct:

- ✓ **Calcite and dolomite are the ORIGINAL carbonate minerals**
- ✓ **Diopside and wollastonite are DIAGNOSTIC calc-silicate minerals**
- ✓ **Forsterite forms from Mg-rich carbonate-silica reactions in metamorphosed dolostones**
- ✓ **Quartz from original clastic silica component**
- ✓ **K-feldspar from clastic feldspar in impure carbonates**
- ✓ **This exact assemblage is described in Winter's textbook as typical of metamorphosed limestone/dolostone**

Perfect match with textbook descriptions of carbonate rock metamorphism.

Key Principle: Carbonate Minerals as Protolith Indicators

The presence of **calcite and dolomite in a metamorphic rock is the single most important clue that the protolith was a carbonate rock** (limestone or dolostone). These minerals:

- Are **exclusive to carbonate protoliths** — no other rock type produces them during metamorphism
- Persist at granulite facies temperatures because they are stable at very high temperatures
- **Directly identify the original rock type** without ambiguity
- Often coexist with **calc-silicate minerals** (diopside, wollastonite) that form from carbonate-silica reactions

Rule: If you see calcite or dolomite in a metamorphic rock, it almost certainly came from a carbonate rock protolith.

5. CMS System and Metamorphosed Carbonates

Winter's textbook describes the metamorphism of impure carbonate rocks using the CMS-HC system (CaO-MgO-SiO₂-H₂O-CO₂). The major minerals in this system include:

Typical Minerals in Metamorphosed Carbonates:

Carbonate minerals: Calcite (CaCO₃), Dolomite (CaMg(CO₃)₂)

Silicates from C-M-S reactions: Diopside (Ca-Mg silicate), Wollastonite (Ca silicate), Forsterite (Mg silicate)

Other minerals: Quartz, K-feldspar, Grossular garnet

The question's assemblage fits perfectly within this system, with no anomalies or unexpected minerals.

Primary Source:

Winter, J. D. (2014). Principles of Igneous and Metamorphic Petrology (3rd ed.). Pearson Education.

Key Sections Referenced:

- Chapter 29: Metamorphism of Calcareous and Ultramafic Rocks
- Calc-silicate Rocks and Their Metamorphism

- Metamorphosed Limestone and Dolostone
 - CMS-HC Chemical System and Phase Equilibria
-

Solution prepared by: Tapaswi Saikia

Geology Student | DMG Entrance Exam Preparation

Based on: Winter, J. D. Principles of Igneous and Metamorphic Petrology

Eclogite Facies Metamorphic Reaction

Grossular Composition Analysis

Q.9: $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$ is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

(b) CaSiO_3

(c) $(\text{Ca}, \text{Mg})\text{Si}_2\text{O}_6$

(d) CaCO_3

✓ CORRECT ANSWER: (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

1. TEXTBOOK REFERENCE & COMPOSITION OF GROSSULAR

According to Perkins (2014), grossular is the aluminum end-member of the ugrandite series of garnets. The definitive composition is explicitly stated as:

"Grossular, $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, has $\text{CaO}:\text{Al}_2\text{O}_3:\text{SiO}_2$ ratio 3:1:3, equivalent to 43% CaO, 14% Al_2O_3 , and 43% SiO_2 ." (Perkins, 2014, p. 107)

2. CHEMICAL VERIFICATION OF THE REACTION

To confirm that option (a) is correct, we can verify the stoichiometry of the given reaction:

Phase	Chemical Formula	Coefficient
Anorthite (plagioclase)	$\text{CaAl}_2\text{Si}_2\text{O}_8$	3×
↓		
Grossular (garnet)	$\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	1×
Kyanite (aluminosilicate)	Al_2SiO_5	2×
Quartz (silica)	SiO_2	1×

Left side (Reactant):

$$3 \times (\text{CaAl}_2\text{Si}_2\text{O}_8) = \text{Ca}_3\text{Al}_6\text{Si}_6\text{O}_{24}$$

Right side (Products):

$$\begin{aligned} &\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + 2(\text{Al}_2\text{SiO}_5) + \text{SiO}_2 \\ &= \text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + \text{Al}_4\text{Si}_2\text{O}_{10} + \text{SiO}_2 \end{aligned}$$



✓ The equation is perfectly balanced, confirming the composition of grossular as $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

3. GARNET CLASSIFICATION & GROSSULAR

Perkins (2014) classifies garnets into two series based on their chemical composition:

"Mineralogists conveniently divide garnets into two series, the pyralspites (pyrope-almandine-spessartine) and the ugrandites (uvarovite-grossular-andradite)." (Perkins, 2014, p. 360)

Series	Composition Formula	End Members
Pyralspites	$(\text{Mg,Fe,Mn})_3\text{Al}_2\text{Si}_3\text{O}_{12}$	Pyrope, Almandine, Spessartine
Ugrandites	$\text{Ca}_3(\text{Cr,Al,Fe})_2\text{Si}_3\text{O}_{12}$	Uvarovite, Grossular, Andradite

"Grossular is the Al end member of the ugrandite series, $\text{Ca}_3[\text{Fe}^{3+},\text{Al}^{3+},\text{Cr}^{3+}]_2\text{Si}_3\text{O}_{12}$, but natural grossular may contain appreciable amounts of Fe^{3+} or Cr^{3+} in solid solution." (Perkins, 2014, p. 364)

4. ECLOGITE FACIES CONTEXT

This reaction is indeed diagnostic of the eclogite facies in metabasic (mafic) rocks. According to the textbook:

"Eclogites are mafic rocks metamorphosed at high pressure and moderate-to-high temperature. They contain the essential minerals pyrope (Mg-rich garnet) and the green Na-rich clinopyroxene called omphacite. Orthopyroxene may also be present in significant quantities. Accessory minerals include kyanite, quartz, spinel, titanite, and many others." (Perkins, 2014, p. 204)

The appearance of kyanite (high-pressure polymorph of Al_2SiO_5) along with garnet and quartz is characteristic of high-pressure metamorphic conditions. Grossular can appear in eclogite-facies rocks when metabasic protoliths are rich in aluminum and calcium-bearing plagioclase.

5. OPTION ANALYSIS

Option	Mineral/Phase	Reasoning
(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	✓ Grossular Garnet	Correct composition of the aluminum end-member of the ugrandite garnet series. Chemically balances the reaction equation perfectly.
(b) CaSiO_3	Wollastonite	A calcium silicate but wrong composition. This mineral forms in lower-pressure marbles, not in the eclogite facies. Lacks aluminum.
(c) $(\text{Ca,Mg})\text{Si}_2\text{O}_6$	Diopside (pyroxene)	A calcium-magnesium silicate. While found in eclogites, it does not balance the reaction equation. Also lacks sufficient aluminum.
(d) CaCO_3	Calcite	A carbonate mineral with no silicate or aluminum content. Chemically unrelated to the anorthite reactant or the products of this metamorphic reaction.

6. SUMMARY

The composition of grossular in this eclogite facies reaction is $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, which is the pure end-member composition of the grossular garnet. This mineral represents one of the garnet phases stable at high pressure and temperature in metamorphic rocks derived from aluminum-rich mafic protoliths.

The reaction demonstrates the transformation of plagioclase-rich igneous rocks (anorthite) into high-pressure metamorphic assemblages dominated by garnet (grossular), the high-pressure aluminosilicate polymorph

(kyanite), and excess silica (quartz), which is characteristic of eclogite-facies metamorphism.

Source: Perkins, D. (2014). *Mineralogy* (3rd ed.). Pearson Education Limited.

Author: By Tapaswi Saikia

Date: June 2026

Metamorphic Mineral Reactions

Question Bank from Perkins, D. (2014). Mineralogy (3rd Ed.)

Comprehensive Exam Preparation

Q.1: ECLOGITE FACIES REACTION

Question: $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$ is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

(b) CaSiO_3

(c) $(\text{Ca}, \text{Mg})\text{Si}_2\text{O}_6$

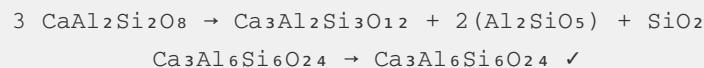
(d) CaCO_3

✓ **CORRECT ANSWER: (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$**

Explanation:

Grossular is the aluminum end-member of the ugrandite garnet series. According to Perkins (2014, p. 107): "*Grossular, $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$, has $\text{CaO}:\text{Al}_2\text{O}_3:\text{SiO}_2$ ratio 3:1:3*"

Chemical Verification:



The equation balances perfectly, confirming the composition. Grossular belongs to the eclogite facies assemblage at high pressure and moderate-to-high temperature.

Q.2: MARBLE METAMORPHIC REACTION (High-Grade)

Question: When a marble (metamorphosed limestone) containing both calcite and quartz is metamorphosed at high temperature, a characteristic reaction occurs. What is the complete chemical equation for the reaction that produces wollastonite?

Hint: The reaction involves calcite and quartz as reactants.

- (a) $\text{CaCO}_3 + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2$
- (b) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + 2 \text{ Quartz}$
- (c) $\text{CaCO}_3 + 2 \text{ SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2 + \text{SiO}_2$
- (d) $2 \text{ CaCO}_3 + \text{SiO}_2 \rightarrow 2 \text{ CaSiO}_3 + 2 \text{ CO}_2$

✓ **CORRECT ANSWER: (a) $\text{CaCO}_3 + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2$**

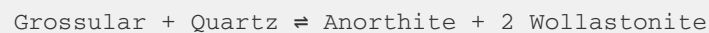
Explanation:

This is a decarbonation reaction occurring at high temperature in marbles. Perkins (2014, p. 193) states: "Calcite (CaCO_3) and quartz (SiO_2) may react to form wollastonite (CaSiO_3) if a limestone containing quartz is metamorphosed at high temperature."

Key Features:

- **Product Minerals:** Wollastonite (CaSiO_3) + CO_2 (volatilized)
- **Rock Type:** Occurs in contact-metamorphosed limestones and calcic marbles
- **Temperature:** High-temperature metamorphism
- **Associated Minerals:** Calcite, dolomite, tremolite, diopside, grossular, epidote

This reaction is shown on a phase diagram in Figure 18 of Perkins (2014) with the full reaction:



Q.3: LOW-GRADE METAMORPHIC REACTION (Kaolinite-Quartz)

Question: In low-temperature metamorphism of clay-rich rocks, kaolinite and quartz react to form a higher-grade mineral. This reaction marks the transition from diagenesis to low-grade metamorphism. What are the products of this reaction?

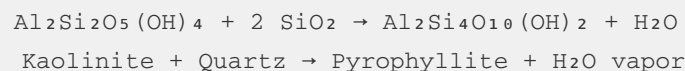


- (a) Pyrophyllite + H₂O vapor
- (b) Kyanite + 3 Quartz + H₂O
- (c) Muscovite + Quartz
- (d) Andalusite + SiO₂ + H₂O

✓ **CORRECT ANSWER: (a) Pyrophyllite + H₂O vapor**

Explanation:

According to Perkins (2014, p. 209), this is Reaction 17:



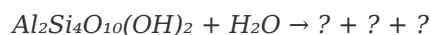
"Reaction 17 is a dehydration reaction that takes place in some metapelitic rocks at low temperature. At low temperature, kaolinite and quartz are stable together, but as temperature increases, kaolinite and quartz react to produce pyrophyllite and H₂O vapor." (Perkins, 2014, p. 209)

Important Note: Pyrophyllite itself breaks down at slightly higher temperatures (Reaction 18) to form kyanite and quartz. Therefore, pyrophyllite may only be stable over a temperature range of about 100°C.

Metamorphic Grade: This reaction is characteristic of the transition zone between diagenesis and very low-grade metamorphism.

Q.4: PYROPHYLLITE BREAKDOWN REACTION (Low-Grade to Medium-Grade Transition)

Question: *Pyrophyllite is unstable at temperatures higher than about 400°C. It breaks down to form the high-pressure aluminosilicate polymorph. What is the balanced equation for this dehydration reaction (Reaction 18)?*

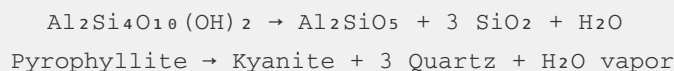


- (a) Kyanite + 3 Quartz + H₂O vapor
- (b) Andalusite + 3 SiO₂ + H₂O
- (c) Sillimanite + 2 SiO₂ + H₂O vapor
- (d) Corundum + 3 Quartz + H₂O

✓ **CORRECT ANSWER: (a) Kyanite + 3 Quartz + H₂O vapor**

Explanation:

This is Reaction 18 from Perkins (2014, p. 209):



"Pyrophyllite itself breaks down at slightly higher temperatures by Reaction 18, so pyrophyllite may only be stable over a range of about 100 °C" (Perkins, 2014, p. 209)

Key Points:

- **Kyanite is the high-pressure polymorph** of Al₂SiO₅
- **Temperature Range:** Pyrophyllite stability: 300-400°C (with H₂O present)
- **Metamorphic Context:** Marks transition from low-grade to medium-grade metamorphism
- **Phase Diagram:** Shown in Figure 13 of Perkins (2014, p. 209)

Pressure-Temperature Significance: These reactions demonstrate how the Al₂O₃-SiO₂-H₂O system controls metamorphic grade in pelitic rocks.

Q.5: MEDIUM-GRADE METAMORPHIC GARNET REACTION

Question: In medium-grade metamorphism of pelitic rocks, almandine garnet ($\text{Fe}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) is a key index mineral. It represents an important metamorphic reaction. What is the lowest temperature at which garnet becomes stable in pelitic rocks?

- (a) $\sim 200^\circ\text{C}$ (greenschist facies)
- (b) $\sim 500^\circ\text{C}$ (medium-grade metamorphism)
- (c) $\sim 700^\circ\text{C}$ (high-grade metamorphism)
- (d) Garnet has no lower temperature limit

✓ **CORRECT ANSWER: (b) $\sim 500^\circ\text{C}$ (medium-grade metamorphism)**

Explanation:

According to Perkins (2014, p. 210):

"Consider a rock that contains garnet. The phase diagram shows that the rock must have formed at temperatures greater than about 500°C (930°F). Garnet does not, however, have an upper temperature limit."

Garnet as an Index Mineral:

- **Lower Limit:** $\sim 500^\circ\text{C}$ (entrance to garnet zone)
- **Upper Limit:** No upper limit (stable to very high grades)
- **Composition:** In metapelites, almandine ($\text{Fe}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) is most common
- **Associations:** Often found with staurolite, kyanite, sillimanite

Garnet Series: Perkins (2014, p. 360) classifies garnets as:

- **Pyrospites:** $(\text{Mg,Fe,Mn})_3\text{Al}_2\text{Si}_3\text{O}_{12}$ (pyrope, almandine, spessartine)
- **Ugrandites:** $\text{Ca}_3(\text{Cr,Al,Fe})_2\text{Si}_3\text{O}_{12}$ (uvarovite, grossular, andradite)

COMPARATIVE METAMORPHIC REACTIONS SUMMARY

Reaction #	Mineral System	Reactants	Products	Metamorphic Grade
Q.1	Eclogite (Mafic)	3 Anorthite	Grossular + 2 Kyanite + Qz	High-P, Mod-T
Q.2	Marble (Carbonate)	Calcite + Quartz	Wollastonite + CO ₂	High-T
Q.3	Pelite (Clay-rich)	Kaolinite + 2 Quartz	Pyrophyllite + H ₂ O	Low-grade
Q.4	Pelite (Clay-rich)	Pyrophyllite + H ₂ O	Kyanite + 3 Quartz + H ₂ O	Low-Med transition
Q.5	Pelite (Clay-rich)	Various precursors	Almandine Garnet	Medium-grade (>500°C)

METAMORPHIC FACIES CONTEXT (from Perkins, 2014)

Low-Grade Facies:

Zeolite → Prehnite-Pumpellyite → Greenschist → Blueschist/Lawsonite-Albite
Temperature Range: 0-300°C | Contains: Chlorite, Pyrophyllite, Kaolinite

Medium-Grade Facies:

Amphibolite (Regional Metamorphism)
Temperature Range: 500-700°C | Contains: Garnet, Staurolite, Kyanite, Sillimanite

High-Grade Facies:

Granulite and Eclogite (High-P/High-T)
Temperature Range: >700°C | Pressure: >10 kbar for Eclogite
Contains: Garnet + Pyroxene ± Kyanite ± Quartz

STUDY TIPS FOR METAMORPHIC REACTIONS

Key Concepts to Remember:

1. **Always Balance Chemical Equations:** Check that atoms on both sides match
2. **Recognize Reaction Types:** Dehydration (loses H₂O), Decarbonation (loses CO₂), Solid-Solid
3. **Know Mineral Compositions:** Learn the formulas of common metamorphic minerals
4. **Understand P-T Conditions:** Different reactions occur at different temperature/pressure combinations
5. **Index Minerals Matter:** Garnet (~500°C), Staurolite (460-700°C), Kyanite (high-P), Sillimanite (high-T)
6. **Rock Type Context:** Marbles (carbonate), Pelites (clay-rich), Metabasites (mafic)
7. **Reference Key Pages:** Perkins (2014) Figures 13, 14, 17, 18 show phase diagrams

Source: Perkins, D. (2014). *Mineralogy* (3rd ed.). Pearson Education Limited.

Author: By Tapaswi Saikia

Compiled: June 2026

This question bank provides comprehensive exam preparation based on textbook content. All reactions and citations are directly from Perkins (2014).

Q.10 Granoblastic polygonal texture is an example of

Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* | Chapter: Structures and Textures of Metamorphic Rocks

ORIGINAL QUESTION — DMG 2019

Q.10. Granoblastic polygonal texture is an example of

- (a) Texture of recrystallization
- (b) Texture of crystallization
- (c) Texture of metasomatism
- (d) Texture of mylonitization

✓ Correct Answer: (a) Texture of Recrystallization



Granoblastic polygonal texture is produced by static recrystallization (annealing) — not primary crystallization, metasomatism, or mylonitization.

DIRECT TEXTBOOK EVIDENCE

The following passages from Winter (2010) directly establish granoblastic polygonal texture as a product of **recrystallization**:

*"In cases of advanced recrystallization, grain boundary area reduction straightens the boundaries further, **eventually producing a granoblastic polygonal texture that is completely recrystallized (annealed).**"*

— Winter, J.D., *Principles of Igneous and Metamorphic Petrology*, p. 509 (Section 3: High-Strain Metamorphic Textures)

*"The hallmark texture of contact metamorphism in a low-pressure relatively static environment is **granoblastic polygonal**. It develops best in monomineralic quartz or feldspar regions and is characterized by polygonal grains with straight boundaries, typically meeting at three-grain boundaries in thin section at 120° angles."*

— Winter, J.D., *Principles of Igneous and Metamorphic Petrology*, p. 536 (Summary section)

*"Static recrystallization occurs after deformation ceases at elevated temperatures or when a thermal disturbance occurs in a low-stress environment... The texture is called **granoblastic polygonal** (or polygonal mosaic), and grains appear in two-dimensional thin sections as equidimensional polygons with grain boundaries that meet in triple junctions with approximately 120° between them."*

— Winter, J.D., *Principles of Igneous and Metamorphic Petrology*, pp. 504–505 (Section 2: Textures of Contact Metamorphism)

*"Table 1 – Textures of Metamorphic Rocks: **Non-Foliated Texture: Granoblastic Polygonal (polygonal mosaic)** — Equidimensional generally anhedral crystals of approximately equal size with regular polygonal shapes (usually 5- or 6-sided) and fairly straight boundaries meeting at triple junctions."*

— Winter, J.D., *Principles of Igneous and Metamorphic Petrology*, Table 1, p. 537

VISUAL: GRANOBLASTIC POLYGONAL TEXTURE

Formation stages of granoblastic polygonal texture by progressive static recrystallization. Adapted from Winter (2010), Figures 9 & 16.

OPTION-BY-OPTION ANALYSIS

✓ (a) Texture of Recrystallization

CORRECT

Granoblastic polygonal texture is the direct product of **static recrystallization** (also called annealing). When deformation ceases at elevated temperatures, grain boundary migration reduces both lattice strain energy and surface energy, producing equidimensional polygonal grains with straight boundaries meeting at $\sim 120^\circ$ triple junctions. Winter explicitly states this texture is "completely recrystallized (annealed)" (p. 509) and lists it under annealing/static recrystallization in his summary (p. 536).

✗ (b) Texture of Crystallization

WRONG

Primary crystallization refers to the first-time growth of minerals from a melt (igneous) or supersaturated fluid (hydrothermal). Granoblastic polygonal texture is a *solid-state* reorganization of pre-existing grains — not growth from melt. Winter distinguishes recrystallization from primary crystallization explicitly: the process acts to "reduce both lattice strain energy and overall surface energy" in already-crystalline material. Option (b) conflates two fundamentally different processes.

✗ (c) Texture of Metasomatism

WRONG

Metasomatism involves **chemical replacement** of one mineral by another through fluid infiltration and ion exchange — it changes the bulk composition of a rock. Granoblastic polygonal texture involves no compositional change; it is a purely mechanical reorganization of grain boundaries. Metasomatic textures include pseudomorphs, symplectites at reaction fronts, and corona/kelyphite textures — not polygonal mosaics.

✗ (d) Texture of Mylonitization

WRONG

Mylonitization is the result of **high-strain ductile deformation** in shear zones and produces foliated rocks with ribbon quartz, σ - or δ -clasts, S-C fabrics, and fine-grained matrix from grain-size reduction. This is the exact *opposite* of granoblastic polygonal texture. Winter's Table 1 (p. 537) places granoblastic polygonal under

"Non-Foliated Texture" and mylonitization textures (ribbon, mortar, augen) under "Textures of Dynamic Metamorphism" — completely separate categories.

FULL CONCEPT: GRANOBLASTIC POLYGONAL TEXTURE

What is it? Granoblastic polygonal texture (also called *polygonal mosaic*) is a non-foliated metamorphic texture characterized by equidimensional, anhedral grains with straight boundaries that meet in triple junctions at approximately 120°.

4.1 How Does it Form?

It forms by **static recrystallization** (annealing) through two driving forces:

- **Lattice strain energy reduction** — deformed, dislocation-rich crystals are unstable and nucleate low-strain sub-grains.
- **Surface energy reduction** — grain boundary migration minimizes total grain boundary area. For isotropic minerals (quartz, calcite), this produces straight boundaries.

The process progresses: **wavy strained grains** → **sutured-serrated boundaries** → **polygonized sub-grains** → **straight-boundary granoblastic polygonal mosaic**

4.2 Geometric Rule: 120° Triple Junctions

When grain boundary energies are equal in all directions (isotropic surface energy, as in quartz and calcite), the minimum-energy configuration requires grain boundaries to meet at **exactly 120°** at triple junctions (monomineralic case). This is a consequence of surface tension equilibrium, analogous to soap bubbles.

4.3 Where Does it Occur?

- **Contact metamorphism** — the hallmark texture; forms in thermal aureoles at low pressure / static conditions (hornfels)
- **Post-kinematic regional metamorphism** — after deformation ceases, rocks anneal to polygonal texture
- **Monomineralic rocks** — best developed in quartzites and marbles; polymineralic rocks show modified angles due to differing grain boundary energies

4.4 Grain Size Control

"The size of the polygonal grains depends mainly on temperature and the presence of fluids (higher temperatures and aqueous fluids promote larger crystals)."

— Winter (2010), p. 505

4.5 Polymineralic Modification

In rocks with two or more minerals (e.g., quartz–pyroxene hornfels), the 120° rule is modified. A–B grain boundaries have *lower* energy than A–A boundaries, so the dihedral angle θ at A–A–B junctions decreases to maximize A–B contact area. True 120° angles appear only at A–A–A triple junctions.

COMPARISON: KEY NON-FOLIATED & DYNAMIC METAMORPHIC TEXTURES

Texture	Type	Key Features	Environment	Minerals
Granoblastic Polygonal	Static Recrystallization	Equidimensional polygons, straight boundaries, 120° triple junctions	Contact aureoles, post-kinematic annealing	Quartz, calcite (isotropic surface energy)
Decussate	Static Recrystallization	Randomly oriented subhedral prisms, no preferred orientation	Contact metamorphism	Mica, amphibole, pyroxene (anisotropic)
Porphyroblastic	Crystallization (prograde)	Large crystals in finer groundmass; may be pre/syn/post-kinematic	Regional and contact metamorphism	Garnet, staurolite, andalusite, cordierite

Texture	Type	Key Features	Environment	Minerals
Mylonitic	Dynamic recrystallization (high strain)	Foliation, ribbon quartz, porphyroclasts, C-S fabrics	Shear zones (ductile)	Quartz ribbons, feldspar augen
Cataclastic	Brittle deformation	Crushed, broken, cracked grains; no recrystallization	Shallow fault zones	Any; feldspars, quartz
Mortar Texture	Dynamic metamorphism	Rounded porphyroclasts rimmed by fine crush matrix	Fault/shear zones	Feldspar, quartz
Symplectite	Reaction texture	Intimate intergrowth of two minerals grown simultaneously	Retrograde reactions, decompression	Kelyphite, myrmekite
Corona / Moat	Reaction / Metasomatic	Rim of one mineral around another	Retrograde reactions	Garnet coronas on olivine

Source: Winter (2010), Table 1, pp. 537–538

⚙️ THREE TYPES OF RECRYSTALLIZATION: A QUICK REFERENCE

Type	When	Mechanism	Resulting Texture
Static Recrystallization (Annealing)	After deformation ceases, at high T in a low-stress environment	Grain boundary migration driven by surface + strain energy reduction	Granoblastic polygonal , decussate

Type	When	Mechanism	Resulting Texture
Dynamic Recrystallization	During deformation at high T	Simultaneous deformation + recovery; subgrain rotation + GBM	Schistosity, ribbon texture, sutured grains
Primary Crystallization	First growth from melt or fluid	Nucleation + growth from liquid or supersaturated solution	Igneous textures, hydrothermal vein minerals

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

Q1. The approximate angle at which grain boundaries meet at triple junctions in granoblastic polygonal texture is: (a) 60° (b) 90° (c) 120° (d) 180°

→ Answer: (c) 120°

Q2. Which rock type best exhibits granoblastic polygonal texture? (a) Phyllite (b) Schist (c) Quartzite (d) Ultramylonite

→ Answer: (c) Quartzite — monomineralic, isotropic surface energy

Q3. Granoblastic polygonal texture is the hallmark texture of: (a) Regional metamorphism (b) Dynamic metamorphism (c) Contact metamorphism (d) Metasomatism

→ Answer: (c) Contact metamorphism

Q4. Decussate texture differs from granoblastic polygonal texture in that: (a) It forms at higher pressures (b) It forms in minerals with strong anisotropy of surface energy (c) It is produced by cataclasis (d) It occurs only in carbonates

→ Answer: (b) — e.g., micas, amphiboles, pyroxenes

Q5. Which of the following textures is NOT a texture of recrystallization? (a) Decussate (b) Granoblastic polygonal (c) Ribbon texture (d) Polygonal mosaic

→ Answer: (c) Ribbon texture is a high-strain deformation texture (mylonitization)

Q6. In a polyminerale hornfels, grain boundaries meeting at triple junctions between minerals A and B will deviate from 120° because: (a) Temperature is too high (b) A–A boundaries have higher energy than A–B boundaries (c) Pressure is insufficient (d) Fluid infiltration changes compositions

→ Answer: (b) — *dihedral angle modification due to differential grain boundary energies*

Q7. The size of polygonal grains in granoblastic texture is primarily controlled by: (a) Original protolith grain size only (b) Deformation intensity (c) Temperature and presence of aqueous fluids (d) Pressure and burial depth

→ Answer: (c) — *higher T and fluids → larger crystals (Winter, p. 505)*

Q8. Post-kinematic recrystallization in a shear zone, after deformation has ceased, will produce: (a) Mylonite foliation (b) Cataclastic texture (c) Granoblastic or decussate texture (d) Pseudotachylite

→ Answer: (c) — *annealing after deformation produces polygonal texture*

QUESTION 11

In metamorphism of calc-silicate rocks, which assemblage can be considered equivalent to the sillimanite zone of pelitic rocks?

(a) Anorthite-pyroxene

(b) Zoisite-calcite-biotite

(c) Diopside-anorthite

✓ (d) Wollastonite-grossular

✓ CORRECT ANSWER: (d) Wollastonite-grossular

OPTION-BY-OPTION ANALYSIS

A ✗ INCORRECT

Anorthite-pyroxene

While anorthite does appear in calc-silicate rocks at higher grades, the "anorthite-pyroxene" assemblage is more characteristic of the

granulite facies

in mafic metamorphic rocks (amphibolite → granulite transition involves pyroxene replacing hornblende). It is not the recognised isograd-equivalent for the sillimanite zone in Kennedy's (1949) Scottish Highlands correlation. This assemblage is incomplete and does not correspond to any standard zone boundary in calc-silicates.

B ✗ INCORRECT

Zoisite-calcite-biotite

This is a

lower-grade assemblage

typical of the zoisite zone in calc-silicate rocks. In Ferry's (1983) Augusta-Waterville study, zoisite appears in the "zoisite zone" which corresponds to a grade *below* the diopside zone. The zoisite zone broadly correlates to the kyanite or lower-staurolite grade in pelites — well below the sillimanite zone. The presence of calcite also indicates lower metamorphic grade (calcite is consumed in high-grade reactions).

C ✗ INCORRECT

Diopside-anorthite

Diopside appears at the

diopside isograd

which, as Winter explicitly states, "occurs at about the grade of the sillimanite isograd in the pelites." While diopside is indeed a high-grade marker, the assemblage "diopside-anorthite" describes a *binary system* from igneous petrology (the classic Di-An eutectic system), not a metamorphic zone assemblage. Furthermore, in the Scottish Highlands correlation (Kennedy 1949), the sillimanite-equivalent in calc-silicates is specifically

wollastonite-grossular

, not diopside-anorthite. The diopside zone corresponds more broadly to the sillimanite zone onset, but "diopside-anorthite" is too generic and not the correct zone assemblage cited in the textbook.

D ✓ CORRECT

Wollastonite-grossular

This is the highest-grade calc-silicate assemblage that forms the **Garnet Zone**

in the Crestmore-type progression, with the reaction: vesuvianite + monticellite + diopside + wollastonite → grossular + diopside + wollastonite. Wollastonite forms from Cal + Qtz at temperatures >550°C at low P (and >750°C at higher P in closed systems), while grossular garnet (a Ca-Al garnet) is its highest-grade companion. Together, wollastonite + grossular characterise the peak metamorphic assemblage in calc-silicate rocks, directly equivalent to the sillimanite zone in pelites. Kennedy's (1949) Scottish Highlands study is the foundational reference establishing this correlation.

CONCEPT: METAMORPHISM OF CALC-SILICATE ROCKS — FULL CHAPTER OVERVIEW

Calc-silicate rocks are metamorphosed impure carbonates (limestones/dolostones with clay, quartz, feldspar). They develop a diagnostic sequence of mineral zones with increasing grade, just as pelitic rocks develop Barrovian zones. The correlation between pelitic and calc-silicate zones was first systematically established by **W. Q. Kennedy (1949)** in the Scottish Highlands, and later expanded by John Ferry (1983) in Maine.

CORE PRINCIPLE

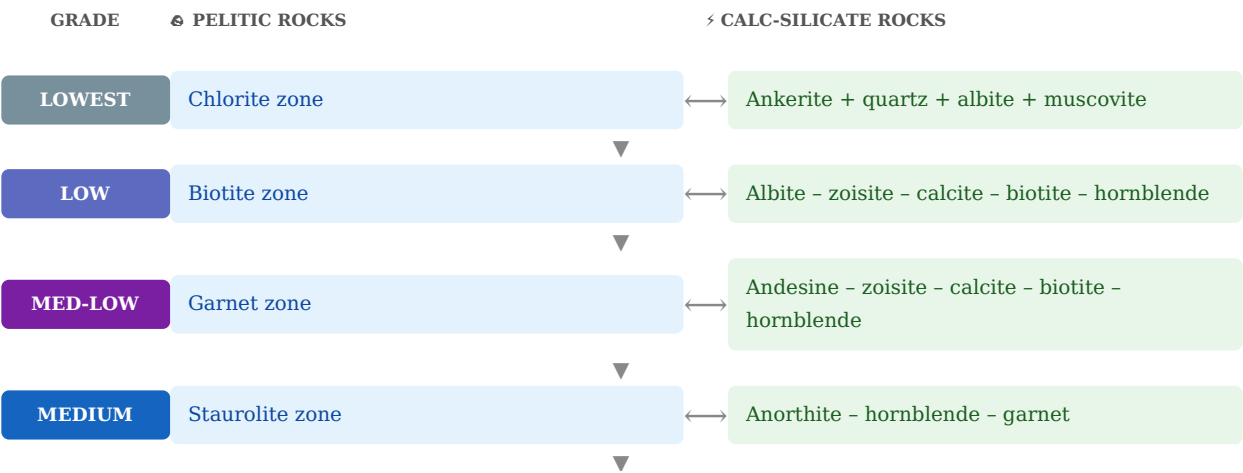
Because calc-silicate rocks and pelitic rocks co-exist in the same metamorphic terrane and experience the same P-T conditions, their mineral zones must be **isograde-equivalent** — they just reflect different bulk compositions responding to the same thermal gradient.

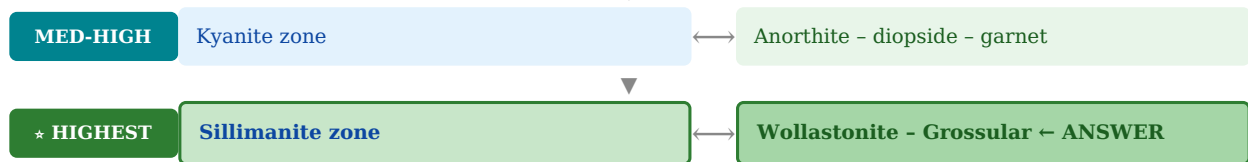
Kennedy's (1949) Pelitic ↔ Calc-Silicate Zone Correlation

After Kennedy (1949) and Tanner (1976), as cited in Winter (2010), Chapter 29, Table 1, p. 674

PELITIC ZONE (BARROVIAN)	CALC-SILICATE MINERAL ASSEMBLAGE	GRADE
Chlorite	—	Lowest
Biotite	Albite - zoisite - calcite - biotite - hornblende	Low
Garnet	Andesine - zoisite - calcite - biotite - hornblende	Lower-medium
Staurolite	Anorthite - hornblende - garnet	Medium
Kyanite	Anorthite - diopside - garnet	Medium-high
Sillimanite	Wollastonite - grossular	Highest

PROGRESSIVE METAMORPHISM: GRADE LADDER (CALC-SILICATE VS. PELITIC)

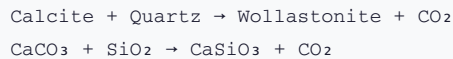




HOW WOLLASTONITE AND GROSSULAR FORM (KEY REACTIONS)

Wollastonite Formation

Wollastonite (CaSiO_3) is produced by the decarbonation of calcite with quartz — the "wollastonite isograd reaction":



This reaction occurs at $T > 550^\circ\text{C}$ at $P = 0.1 \text{ GPa}$ (if $X_{\text{CO}_2} > 0.2$), but may require $>750^\circ\text{C}$ in closed systems buffered to high X_{CO_2} . The wollastonite isograd is a classic **decarbonation reaction**.

Grossular Formation

Grossular garnet ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$) forms at high-grade conditions in calc-silicate rocks, as the highest-grade zone marker:

Grossular + Diopside + Wollastonite — Zone 10 (Garnet Zone)
 At Crestmore: this is the innermost and highest-grade assemblage

*"In this progression of zones, we can again see the sequential development of index minerals toward the igneous contact, such as clinohumite, followed by forsterite, clintonite, monticellite, melilite, spurrite/tilleyite, merwinite, vesuvianite, diopside, **wollastonite, and finally grossular garnet.**"*

— Winter (2010), Principles of Igneous and Metamorphic Petrology, Chapter 29 (Crestmore Aureole discussion, p. 486)

*"The diopside isograd in the calcareous rocks occurs at about the grade of the **sillimanite isograd in the pelites**. Diopside growth results from dehydration of amphibole at high grade..."*

— Winter (2010), Chapter 29, §1.4.5 Diopside Zone, p. 674

"Wollastonite may form in the upper amphibolite facies or in the granulite facies. Because of the high temperatures required, it is not common in regional metamorphic rocks."

— Winter (2010), Chapter 29, §1.2 Regional Metamorphism, p. 670

CONTACT VS. REGIONAL METAMORPHISM OF CALC-SILICATES

ASPECT	CONTACT METAMORPHISM	REGIONAL METAMORPHISM
Heat source	Igneous intrusion (pluton)	Geothermal/orogenic burial
Typical example	Crestmore, California; Alta, Utah	Scottish Highlands, Maine (Waterville)
Zone width	Narrow aureole (tens of metres to km)	Broad regional zones (tens of km)
Low-grade minerals	Talc, tremolite (in dolostones)	Tremolite (talc less common)
Highest-grade minerals	Wollastonite, grossular, periclase, monticellite	Wollastonite, diopside, grossular
Fluid system	Often open (H_2O infiltration from pluton)	Often closed (internally buffered)

P-T diagram used	T-X _{CO₂} at low P (0.1 GPa)	T-X _{CO₂} at higher P (0.5 GPa)
Sillimanite-equivalent zone	Wollastonite - Grossular (BOTH cases)	

CLASSIC CALC-SILICATE ZONE SEQUENCE — CRESTMORE, CALIFORNIA

Burnham (1959) mapped the following zones in the carbonate aureole at Crestmore, CA. This is the classic reference sequence cited by Winter:

ZONE NO.	ZONE NAME	ASSEMBLAGE
1-3	Forsterite Zone	Calcite + brucite/clinochumite/forsterite + spinel
4-7	Monticellite Zone	Calcite + forsterite/monticellite + spurrite/merwinite + melilite
8	Vesuvianite Zone	Vesuvianite + monticellite + spurrite + merwinite + melilite
9	(Upper Vesuvianite)	Vesuvianite + monticellite + diopside + wollastonite
10	Garnet Zone	Grossular + diopside + wollastonite ← HIGHEST GRADE

🔑 EXAM KEY POINT

Zone 10 (Grossular + Diopside + Wollastonite) is the innermost, highest-grade zone at Crestmore. Wollastonite and grossular are the diagnostic minerals of the peak calc-silicate assemblage, equivalent to sillimanite in pelites.

QUICK MINERAL REFERENCE FOR CALC-SILICATE ROCKS

MINERAL	FORMULA	GRADE INDICATOR	IN PELITIC EQUIVALENT?
Calcite	CaCO ₃	All grades; consumed at high grade	Not a pelitic mineral
Talc	Mg ₃ Si ₄ O ₁₀ (OH) ₂	Very low grade (in dolostones)	Chlorite zone analogue
Tremolite	Ca ₂ Mg ₅ Si ₈ O ₂₂ (OH) ₂	Low-medium grade	Biotite zone analogue
Zoisite / Epidote	Ca ₂ Al ₃ Si ₃ O ₁₂ (OH)	Lower-medium grade (H ₂ O-rich)	Garnet zone analogue
Diopside	CaMgSi ₂ O ₆	Medium-high grade (key isograd)	Kyanite-sillimanite zone
Wollastonite	CaSiO₃	High grade (T > 550°C); peak marker	Sillimanite zone
Grossular	Ca₃Al₂Si₃O₁₂	High grade; innermost zone	Sillimanite zone
Vesuvianite	Complex Ca-Mg-Al silicate	High-T contact aureoles (sanidinite)	No exact pelitic equivalent
Monticellite	CaMgSiO ₄	Very high T (> 800°C); rare	Granulite facies analogue

THE CAO-MGO-SIO₂ (CMS) SYSTEM — CHEMICAL FRAMEWORK

Most calc-silicate mineral assemblages are understood through the **CaO-MgO-SiO₂-H₂O-CO₂ (CMS-HC)** system. When projected from CO₂ and H₂O, the minerals plot in a triangular CMS diagram, and the progressive zones correspond to increasing SiO₂ activity (from intrusion-derived fluids or original impurities).

CaO corner → Calcite, Dolomite

MgO corner → Periclase, Brucite

SiO₂ corner → Quartz

Wollastonite → near CaO-SiO₂ join

Diopside → CaO-MgO-SiO₂ midfield

"Wollastonite occurs only in limestones and very siliceous dolostones and is separated from virtually all of the dark shaded common dolostone area by the Cal-Di and Cal-Tr tie-lines. In most dolostones, quartz is consumed by early reaction with dolomite to create these tie-lines."

— Winter (2010), Chapter 29, p. 667

The **T-X_{CO₂} diagram** is the primary tool used for understanding calc-silicate metamorphism. Because both CO₂ and H₂O are released in decarbonation and dehydration reactions, the fluid composition (X_{CO₂}) is a critical variable, unlike in pelitic systems where only H₂O is dominant.

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

Q1. The wollastonite isograd in calc-silicate rocks is defined by which reaction?

→ **Answer: Calcite + Quartz → Wollastonite + CO₂. This is a decarbonation reaction requiring T > 550°C at low P.**

Q2. In which metamorphic facies would you expect to find both wollastonite and grossular garnet in a calc-silicate rock?

→ **Answer: Upper amphibolite facies to granulite facies (equivalent to sillimanite zone in pelites).**

Q3. Which zone in calc-silicate rocks is equivalent to the kyanite zone in pelitic rocks?

→ **Answer: Anorthite-diopside-garnet (Kennedy 1949; Scottish Highlands correlation).**

Q4. What are the five progressive zones recognised by Ferry (1983) in calcareous rocks of the Augusta-Waterville area, Maine?

→ **Answer: (1) Ankerite zone, (2) Biotite zone, (3) Amphibole zone, (4) Zoisite zone, (5) Diopside zone. The diopside zone corresponds to the sillimanite isograd in pelites.**

Q5. In a siliceous dolostone contact aureole, which mineral is produced by the reaction: 11 Dolomite + Tremolite → 8 Forsterite + 13 Calcite + H₂O + 9 CO₂?

→ **Answer: Forsterite. This is the forsterite-in isograd reaction in dolostones (Reaction 9, Winter Chapter 29).**

Q6. A T-X_{CO₂} diagram is preferred over a T-P diagram for studying metamorphism of which rock type, and why?

→ **Answer: Calcareous/calc-silicate rocks. Because fluid composition (X_{CO₂}) is a critical variable — both CO₂ and H₂O are generated by reactions, making fluid composition as important as temperature.**

Q7. Which study by Kennedy (1949) is classically cited for establishing the equivalence of calc-silicate mineral zones with Barrovian pelitic zones?

→ **Answer: Kennedy's study of metamorphosed calc-silicate rocks in the Scottish Highlands, summarised in Table 1 of Winter (2010), Chapter 29.**

Q8. In a closed metamorphic system, why does the wollastonite isograd occur at higher temperatures than in an open system?

→ **Answer: Because earlier reactions buffer the fluid to high X_{CO₂} values (>0.62), and since Cal + Qtz → Wo + CO₂ requires low X_{CO₂} to proceed, the high CO₂ activity shifts the reaction to much higher temperatures (>750°C vs ~550°C in open systems).**

Q9. The assemblage at Crestmore Zone 10 (innermost) is grossular + diopside + wollastonite. What does this represent petrogenetically?

→ **Answer: The highest-grade, innermost calc-silicate zone in the contact aureole, equivalent to the sillimanite zone in pelites. Grossular garnet is the last index mineral to appear in the progressive sequence.**

Q10. Which calc-silicate assemblage is characteristic of the biotite-grade equivalent zone?

→ **Answer: Albite - zoisite - calcite - biotite - hornblende (Kennedy 1949).**

SUMMARY: WHY EACH OPTION IS RIGHT/WRONG

OPTION	ASSEMBLAGE	GRADE IN CALC-SILICATES	PELITIC EQUIVALENT	VERDICT
(a)	Anorthite-pyroxene	Granulite facies (mafic rocks)	Not a standard Barrovian zone marker	✗ Wrong
(b)	Zoisite-calcite-biotite	Low-medium grade (Zoisite Zone)	Garnet / lower staurolite zone	✗ Wrong
(c)	Diopside-anorthite	A binary igneous system, not a defined calc-silicate zone	Kyanite-sillimanite boundary (partial)	✗ Wrong
(d)	Wollastonite-grossular	Highest grade — Garnet/Wollastonite Zone	Sillimanite zone ✓	✓ Correct

Tapaswi Saikia | Geology Student | DMG Entrance Exam Preparation

Source: Winter, J.D. (2010). *Principles of Igneous and Metamorphic Petrology*, 2nd Ed. Chapter 29 — Metamorphism of Calcareous and Ultramafic Rocks. Table 1 (Kennedy 1949) · §1.4.5 · p. 674, 670, 486

DMG 2019 | Question 11 | Metamorphic Petrology — Calc-Silicate Rocks

THE QUESTION

Q.12 · DMG 2019

The hinge line of a fold is

- a** A line that divides the fold into symmetrical parts
- b** A line that is always parallel to fold axis
- c** A line of maximum curvature in a folded bed
- d** A line of minimum curvature in a folded bed



CORRECT ANSWER

A line of maximum curvature in a folded bed

FIGURES FROM THE TEXTBOOK

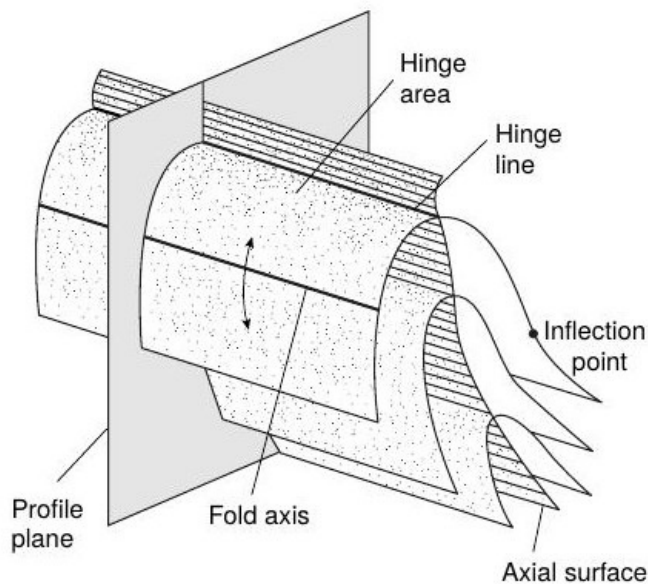


FIGURE 10.2 The terminology of a fold.

10.2 ANATOMY OF A FOLDED SURFACE 239

FIGURE 10.2 — The terminology of a fold. Note the **Hinge line** (line of maximum curvature), **Hinge area**, **Inflection point**, **Fold axis**, **Axial surface**, and **Profile plane**.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 239

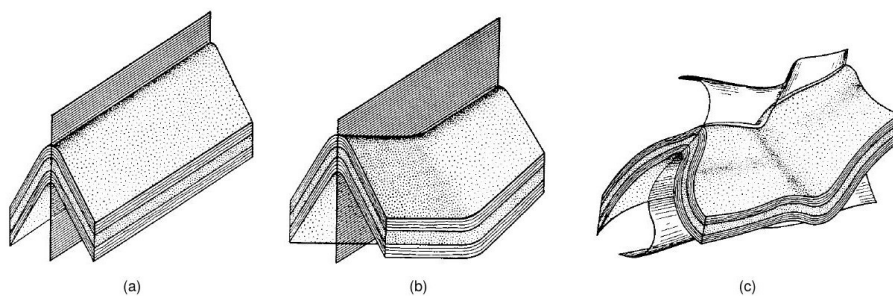


FIGURE 10.3 A cylindrical fold (a) is characterized by a straight hinge line and a noncylindrical fold (b) by a curved hinge line. The axial surface may be planar, as in (a) and (b), or curved (c).

surface can wrap partway around a cylinder. If this is

L_{∞}

FIGURE 10.3 — (a) A cylindrical fold with a **straight hinge line** parallel to the fold axis. (b) A noncylindrical fold with a **curved hinge line**. (c) A fold with a curved axial surface. This figure directly explains why option (b) is wrong: the hinge line is **NOT** always parallel to the fold axis.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 240

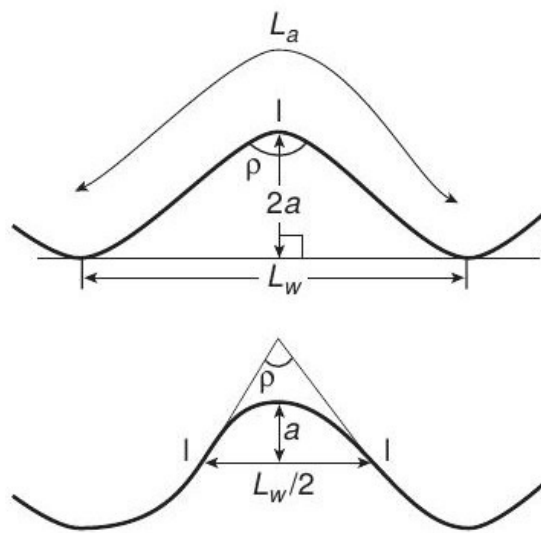


FIGURE 10.4 The interlimb angle (ρ), the wavelength (L_w), the amplitude (a), and the arc length (L_a) of a fold system in profile.

the cross-sectional plane, which has implication for the

FIGURE 10.4 — The interlimb angle (ρ), wavelength (L_w), amplitude (a), and arc length (L_a) of a fold system in profile.

SOURCE: VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · PAGE 240

TEXTBOOK DEFINITIONS

VAN DER PLUIJM & MARSHAK · §10.2 ANATOMY OF A FOLDED SURFACE · PAGE 239

"The hinge area is the region of greatest curvature and separates the two limbs. **The line of greatest curvature in a folded surface is called the hinge line.** You may think of a limb as the less curved portion of a fold. In a limb there is a point where the sense of curvature changes, called the **inflection point.**"

🔊 Page 239

TABLE 10.1 — VOCABULARY OF A FOLD · PAGE 241

Hinge line: "The line of greatest curvature."

Inflection point: "The position in a limb where the sense of curvature changes."

Axial surface: "The surface containing the hinge lines from consecutive folded surfaces."

Fold axis: "Fold generator in cylindrical folds."

🔊 Table 10.1 · Page 241

ON HINGE LINE VS. FOLD AXIS — PAGE 240, FOOTNOTE 1

"In the case of cylindrical folds the fold axis is of course parallel to the hinge line.¹"

¹ "Sometimes fold axis is used as a synonym for hinge line, but this is **not correct.**"

🔊 Page 240

✓ Hinge Line

The line of **greatest (maximum) curvature** in a folded surface. Connects all hinge points. Visible in Figure 10.2 running along the crest of the fold.

Hinge Area

The **region** of greatest curvature separating the two limbs. The hinge line is the mathematical line of maximum curvature within this zone.

Inflection Point / Line

The point where **sense of curvature changes** (concave → convex). Located on the limbs. Represents *minimum/zero* curvature — the exact opposite of the hinge line.

Axial Surface

Surface **containing hinge lines** of all successive folded layers. Not necessarily planar (Fig. 10.3c). Does *not* necessarily divide fold symmetrically.

Fold Axis

Theoretical **generator line** in cylindrical folds only. When moved parallel to itself, sweeps out the folded surface. Parallel to hinge line *only* in cylindrical folds.

Profile Plane

The plane **perpendicular to the hinge line**. The reference plane for describing fold shape in cross-section.

Limb

The **less curved portion** of a fold, between the hinge zone and inflection point.

Interlimb Angle (ρ)

Angle between limbs measured in the profile plane. Determines fold tightness: Isoclinal (0–10°) → Tight (10–60°) → Open (60–120°) → Gentle (120–180°).

CRITICAL DISTINCTION — HINGE LINE VS. FOLD AXIS (FIGURE 10.3)

FEATURE	HINGE LINE	FOLD AXIS
Nature	Real geometric line on folded surface	Theoretical generator line (conceptual)
Definition	Line of maximum curvature	Line that, moved parallel to itself, generates the folded surface
Fold types	Exists in ALL folds	Defined only for <i>cylindrical</i> folds
Always parallel?	Only in cylindrical folds. In noncylindrical folds, hinge line curves → option (b) "always parallel" is WRONG.	

OPTION-BY-OPTION ANALYSIS

a

✗ INCORRECT

"A line that divides the fold into symmetrical parts"

This describes the **axial surface**, not the hinge line. Moreover the textbook explicitly states the axial surface "does not necessarily divide the fold into equal halves that are mirror images of one another" (p. 240). So this is doubly wrong — wrong structure named, and the description itself is not even always true.

→ What this refers to: **Axial Surface** (loosely, for symmetric folds only)

b

✗ INCORRECT — CLASSIC TRAP

"A line that is always parallel to fold axis"

True *only* for cylindrical folds. In noncylindrical folds (Figure 10.3b), the hinge line **curves** — so the fold axis concept does not even apply as a single line. The textbook's own footnote calls this confusion "not correct." The word **"always"** makes this option definitively wrong.

→ True only for: **Cylindrical folds** — not universally

c

✓ CORRECT ANSWER

"A line of maximum curvature in a folded bed"

Exact match to the textbook: "*The line of greatest curvature in a folded surface is called the hinge line*" (p. 239). Table 10.1 (p. 241) gives the same definition: **Hinge line** = **"The line of greatest curvature."** Visible in Figure 10.2 running along the apex of the folded layer.

d

✗ INCORRECT — EXACT OPPOSITE

"A line of minimum curvature in a folded bed"

This is the **inflection point/line** — where the sense of curvature reverses, located on the *limbs* of the fold. Table 10.1 (p. 241): "*Inflection point: The position in a limb where the sense of curvature changes.*" This is the geometric opposite of the hinge line.

→ What this refers to: **Inflection Point / Inflection Line**

🧠 EXAM MEMORY AID

Hinge line = Maximum curvature = Tightest bend (crest/trough zone)

Inflection line = Minimum/zero curvature = Curvature reversal on limb

Axial surface = Contains ALL hinge lines from successive layers

Fold axis || Hinge line → ONLY in *cylindrical* folds, NOT always

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The inflection point in a folded layer is defined as

- (a) The point of maximum curvature
- (b) The highest topographic point of the fold
- (c) The position where the sense of curvature changes
- (d) A point lying on the axial surface

✓ **Answer: (c) — Table 10.1, p. 241**

PROBABLE Q · 2

The axial surface of a fold is best defined as

- (a) A plane that always bisects the fold into equal mirror-image halves
- (b) The surface containing hinge lines of successive folded layers
- (c) Always a vertical plane through the fold
- (d) The plane perpendicular to the hinge line

✓ **Answer: (b) — p. 240 & Table 10.1, p. 241**

PROBABLE Q · 3

In a cylindrical fold, the fold axis is

- (a) Perpendicular to the hinge line
- (b) Contained within the limb of the fold
- (c) Parallel to the hinge line
- (d) Always horizontal

✓ **Answer: (c) — p. 240**

PROBABLE Q · 4

A fold with an interlimb angle of 45° is classified as

- (a) Gentle (120° – 180°)
- (b) Open (60° – 120°)
- (c) Tight (10° – 60°)
- (d) Isoclinal (0° – 10°)

✓ **Answer: (c) — Table 10.3, p. 246**

PROBABLE Q · 5

The profile plane of a fold is defined as the plane that is

- (a) Parallel to the hinge line
- (b) Perpendicular to the hinge line
- (c) Any vertical section through the fold
- (d) Parallel to the axial surface

✓ **Answer: (b) — p. 239–240**

PROBABLE Q · 6

Which of the following statements about a noncylindrical fold is correct?

- (a) It has a straight hinge line parallel to the fold axis
- (b) Its folded surface wraps around a cylinder
- (c) It has a curved hinge line
- (d) Its axial surface is always planar

✓ **Answer: (c) — p. 239–240 & Figure 10.3b**

THE QUESTION

Q.13 · DMG 2019

In ocean, regional strike-slip faults associated with lithospheric plate boundaries are known as

a Transform faults

b Tear faults

c Wrench faults

d Scissor faults

CORRECT ANSWER

A

Transform faults — plate boundaries where lithosphere is neither created nor destroyed

TEXTBOOK DEFINITIONS — THE AUTHORITY

TABLE 8.2 — TYPES OF FAULTS · VAN DER PLUIJM & MARSHAK · PAGE 171

Transform fault: "In the preferred sense, transform faults are **plate boundaries at which lithosphere is neither created nor destroyed**. In a general sense, a transform fault links two other faults and accommodates the relative motion between the blocks of rock that move because of the displacement on the other two faults."

📖 Table 8.2, Page 171 — Chapter 8

TABLE 19.1 — TERMINOLOGY FOR STRIKE-SLIP FAULTS · PAGE 478

Transform fault: "A strike-slip fault that has the following characteristics: once formed, displacement across it can be constant along the length of the fault; displacement across it can be much greater than the length of the active fault; its length can be constant, increase, or decrease with time; it terminates at another fault. **Transform faults can be lithosphere plate boundaries**, in which case they terminate at intersections with other plate boundaries."

📖 Table 19.1, Page 478 — Chapter 19: Strike-Slip Tectonics

§19.2.1 TRANSFORM FAULTS · PAGE 479

"J. Tuzo Wilson introduced the term 'transform faults' to the geologic literature in the early 1960s, in reference to **the third category of plate boundary** (distinct from convergent and divergent boundaries) ... Strike-slip faults in oceanic crust most commonly occur along **mid-ocean ridges**, where they trend perpendicular to the ridge axis and link segments of the ridge."

"At **transform plate boundaries**, one plate slides past another along a strike-slip fault. Since no new plate is created and no old plate is consumed along a transform, such a boundary can also be called a 'conservative boundary.' Transform plate boundaries can occur either in **continental or oceanic lithosphere**."

FIGURES FROM THE TEXTBOOK

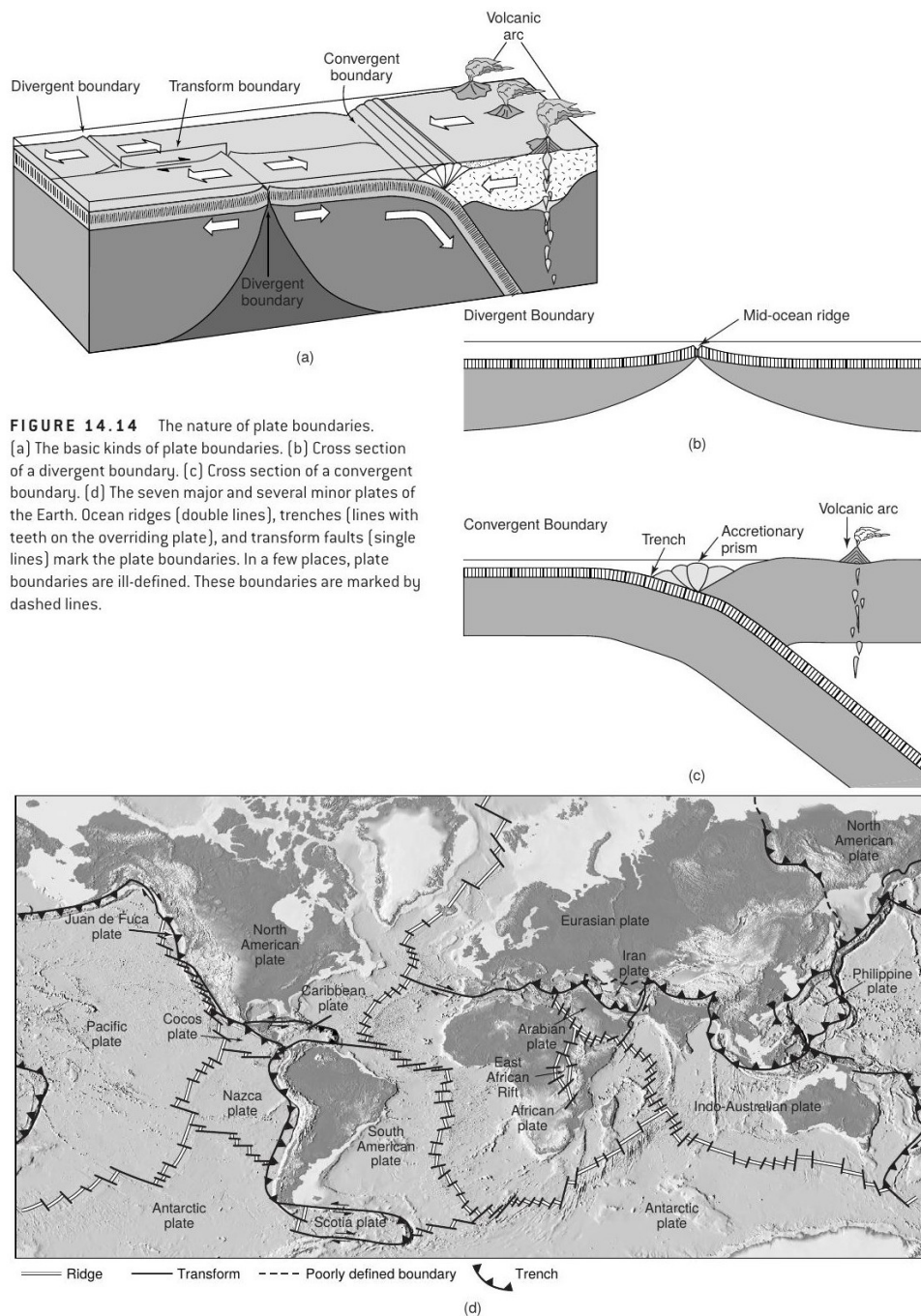


FIGURE 14.14 — The nature of plate boundaries. (a) The three types: divergent boundary (mid-ocean ridge), convergent

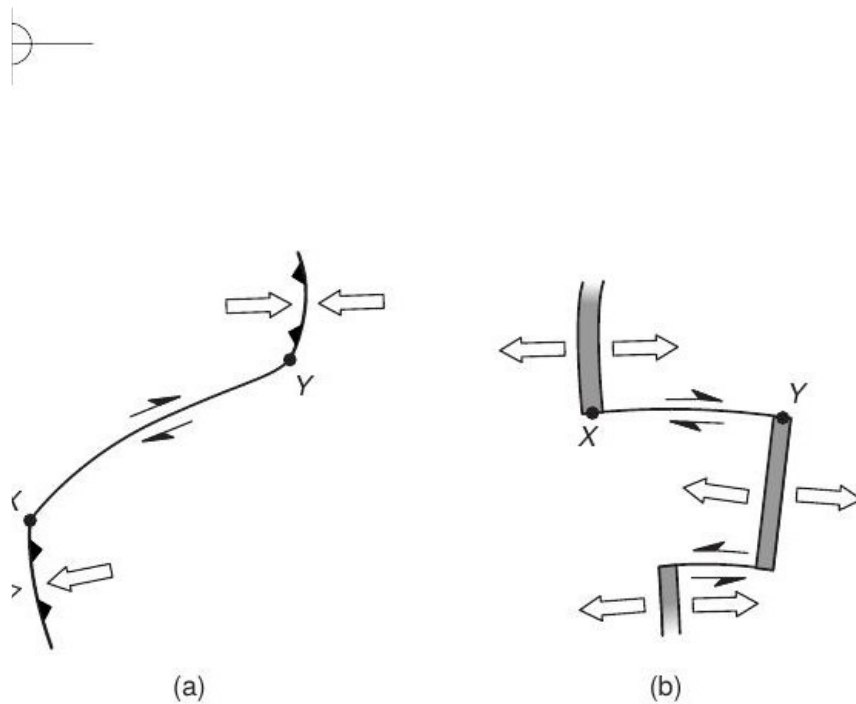
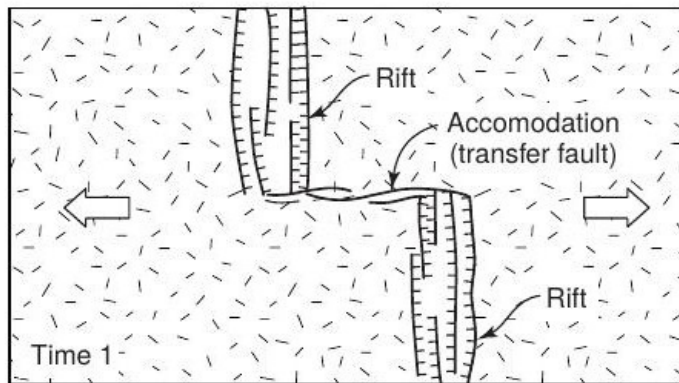


FIGURE 19.3 (a) Map sketch showing a transform fault linking convergent plate boundaries. The fault trace occurs only between X and Y. (b) Map sketch showing a transform fault linking two ridge segments.

FIGURE 19.3 — (a) Transform fault linking two convergent plate boundaries. (b) Transform fault linking two ridge segments. The active fault trace exists only between X and Y.



(a)

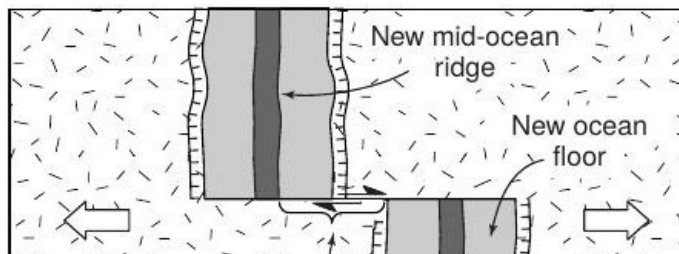
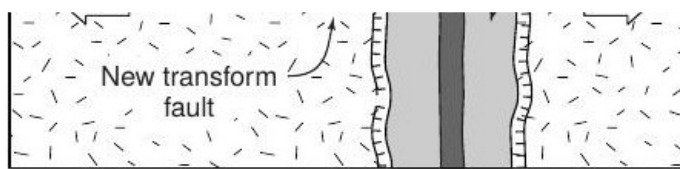
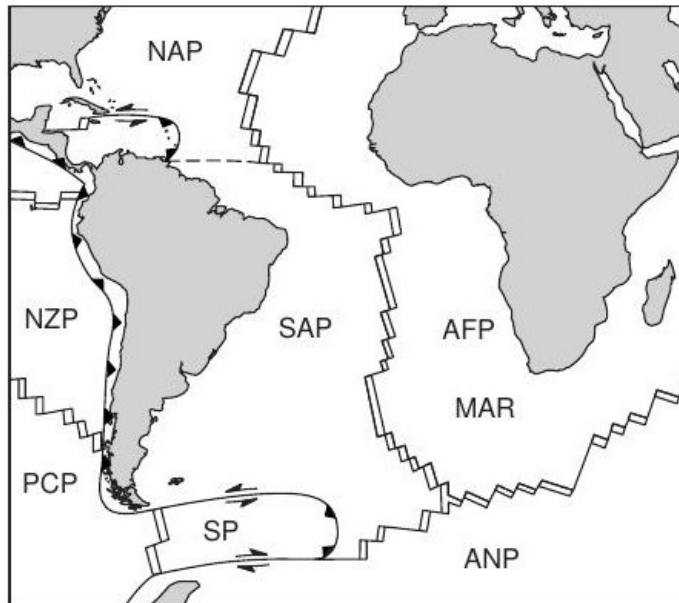


FIGURE 19.7a-b — (a) Transfer fault linking two rift segments. (b) As rifting succeeds, a new mid-ocean ridge forms and the transfer fault **evolves into a classic oceanic transform fault**.



(b)



(c)

FIGURE 19.7 [a] Sketch map showing a transfer fault linking two rift segments. [b] Later, the rift segments have evolved into new mid-ocean ridges, and the transfer fault has evolved into an oceanic transform fault. Note that this fault still has the same length as when it originated. [c] Simplified map showing the transform faults along the Mid-Atlantic Ridge in the South Atlantic Ocean. SAP = South American Plate, NAP = North American Plate, AFP = African Plate, ANP = Antarctic Plate, NZP = Nasca Plate, CS = Caribbean Sea, SS = Scotia Sea, MAR = Mid-Atlantic Ridge.

9.2 TRANSFORM VERSUS TRANSCURRENT FAULTS 481

FIGURE 19.7c — Simplified map showing transform faults along the Mid-Atlantic Ridge in the South Atlantic Ocean, linking segments of the ridge. SAP = South American Plate, AFP = African Plate, NAP = North American Plate, ANP = Antarctic Plate, MAR = Mid-Atlantic Ridge. These are classic examples of **oceanic regional strike-slip faults at lithospheric plate boundaries** — i.e., **transform faults**.

★ J. TUZO WILSON — THE FATHER OF TRANSFORM FAULTS (1960S)

J. Tuzo Wilson coined the term "**transform faults**" to describe the third type of plate boundary — distinct from divergent (mid-ocean ridges) and convergent (subduction zones). He recognized that in ocean basins, mid-ocean ridges are not continuous features but are offset by **regional strike-slip faults** that link ridge segments. These faults are plate boundaries where plates simply *slide past* each other — no new crust is created, and no old crust is destroyed. This makes them "**conservative boundaries**."

The table below (directly from Van der Pluijm & Marshak, Table 8.2, p. 171) gives all fault types — reproduced here for reference:

TABLE 8.2	TYPES OF FAULTS
Allochthon	The thrust sheet above a detachment is the allochthon (meaning that it is composed of <i>allochthonous</i> rock; i.e., rock that has moved substantially from its place of origin).
Autochthon	The footwall below a detachment is the autochthon; it is composed of <i>autochthonous</i> rock, or rock that is still in its place of origin.
Contractional fault	A contractional fault is one whose displacement results in shortening of the layers that the fault cuts, regardless of the orientation of the fault with respect to horizontal.
Décollement	The French word for detachment.
Detachment fault	This term is used for faults that initiate as a horizontal or subhorizontal surface along which the hanging-wall sheet of rock moved relative to the footwall. An older term "overthrust" is a regional detachment fault on which there has been a thrust sense of movement. Some detachments are listric, and on some detachments, regional normal-sense displacement occurs.
Dip-slip fault	The slip direction on a dip-slip fault is approximately parallel to the dip of the fault (i.e., has a rake between -80° and 90°).
Extensional fault	An extensional fault is one whose displacement results in extension of the layers that the fault cuts, regardless of the orientation of the fault with respect to horizontal.
Normal fault	A normal fault is a dip-slip fault on which the hanging wall has slipped down relative to the footwall.
Oblique-slip fault	The slip direction on an oblique-slip fault has a rake that is not parallel to the strike or dip of the fault. In the field, faults with a slip direction between -10° and -80° are generally called oblique-slip.
Overthrust fault	This is an older term that you may find in older papers on faults, but is no longer used much today. The term is used for thrust faults of regional extent. In this context, "regional extent" means that the thrust sheet has an area measured in tens to hundreds of square km, and the amount of slip on the fault is measured in km or tens of km. Today, such faults are generally called regional detachments.
Par-autochthonous	If a fault block has only moved a small distance from its original position, the sheet is par-autochthonous (literally, relatively in place).
Reverse fault	A reverse fault is a dip-slip fault on which the hanging wall has slipped up relative to the footwall.
Scissors fault	On a scissors fault, the amount of slip changes along strike so that the hanging-wall block rotates around an axis that is perpendicular to the fault surface (Figure 8.4i).
Strike-slip fault	The slip direction on a strike-slip fault is approximately parallel to the fault strike (i.e., the line representing slip direction has a rake [pitch] in the fault plane of less than -10°). Strike-slip faults are generally steeply dipping to vertical.
Transfer fault	A transfer fault accommodates the relative motion between blocks of rock that move because of the displacement on other faults.
Transform fault	In the preferred sense, transform faults are plate boundaries at which lithosphere is neither created nor destroyed. In a general sense, a transform fault links two other faults and accommodates the relative motion between the blocks of rock that move because of the displacement on the other two faults. However, we reserve the term <i>transfer fault</i> for this general type of displacement, independent of scale.

TABLE 8.2 — Types of Faults. Note the precise definitions of **Transform fault**, **Scissors fault**, and **Strike-slip fault**. The Tear fault definition is given in Chapter 18.

TABLE 19.1	TERMINOLOGY FOR STRIKE-SLIP FAULTS
Flower structure	An array of faults in a strike-slip fault zone that merges at depth into a near-vertical fault plane, but near the ground surface diverges so as to have shallower dips. In a positive flower structure, there is a component of thrusting on the faults, and in a negative flower structure, there is a component of normal faulting.
Horsetail	A group of fault splays at the endpoint (fault tip) of a fault; the splays diverge to define a fanlike array in map view.
Lateral escape	The strike-slip movement of a crustal block in a direction perpendicular to the regional convergence direction in a collisional orogen; the block is essentially squeezed laterally out of the way of the colliding blocks.
Lateral ramp	A surface connecting two non-coplanar parts of a thrust fault or normal fault (i.e., a ramp that is roughly parallel to the transport direction). Strike-slip movement occurs on a lateral ramp.
Non-coplanar	An adjective used to describe a geometric arrangement of two planes in which the planes have the same attitude, but are not aligned end-to-end to form a single plane.
Pull-apart basin	A sedimentary basin that forms at a releasing bend along a strike-slip fault.
Releasing bend	In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pulls away from the block on the other, causing transtensional deformation.
Restraining bend	In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pushes against the block on the other, causing transpressional deformation.
Stepover	A geometry that occurs where the end of one fault trace overlaps the end of another, non-coplanar fault trace.
Strike-slip duplex	An arrangement of sigmoidal-shaped fault splays (in map view) at a stepover, a restraining bend, or a releasing bend, whose geometry in map view resembles the cross-sectional geometry of the duplexes that occur in thrust-fault or normal-fault systems.
Strike-slip fault	Any fault on which displacement vectors are parallel to the strike of the fault, in present-day surface coordinates. The use of this term is purely geometric, and has no genetic, tectonic, or size connotation.
Tear fault	A traditional term for strike-slip faults that occur in a thrust sheet and accommodate differential displacement of one part of a thrust sheet relative to an adjacent part; in more recent jargon, a tear fault is a vertical lateral ramp.
Transcurrent fault	A strike-slip fault that has the following characteristics: it dies out along its length; the displacement across it is less than the length of the fault; the length of the fault increases with time and continued movement; displacement on the fault is greatest at the center of the fault trace and decreases toward the endpoints (tips) of the fault.
Transfer fault	A fault trending roughly perpendicular to the axis of a rift that links together two non-coplanar parts of the rift; transfer faults accommodate map-view offset of the locus of extension. Kinematically, transfer faults are a type of transform fault. If a rift succeeds in evolving into a mid-ocean ridge, the transfer fault becomes an oceanic transform fault.
Transform fault	A strike-slip fault that has the following characteristics: once formed, displacement across it can be constant along the length of the fault; displacement across it can be much greater than the length of the active fault; its length can be constant, increase, or decrease with time; it terminates at another fault. Transform faults can be lithosphere plate boundaries, in which case they terminate at intersections with other plate boundaries.
Transpression	A combination of strike-slip and compressional deformation; transpression occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of compression across the fault (leading to shortening).
Transtension	A combination of strike-slip and tension; transtension occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of tension across the fault (leading to extension).
Wrench fault	A traditional synonym for a strike-slip fault. It was commonly used in the petroleum geology literature, typically in reference to a regional-scale continental strike-slip fault. We abandon the term, as it tends to be used inconsistently.

TABLE 19.1 — Complete terminology for strike-slip faults (Chapter 19). Note especially the definitions of **Transform fault**, **Tear fault**, **Transcurrent fault**, and **Wrench fault** — these directly answer why options (b), (c), and (d) are wrong.

SOURCE: VAN DER PLUIJM & MARSHAK • CHAPTER 19 • PAGE 478

KEY FAULT TYPES — CONCEPT CARDS

✓ Transform Fault

Plate boundary strike-slip fault. Lithosphere neither created nor destroyed. Occurs in ocean basins linking mid-ocean ridge segments. Coined by J. Tuzo Wilson. Classic examples: Mid-Atlantic Ridge transforms, San Andreas (continental).

✗ **Tear Fault**

A strike-slip fault **within a thrust sheet** that accommodates differential displacement of one part of the sheet relative to another. Nearly vertical, parallel to transport direction. Found in thrust belts — NOT at plate boundaries.

✗ **Wrench Fault**

Traditional synonym for a continental strike-slip fault, used mainly in petroleum geology. Textbook explicitly states: "We abandon the term, as it tends to be used inconsistently." NOT an oceanic plate boundary feature.

✗ **Scissor (Scissors) Fault**

A fault where **slip changes along strike**, so the hanging wall block **rotates** around an axis perpendicular to the fault surface. A specific kinematic type — not a regional oceanic plate boundary.

COMPARISON TABLE — ALL FOUR FAULT TYPES

FAULT TYPE	NATURE OF SLIP	TECTONIC SETTING	PLATE BOUNDARY?	OCEAN FLOOR?
Transform fault ✓	Strike-slip; constant displacement	Links ridge segments, trench segments, or other plate boundaries	YES — conservative boundary	YES — most common in ocean
Tear fault	Dominantly strike-slip; vertical	Within thrust sheets; accommodates differential displacement	No	No — thrust belt feature
Wrench fault	Strike-slip (continental)	Continental crust; petroleum geology term	Sometimes, but term abandoned	No — continental usage
Scissor fault	Slip changes along strike; rotation	Any setting with rotational block motion	No	No — specific kinematic type

OPTION-BY-OPTION ANALYSIS

a ✓ **CORRECT ANSWER**

"Transform faults"

Directly defined by the textbook as plate boundaries where lithosphere is *neither created nor destroyed* (Table 8.2, p. 171). In ocean basins, they most commonly link segments of mid-ocean ridges, trending perpendicular to the ridge axis (§19.2.1, p. 479). These are the premier example of **regional oceanic strike-slip faults at lithospheric plate boundaries**. First described by J. Tuzo Wilson in the 1960s, their existence is one of the key proofs of plate tectonics.

b ✗ **INCORRECT**

"Tear faults"

Tear faults are **intra-thrust sheet** structures — they occur within thrust sheets to accommodate differences in how much one part of the sheet has moved versus another. Table 19.1 (p. 478) defines them as: "A traditional term for strike-slip faults that occur in a thrust sheet and accommodate differential displacement... in more recent jargon, a tear fault is a vertical lateral ramp." They are found in fold-thrust belts, not in the ocean at plate boundaries.

→ Correct setting for tear faults: **Fold-thrust belts** (e.g., lateral ramps in Appalachians, Himalayas)

c ✗ **INCORRECT**

"Wrench faults"

Wrench fault is defined by Table 19.1 (p. 478) as simply: "A traditional synonym for a strike-slip fault. It was commonly used in the petroleum geology literature, typically in reference to a **regional-scale continental** strike-slip fault. We abandon the term, as it tends to be used inconsistently." While all transform faults are strike-slip faults, calling oceanic plate-boundary strike-slip faults "wrench faults" is imprecise and the term is explicitly abandoned by the textbook. The specific, correct, formally defined term for oceanic plate boundary strike-slip faults is *transform fault*.

→ Wrench fault = outdated/imprecise term for **continental** strike-slip faults

d ✖ INCORRECT

"Scissor faults"

A scissor (scissors) fault is a specific kinematic type where **slip magnitude changes along strike**, causing one block to rotate around an axis perpendicular to the fault surface (Table 8.2, p. 171; Figure 8.4i). This is a purely geometric description — it says nothing about plate boundaries, ocean floors, or regional scale. Scissor faults are not a category of plate boundary.

→ Scissor fault = **rotational slip** type, not a plate boundary concept

🧠 EXAM MEMORY AID

Transform fault = Ocean + plate boundary + strike-slip → "*conservative boundary*"

Tear fault = Inside thrust sheets → differential displacement of sheet

Wrench fault = Abandoned term → used only for *continental* strike-slip

Scissor fault = Slip changes along strike → block *rotates*

Key phrase from the question: "oceanic + plate boundary + regional" → only Transform fits

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

Transform faults are also called "conservative plate boundaries" because

- (a) They conserve the kinetic energy of plate motion
- (b) They are ancient and geologically stable
- (c) Lithosphere is neither created nor destroyed along them
- (d) They conserve the orientation of the stress field

✓ **Answer: (c) — Van der Pluijm & Marshak, p. 357 (Ch.14) & Table 8.2, p. 171**

PROBABLE Q · 2

The term "transform fault" was introduced by

- (a) Harry Hess
- (b) Alfred Wegener
- (c) J. Tuzo Wilson
- (d) Lynn Sykes

✓ **Answer: (c) — Van der Pluijm & Marshak, §19.2.1, p. 479**

PROBABLE Q · 3

Which of the following is NOT a characteristic of a transform fault?

- (a) Displacement can exceed the length of the fault
- (b) It terminates at another fault or plate boundary
- (c) Displacement increases from center toward endpoints
- (d) Length can remain constant with time

✓ **Answer: (c) — That describes a transcurrent fault. Table 19.1, p. 478**

PROBABLE Q · 4

A tear fault is characteristically found in

- (a) Mid-ocean ridges linking two ridge segments
- (b) Transform plate boundaries in oceanic crust
- (c) Thrust sheets, accommodating differential displacement
- (d) Rift zones during early continental breakup

✓ **Answer: (c) — Table 19.1, p. 478 & §18.1, p. 447**

PROBABLE Q · 5

Which of the following pairs represents the three types of plate boundaries?

- (a) Normal, Reverse, Strike-slip
- (b) Divergent, Convergent, Transform
- (c) Rift, Subduction, Collision
- (d) Transcurrent, Transpressional, Transtensional

✓ **Answer: (b) — Van der Pluijm & Marshak, §14.8, p. 357 & Figure 14.14**

PROBABLE Q · 6

On a scissors fault, the distinctive kinematic feature is that

- (a) Strike-slip displacement is constant along the fault length
- (b) The hanging wall rotates around an axis perpendicular to the fault surface
- (c) It always occurs at mid-ocean ridges
- (d) Slip direction is parallel to the dip of the fault

✓ **Answer: (b) — Table 8.2, p. 171**

THE QUESTION

Q.14 · DMG 2019

In a fold, rock salt has broken through the more brittle overlying rocks and occupied its core. The structure is known as

- a** Disharmonic fold
- b** Drag fold
- c** Box fold
- d** **Diapir fold**

D

CORRECT ANSWER

Diapir fold — salt that pierces bedding in overlying strata and occupies the core

TEXTBOOK DEFINITIONS — THE AUTHORITY

TABLE 2.5 — TERMINOLOGY OF SALT STRUCTURES · PAGE 28 · VAN DER PLUIJM & MARSHAK

Salt diapir: "A salt structure that **pierces bedding in overlying strata.**"

Salt dome: "An upward salt bulge relative to the source layer that is roughly symmetric in plan view. Strata overlying these structures conformably warp around the structure."

📖 Table 2.5, Page 28 — Chapter 2: Primary and Nontectonic Structures

§2.3 SALT STRUCTURES · PAGE 26 · VAN DER PLUIJM & MARSHAK

"Salt is a nonporous and essentially incompressible material... salt is much weaker and, as a consequence, is able to **flow like a viscous fluid** under conditions in which other sedimentary rocks behave in a brittle fashion... If gravity is the only reason for salt movement, the deformation resulting from its movement is called **halokinesis.**"

📖 Page 26 — §2.3

§2.3.2 GEOMETRY OF SALT STRUCTURES · PAGE 27-28

"If the salt rises after the overlying strata have already been deposited, then the rising salt will **warp and eventually break through the overlying strata.** This process is called **upbuilding**... When salt diapirs approach the surface, the overlying strata are arched up and therefore are locally stretched, resulting in the development of normal faults in a complex array over the crest of the salt structure."

📖 Pages 27-29 — §2.3.2

TABLE 2.5	TERMINOLOGY OF SALT STRUCTURES
Detached bulb	Stems or walls connecting salt stocks to the source bed may pinch out, so that salt that was originally separated by the stem or wall becomes juxtaposed. Flow of the salt into bulbs results in folds that wrap around themselves and thus have circular profiles.
Detached canopy	Stems or walls connecting salt canopies to the source bed may pinch out, so that salt that was originally separated by the stem or wall becomes juxtaposed.
Salt anticline	An upward salt bulge relative to the source layer that is elongate in plan view. Strata overlying these structures conformably warp around the structure.
Salt canopy	In regions where the source bed was quite thick so that many diapirs form, salt walls or salt stocks may spread out and merge at a higher stratigraphic level. Salt canopies may flow dominantly in one direction in response to gravity, and if so are sometimes called <i>salt-tongue canopies</i> .
Salt diapir	A salt structure that pierces bedding in overlying strata.
Salt dome	An upward salt bulge relative to the source layer that is roughly symmetric in plan view. Strata overlying these structures conformably warp around the structure.
Salt glacier	Salt that flows out over the land when the salt diapir pierces ground surface (Figure 2.17).
Salt pillow	Same as <i>salt dome</i> .
Salt roller	An asymmetric salt dome that resembles an ocean wave.

TABLE 2.5 — Terminology of Salt Structures. Note **Salt diapir**: "A salt structure that pierces bedding in overlying strata" — directly matching the question description. *Salt dome merely warps strata; the diapir breaks through.*

VAN DER PLUIJM & MARSHAK • CHAPTER 2 • PAGE 28

Salt stock	Equant salt diapir in plan view. Mature salt diapirs are generally narrower at depth and broader at the top; the lower part is called the <i>stem</i> and the upper part is called the <i>bulb</i> .
Salt wall	Elongate salt diapir in plan view.

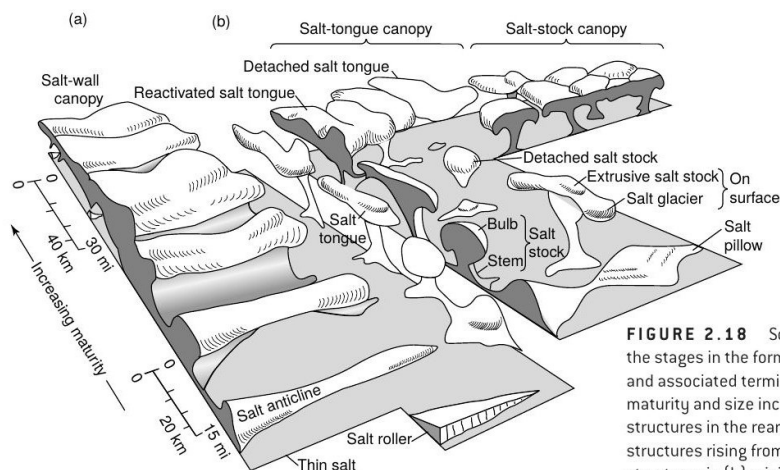


FIGURE 2.18 Schematic diagram showing the stages in the formation of salt structures and associated terminology. Structural maturity and size increase toward the structures in the rear. Sequence in (a) shows structures rising from line sources, whereas structures in (b) originate from point sources.

FIGURE 2.18 — Schematic diagram showing the stages in the formation of salt structures. Note the progression from salt anticline → salt roller → salt pillow → salt wall → salt stock (diapir). The diapir stage is where the salt **pierces through** overlying brittle strata.

VAN DER PLUIJM & MARSHAK • CHAPTER 2 • PAGE 28

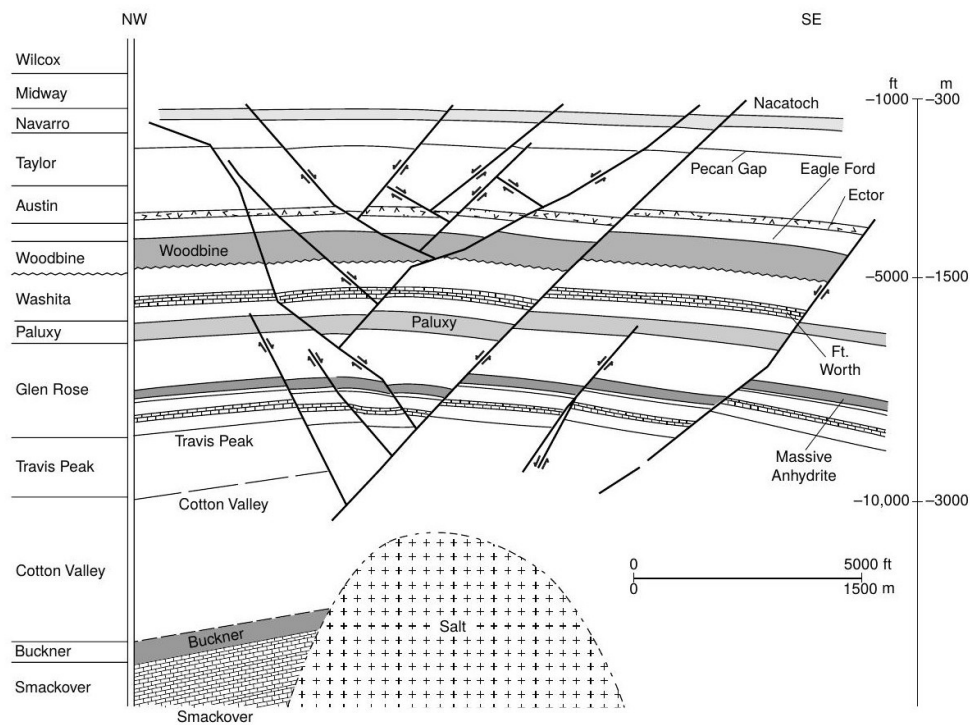


FIGURE 2.19 — Cross section illustrating a normal fault array over the top of a salt dome in Texas. Note the upward-piercing salt body (cross-hatched, center) that has broken through the overlying sedimentary strata — this is the diapir geometry described in the question. The overlying brittle strata are arched and faulted above.

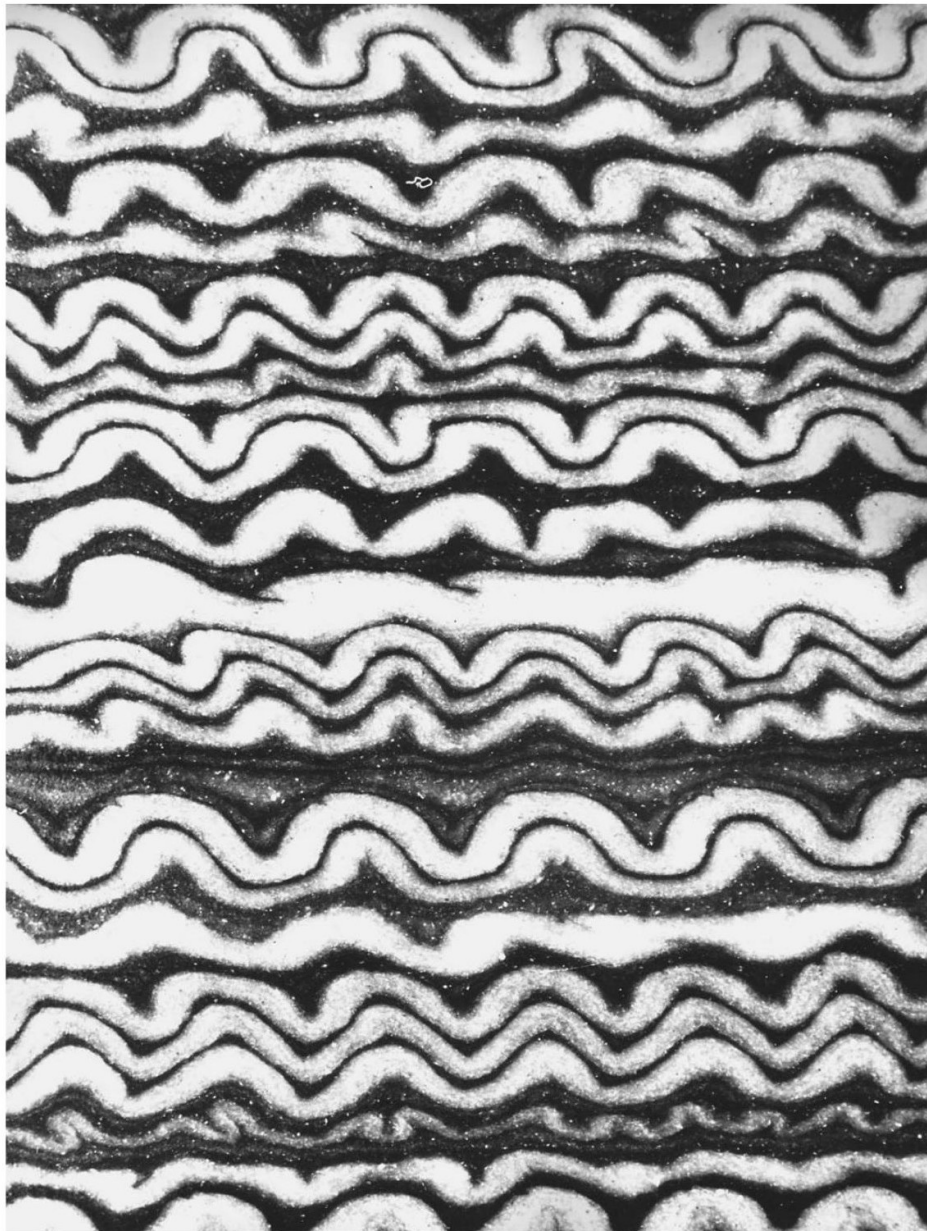


FIGURE 10.5 Small-scale disharmonic folds in anhydrite of the Permian Castile Formation in the Delaware Basin of Texas. White layers are anhydrite; dark layers consist of calcite that is rich in organic material (hence the dark color). Note that detachments occur in the organic-rich calcite layers and that the fold shapes (including box folds) in the anhydrite vary as a function of layer thickness.

242 FOLDS AND FOLDING

FIGURE 10.5 — Small-scale **disharmonic folds** in anhydrite of the Permian Castile Formation, Texas. White = anhydrite; dark = organic-rich calcite. Note that different layers fold with different wavelengths/amplitudes (disharmonic). Box folds are also visible — layer-detached, flat-topped geometries. This directly illustrates options (a) and (c).



FIGURE 10.18 Monocline in the Big Horn Mountains (Wyoming, USA).

10.5 SOME SPECIAL FOLD GEOMETRIES

We end the descriptive part of this chapter with a few special fold geometries that you may encounter in your field career. **Monoclines** are fold structures with only one tilted limb; the beds on either side of the tilted limb are horizontal. Monoclines typically result from a vertical offset in the subsurface near the tilted portion of the structure. The fault uplifts a block of relatively rigid igneous or metamorphic rock, and the overlying sedimentary layers drape over the edge of the uplifted block to form the monocline (Figure 10.18). Spectacular examples are found in the Colorado Plateau of the western United States. **Kink folds** are small folds (less than a meter) that are characterized by straight limbs and sharp hinges. Typically they occur in finely laminated (that is, strongly anisotropic) rocks, such as shales and slates (Figure 10.19). Sharply bending a deck of cards is a good analogy for the kinking process, because kink folds are formed by displacements between individual laminae (individual cards in the analogy). **Chevron folds** (Figure 10.8) are the larger-scale equivalent of kink folds.

The term **box fold** describes a geometry that is pretty self-explanatory (Figure 10.5). In order for a box fold to form, a layer must be detached from the underlying and overlying layers. They are therefore common in areas with weak basal layers, such as in the Jura Mountains of Switzerland. **Ptygmatic folds** are irregular and isolated



FIGURE 10.19 Kink folds in mica-rich portion of greywackes of the Cantabrian Mountains (northern Spain).

§10.5 Some Special Fold Geometries — Textbook page covering box folds, kink folds, monoclines, and ptygmatic folds. The **box fold** definition is here: "describes a geometry that is pretty self-explanatory... a layer must be detached from the underlying and overlying layers."

HOW A DIAPIR FORMS — STEP BY STEP

1 Salt Deposition in Passive-Margin Basin

Halite (NaCl) evaporites are deposited in rift basins during continental breakup. Thick salt layers accumulate at the base of the sedimentary sequence.

2 Burial and Density Inversion

Overlying sediments compact and become denser ($\approx 2500 \text{ kg/m}^3$). Salt stays at $\approx 2200 \text{ kg/m}^3$ and does not compact. At depths $>6 \text{ km}$, denser rock overlies less dense salt → **density inversion**.

3 Halokinesis Begins — Positive Buoyancy

The light salt is positively buoyant. Differential loading creates pressure gradients. Salt flows plastically upward from the **source layer** — the weak salt flows while surrounding rocks behave brittly.

4 Salt Dome / Pillow Stage

Salt bulges upward as a salt anticline or pillow. Overlying strata **conformably warp** around the structure without being pierced. This is the pre-diapir stage.

5 Diapir Stage — Piercing Through ✓ (THE ANSWER)

The buoyant salt breaks through the **more brittle overlying strata** and rises as a diapir (salt stock or salt wall). The salt now **occupies the core** of the structure. Normal faults develop above. This is exactly the scenario described in the question.

6 Salt Glacier (If Extruded to Surface)

If the diapir reaches the surface, salt flows out as a **salt glacier** (extrusive salt stock). Examples in western Iran.

ALL FOUR FOLD TYPES — CONCEPT CARDS

✓ Diapir Fold

A body of **mobile material (rock salt, magma, or overpressured shale)** that has pierced upward through more brittle overlying strata and occupies the core of the structure. Driven by buoyancy (density inversion) and differential loading. Classic example: salt diapirs / salt domes in the Gulf of Mexico.

✗ Disharmonic Fold

Folds in which different layers have **different wavelengths and/or amplitudes** — the fold pattern is not continuous from one layer to the next. Caused by differences in competency between layers. Nothing to do with salt piercing through strata.

✗ Box Fold

A fold with a **flat-topped hinge zone** (box-shaped in cross-section). Requires layer detachment from above and below. Common in weak basal layers (e.g., Jura Mountains). Purely a fold shape — no material piercing through anything.

✗ Drag Fold

An **outdated/discouraged term** for fault-related folds, previously thought to form by "drag" on fault surfaces. Textbook (p. 187): "We discourage and avoid the use of this term." Also = parasitic folds associated with larger folds. Completely unrelated to salt intrusion.

COMPARISON TABLE — ALL FOUR OPTIONS

FOLD TYPE	KEY FEATURE	MECHANISM	MATCHES QUESTION?
Diapir fold ✓	Mobile material pierces through brittle overlying strata; occupies core	Buoyancy + density inversion (halokinesis)	YES — exactly described
Disharmonic fold	Layers fold with different wavelengths/amplitudes	Competency contrast between layers	No — no piercing or core

Box fold	Flat-topped (box-shaped) fold hinge	Layer detachment + compression	No — just a fold shape
Drag fold	Abandoned term ; fault-related asymmetric fold	Parasitic folding near faults	No — discarded term, unrelated

OPTION-BY-OPTION ANALYSIS

a ✗ INCORRECT

"Disharmonic fold"

A disharmonic fold describes variation in fold geometry between layers — some layers fold with large amplitudes and short wavelengths, others with different dimensions. Van der Pluijm & Marshak (p. 241): *"If some layers have different wavelengths and/or amplitudes, the folds are disharmonic."* Figure 10.5 shows disharmonic folds beautifully in alternating anhydrite and calcite layers. **No piercing, no core, no salt intrusion involved.**

→ Disharmonic = different fold shapes in different layers (Figure 10.5, p. 242)

b ✗ INCORRECT — DISCARDED TERM

"Drag fold"

The textbook actively discourages this term. Van der Pluijm & Marshak (p. 187): *"We close with a comment on the term drag fold, which is used to refer to all fault-related folds in older literature. This general application of the term is misleading... Thus we **discourage and avoid the use of this term.**"* In another usage, "drag folds" = parasitic folds on the limbs of larger folds, previously misunderstood. Neither usage involves mobile material breaking through overlying rocks.

→ Drag fold = outdated/avoided term; related to faults or parasitic folding, NOT salt intrusion

c ✗ INCORRECT

"Box fold"

A box fold is purely a **shape descriptor** — the fold hinge is flat and wide, giving a box-like cross-section. Van der Pluijm & Marshak (p. 250): *"The term box fold describes a geometry that is pretty self-explanatory... In order for a box fold to form, a layer must be detached from the underlying and overlying layers."* It is common in the Jura Mountains. The geometry is box-shaped in profile — it says nothing about any material piercing through other rocks.

→ Box fold = flat-topped fold shape caused by layer detachment (Figure 10.5 shows box folds in anhydrite)

d ✓ CORRECT ANSWER

"Diapir fold"

The question describes three key facts: (1) **rock salt** as the mobile material, (2) it has **broken through** the more brittle overlying rocks, and (3) it **occupies the core**. This is the textbook definition of a diapir. Table 2.5 (p. 28): *"Salt diapir: A salt structure that **pierces bedding in overlying strata.**"* The process is halokinesis — buoyancy-driven upward flow of salt, which is ductile while surrounding rocks remain brittle. This is the fundamental contrast that drives piercement: salt flows, overlying rocks break.

→ Diapir = piercing mobile body; halokinesis = the process; "occupies the core" = classic diapir geometry

EXAM MEMORY AID

Diapir = Mobile material **PIERCES** through brittle rocks → occupies core

Disharmonic = Different wavelengths in different layers (no piercing)

Box fold = Flat-topped hinge shape (detached layers, Jura Mountains)

Drag fold = Discouraged/abandoned term (avoid in exams!)

Key words in question → "broken through" + "occupied its core" = **DIAPIR**

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The gravity-driven movement of salt, independent of tectonic forces, is called

- (a) Diapirism
- (b) Halokinesis
- (c) Halotectonics
- (d) Salt tectonics

✓ **Answer: (b) — Van der Pluijm & Marshak, §2.3, p. 26**

PROBABLE Q · 2

A salt diapir differs from a salt dome in that the salt diapir

- (a) Is symmetric in plan view and warps overlying strata
- (b) Pierces the bedding in overlying strata
- (c) Is driven by tectonic compression only
- (d) Only forms in continental rift zones

✓ **Answer: (b) — Table 2.5, p. 28 — salt dome warps; salt diapir PIERCES**

PROBABLE Q · 3

Disharmonic folds develop when adjacent layers have

- (a) The same wavelength but different amplitudes only
- (b) Different wavelengths and/or amplitudes due to competency contrasts
- (c) Been pierced by a mobile core
- (d) Been detached from basal layers

✓ **Answer: (b) — Van der Pluijm & Marshak, p. 241**

PROBABLE Q · 4

Box folds are commonly developed in which of the following settings?

- (a) Deep crustal ductile shear zones
- (b) Regions with weak basal layers allowing layer detachment
- (c) Salt diapir regions where piercement has occurred
- (d) Transform fault zones

✓ **Answer: (b) — Van der Pluijm & Marshak, §10.5, p. 250 — Jura Mountains example**

PROBABLE Q · 5

The primary reason rock salt rises diapirically through overlying denser sedimentary rocks is

- (a) Tectonic compression squeezing salt upward
- (b) Positive buoyancy due to density inversion
- (c) Thermal expansion of salt at depth
- (d) Dehydration reactions increasing salt volume

✓ **Answer: (b)** — §2.3.1, p. 26-27; salt $\approx 2200 \text{ kg/m}^3$, surrounding rocks $\approx 2500 \text{ kg/m}^3$

PROBABLE Q · 6

When a salt diapir pierces through to the surface and flows laterally, the resulting structure is called

- (a) Salt wall
- (b) Salt pillow
- (c) Salt glacier
- (d) Detached canopy

✓ **Answer: (c)** — Table 2.5, p. 28; classic examples in western Iran

PROBABLE Q · 7

Which of the following geological settings most commonly hosts salt diapirs?

- (a) Active collision orogens
- (b) Passive continental margins with thick evaporite sequences
- (c) Mid-ocean ridge systems
- (d) Continental rift shoulders

✓ **Answer: (b)** — §2.3, p. 26; passive-margin basins discussed extensively

THE QUESTION

Q.15 · DMG 2019

A fault having both dip-slip as well as strike-slip components is termed as

- a Strike-slip fault
- b Dip-slip fault
- c Oblique-slip fault**
- d Wrench fault



CORRECT ANSWER

Oblique-slip fault — net slip vector has both a dip-slip AND a strike-slip component

TEXTBOOK DEFINITIONS — TWO AUTHORITIES

FOSSEN · STRUCTURAL GEOLOGY · CHAPTER 9 · §9.1 FAULT TERMINOLOGY

"Many faults show some deviation from true dip-slip and strike-slip displacement in the sense that the **net slip vector is oblique**. Such faults are called **oblique-slip faults**. The degree of obliquity is given by the **pitch (also called rake)**, which is the angle between the strike of the slip surface and the slip vector (striation)."

📖 Chapter 9, p. 153 (Fossen, Structural Geology)

VAN DER PLUIJM & MARSHAK · EARTH STRUCTURE · §8.2 FAULT GEOMETRY AND DISPLACEMENT · PAGE 170

"If the net-slip vector parallels the dip direction of the fault (within $\sim 10^\circ$), the fault is called a **dip-slip fault**; if the vector roughly parallels the strike of the fault, the fault is called a **strike-slip fault**. If the vector is not parallel to either dip direction of the strike, we call the fault an **oblique-slip fault**. As you can see in Figure 8.3, **oblique-slip faults have both a strike-slip and a dip-slip component of movement**."

📖 Page 170 — §8.2, Van der Pluijm & Marshak

FOSSEN · STRUCTURAL GEOLOGY · FIGURE 9.2 CAPTION

"Normal, strike-slip, and reverse faults. **These are end-members of a continuous spectrum of oblique faults**. The spherical projections represent the fault plane (great circle) and the displacement vector."

📖 Figure 9.2, Chapter 9 — Fossen

END-MEMBER 1

Strike-slip fault

Net slip is **horizontal**, parallel to fault strike. No dip-slip component. Vertical or near-vertical fault plane.

Pitch (rake) $\approx 0^\circ$

✔ INTERMEDIATE — THE ANSWER

Oblique-slip fault

Net slip is **oblique** — has BOTH a dip-slip component (up/down) AND a strike-slip component (lateral). The net slip vector is at an intermediate pitch.

Pitch (rake) between 10° and 80°

END-MEMBER 2

Dip-slip fault

Net slip is **in the dip direction**, parallel to the fault dip. No lateral component. Includes normal and reverse faults.

Pitch (rake) $\approx 90^\circ$

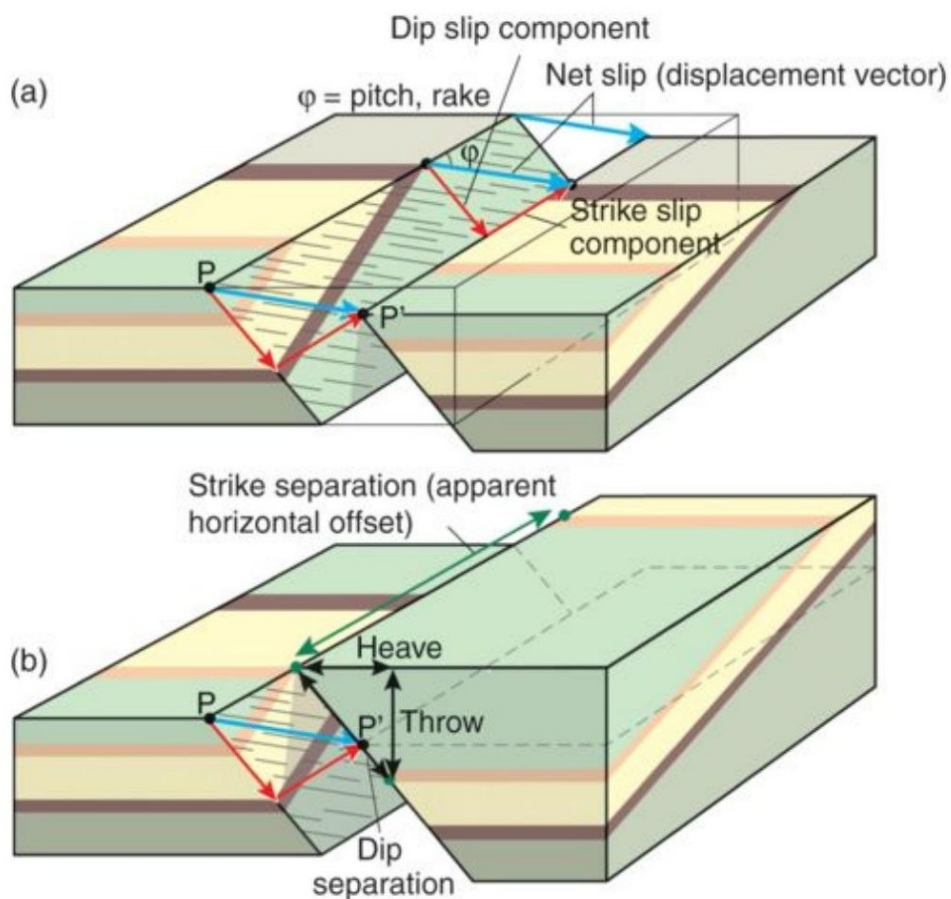


Figure 9.6 Illustration of an oblique-slip fault affecting tilted layers. The fault is normal sinistral and the blue arrows show the true displacement or net slip, which has a dip-slip and a strike-slip component (a). The net slip vector connects points (for example, P and P' in the figure) that were adjacent to each other before faulting. In (b) we have added stratigraphy to the hanging wall up to the level of the footwall. Now the strike and dip separation of the layers, which depend on the orientation of the layers and the nature of the fault displacement,

FIGURE 9.6 (Fossen, Ch. 9) — Illustration of an oblique-slip fault affecting tilted layers. The fault is **normal sinistral** — an oblique-slip type with both a **dip-slip component** (hanging wall down = normal) and a **strike-slip component** (sinistral = left-lateral). The blue arrows show the **net slip (displacement vector)**, which is neither purely dip-parallel nor purely strike-parallel. Note ϕ = pitch (rake) angle between the strike and the slip direction.

SOURCE: FOSSEN — STRUCTURAL GEOLOGY · CHAPTER 9 · FIGURE 9.6

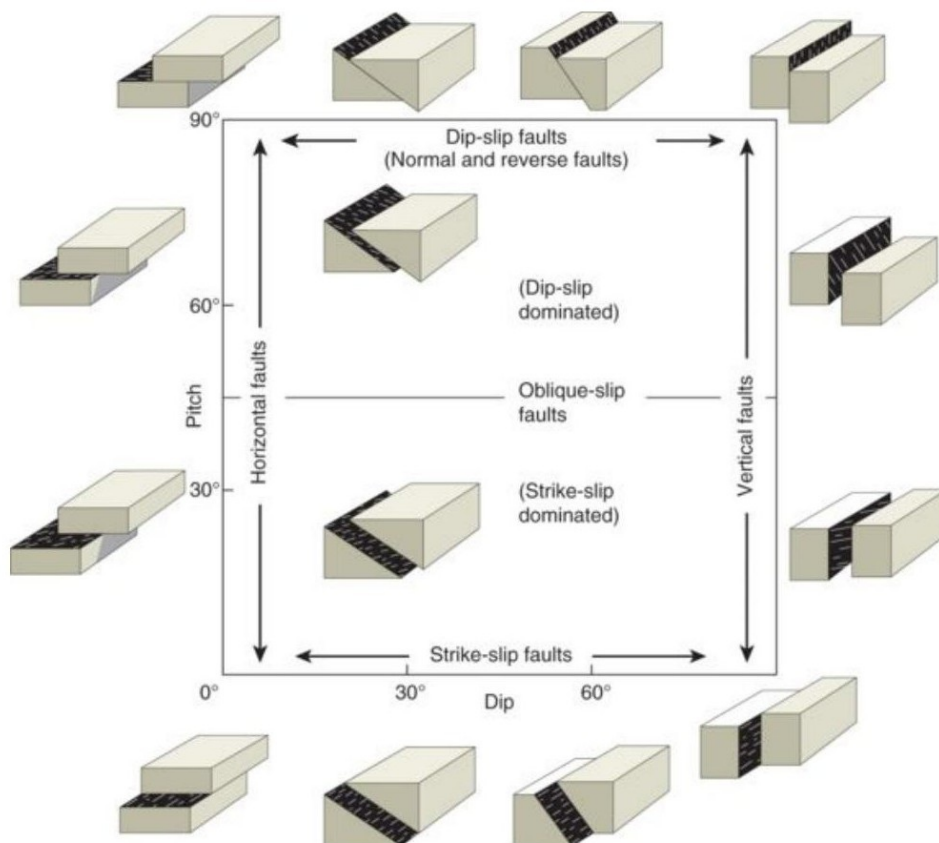


Figure 9.8 Classification of faults based on the dip of the fault plane and the pitch, which is the angle between the slip direction (displacement vector) and the strike.

Based on Angelier ([1994](#)).

FIGURE 9.8 (Fossen, Ch. 9) — Classification of faults based on **dip of the fault plane (x-axis)** and **pitch/rake (y-axis)**. The central zone (intermediate pitch, 30°–60°) is labelled **Oblique-slip faults** — this zone contains all faults with both dip-slip and strike-slip components. Pure dip-slip faults (normal/reverse) plot at the top (pitch 90°); pure strike-slip faults plot at the bottom (pitch 0°). Based on Angelier (1994).

SOURCE: FOSSEN — STRUCTURAL GEOLOGY · CHAPTER 9 · FIGURE 9.8

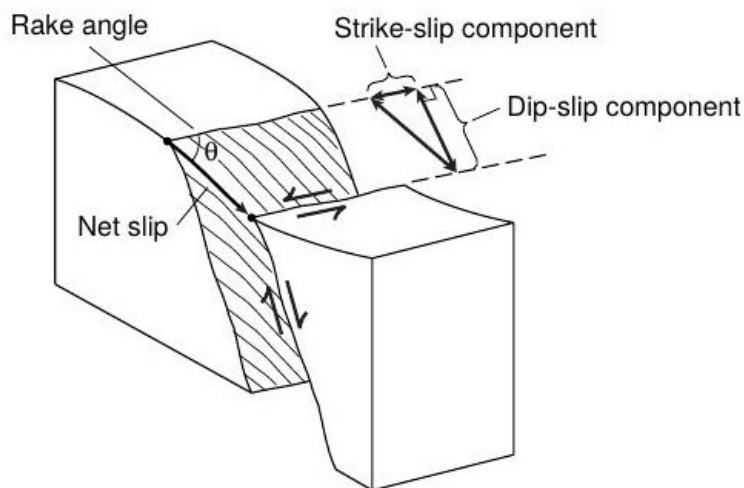


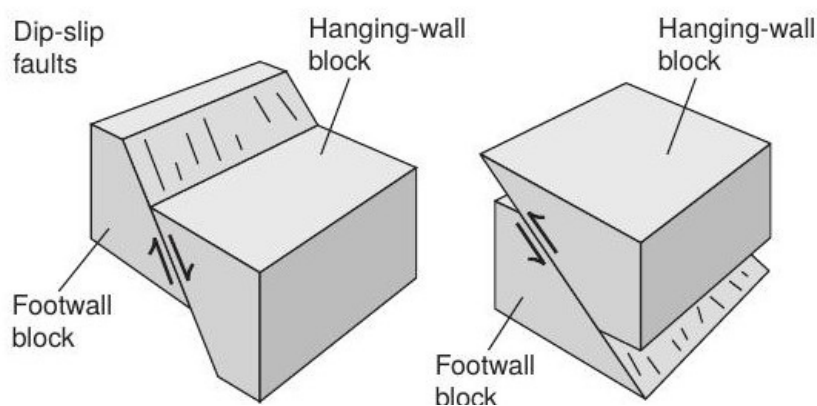
FIGURE 8.3 Block diagram sketch showing the net-slip vector with its strike-slip and dip-slip components, as well as the rake and rake angle.

dip direction, and the **strike-slip component** of net slip in the direction parallel to the strike. If the net-slip vector parallels the dip direction of the fault (within $\sim 10^\circ$), the fault is called a **dip-slip fault**; if the vector roughly parallels the strike of the fault, the fault is called a **strike-slip fault**. If the vector is not parallel to either dip direction or the strike, we call the fault an

8.2 FAULT GEOMETRY AND DISPLACEMENT 169

FIGURE 8.3 (VdP & Marshak, p. 169) — Block diagram showing the **net-slip vector** with its **strike-slip component** and **dip-slip component**, as well as the rake angle. When both components exist simultaneously $\rightarrow$ oblique-slip fault.

VAN DER PLUIJM & MARSHAK · CH. 8 · PAGE 169



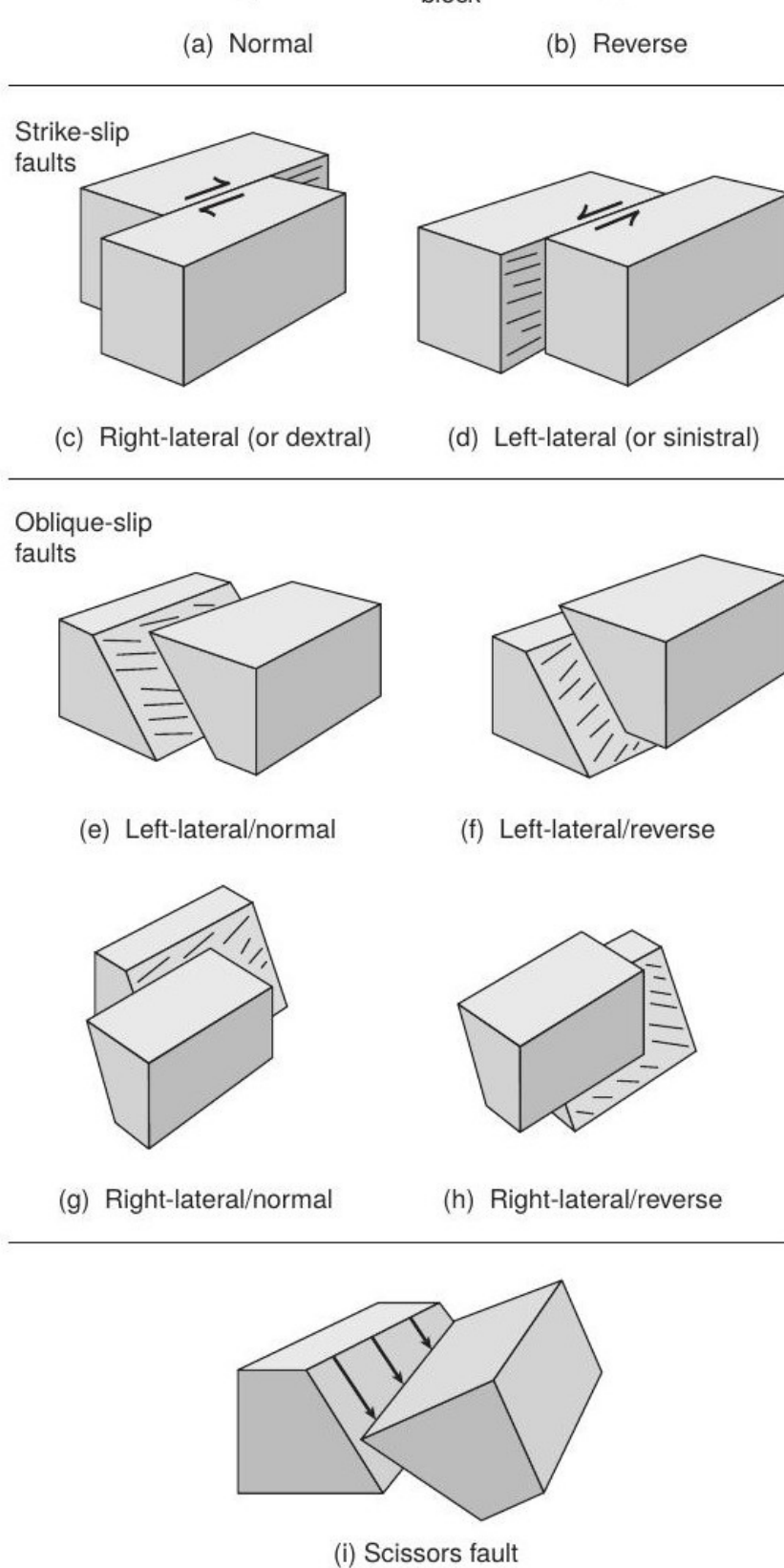


FIGURE 8.4 Block diagram sketches showing the different types of faults.

FIGURE 8.4 (VdP & Marshak, p. 170) — Block diagrams of all fault types. Note (e)–(h) = **oblique-slip faults**: left-lateral/normal, left-lateral/reverse, right-lateral/normal, right-lateral/reverse. These combine dip-slip and strike-slip in one movement.

SUBTYPES OF OBLIQUE-SLIP FAULTS

An oblique-slip fault is described by naming **both components** — the dip-slip sense first, then the strike-slip sense (or vice versa). Four combinations exist:

- Left-lateral / Normal (Sinistral-Normal)**

Hanging wall moves down (normal) AND to the left (sinistral). Example: Figure 9.6 in Fossen.
- Left-lateral / Reverse (Sinistral-Reverse)**

Hanging wall moves up (reverse) AND to the left (sinistral). Common in oblique collision settings.
- Right-lateral / Normal (Dextral-Normal)**

Hanging wall moves down (normal) AND to the right (dextral). Common in pull-apart or transtensional settings.
- Right-lateral / Reverse (Dextral-Reverse)**

Hanging wall moves up (reverse) AND to the right (dextral). Common in oblique convergence / transpressional settings.

ALL FOUR OPTIONS — CONCEPT CARDS

- ✓ **Oblique-slip fault**

Net slip vector is **oblique** — has BOTH a dip-slip component and a strike-slip component. Pitch is between ~10° and ~80°. Fossen: "end-members of a continuous spectrum of oblique faults." Four subtypes (sinistral/dextral × normal/reverse).
- ✗ **Strike-slip fault**

Net slip is **parallel to fault strike** — purely horizontal, no dip-slip component. Pitch ≈ 0°. Can be sinistral (left-lateral) or dextral (right-lateral). This is an *end-member* of the fault spectrum, not a combination.
- ✗ **Dip-slip fault**

Net slip is **parallel to fault dip** — purely in dip direction, no strike-slip component. Pitch ≈ 90°. Includes normal faults (hanging wall down) and reverse faults (hanging wall up). Again, a pure end-member.
- ✗ **Wrench fault**

An **old/informal term for a strike-slip fault** in the petroleum literature. VdP & Marshak (Table 19.1): "A traditional synonym for a strike-slip fault. We *abandon* the term, as it tends to be used inconsistently." A wrench fault is purely strike-slip — no dip-slip component.

SUMMARY COMPARISON TABLE

FAULT TYPE	NET SLIP DIRECTION	DIP-SLIP COMPONENT?	STRIKE-SLIP COMPONENT?	PITCH (RAKE)
Strike-slip fault	Horizontal (parallel to strike)	No	Yes (only)	≈ 0°
Dip-slip fault (normal/reverse)	Down/up the dip direction	Yes (only)	No	≈ 90°

Oblique-slip fault ✓	Oblique (intermediate)	Yes	Yes	10°-80°
Wrench fault	Horizontal (same as strike-slip)	No	Yes (only)	≈ 0°

OPTION-BY-OPTION ANALYSIS

a ✗ INCORRECT

"Strike-slip fault"

A strike-slip fault has movement **only in the horizontal direction**, parallel to the fault's strike. There is no dip-slip component. It is a pure end-member of the fault classification spectrum. Both Fossen (Ch. 9) and Van der Pluijm & Marshak (§8.2, p. 169) explicitly define it as having slip "in the horizontal plane." The question asks for a fault with *both* components — strike-slip alone fails this condition.

→ Strike-slip fault: pitch ≈ 0° — one component only (Fossen Fig. 9.8)

b ✗ INCORRECT

"Dip-slip fault"

A dip-slip fault has movement **only in the dip direction** of the fault surface, resulting in either uplift or subsidence of the hanging wall. There is no lateral (strike-slip) component. Normal faults and reverse faults are the two sub-types. Pitch ≈ 90°. Again, this is a pure end-member — it has only one component of slip, not both.

→ Dip-slip fault: pitch ≈ 90° — one component only (VdP §8.2, p. 169)

c ✓ CORRECT ANSWER

"Oblique-slip fault"

Fossen (Ch. 9): "*Many faults show some deviation from true dip-slip and strike-slip displacement in the sense that the net slip vector is oblique. **Such faults are called oblique-slip faults.***"

Van der Pluijm & Marshak (p. 170): "*If the vector is not parallel to either dip direction of the strike, we call the fault an oblique-slip fault. Oblique-slip faults have **both a strike-slip and a dip-slip component of movement.***"

Both books confirm oblique-slip is the correct term for a fault with both components simultaneously. The slip vector is oblique on the fault surface, giving an intermediate pitch between 0° (pure strike-slip) and 90° (pure dip-slip). Figure 9.8 (Fossen) visually places oblique-slip faults in the center of the classification diagram.

d ✗ INCORRECT — ABANDONED TERM

"Wrench fault"

Wrench fault is an **obsolete term from petroleum geology** for a large-scale continental strike-slip fault. Van der Pluijm & Marshak (Table 19.1, p. 478) state explicitly: "*A traditional synonym for a strike-slip fault... We abandon the term, as it tends to be used inconsistently.*" Critically, a wrench fault has only a *strike-slip* component — it is not a combination of both strike-slip and dip-slip. It therefore cannot answer the question, even before the question of whether the term should be used at all.

→ Wrench fault = informal/abandoned synonym for **strike-slip only**; not both components

EXAM MEMORY AID

Strike-slip = horizontal only → pitch 0° (end-member)

Dip-slip = vertical only → pitch 90° (end-member)

Oblique-slip = BOTH → pitch 10°-80° (intermediate) ✓

Wrench fault = old term for strike-slip → abandoned; not both

Keyword in question: "**both** dip-slip **as well as** strike-slip" = oblique-slip

PROBABLE EXAM QUESTIONS FROM THIS TOPIC

PROBABLE Q · 1

The angle between the slip direction (striation) and the strike of a fault surface is called

- (a) Rake (pitch)
- (b) Plunge
- (c) Trend
- (d) Dip

✓ **Answer: (a)** — Fossen, Ch. 9 Fig. 9.8; VdP & Marshak, Fig. 8.3, p. 169

PROBABLE Q · 2

Normal and strike-slip faults represent which of the following in terms of fault classification?

- (a) Subtypes of oblique-slip faults
- (b) End-members of a continuous spectrum of oblique faults
- (c) Always mutually exclusive categories
- (d) Types classified by dip angle only

✓ **Answer: (b)** — Fossen, Fig. 9.2 caption: "end-members of a continuous spectrum of oblique faults"

PROBABLE Q · 3

A fault where the hanging wall moves downward AND to the left is described as

- (a) Normal sinistral (left-lateral/normal) oblique-slip fault
- (b) Reverse dextral oblique-slip fault
- (c) Strike-slip fault
- (d) Thrust fault

✓ **Answer: (a)** — VdP & Marshak Fig. 8.4e, p. 170; Fossen Fig. 9.6

PROBABLE Q · 4

The term "wrench fault" is considered problematic because

- (a) It implies both dip-slip and strike-slip components
- (b) It is used inconsistently and is now largely abandoned
- (c) It only applies to oceanic plate boundaries
- (d) It refers to a specific kinematic model for thrust faults

✓ **Answer: (b)** — VdP & Marshak, Table 19.1, p. 478: "We abandon the term, as it tends to be used inconsistently"

PROBABLE Q · 5

A fault with a pitch (rake) of 45° would be classified as

- (a) Pure dip-slip
- (b) Pure strike-slip
- (c) Oblique-slip
- (d) Thrust fault

✓ **Answer: (c)** — Fossen, Fig. 9.8; pitch between 10° and 80° = oblique-slip

PROBABLE Q · 6

The vector that connects two formerly adjacent points on either side of a fault is called

- (a) Slip component
- (b) Net slip (displacement vector)
- (c) Throw
- (d) Heave

✓ **Answer: (b)** — Fossen, Ch. 9, p. 153; VdP & Marshak, p. 168

Crystal Symmetry — Complete Concept Guide

Symmetry Elements · Point Groups · The Six Crystal Systems · Crystallographic Axes

Reference: Perkins, *Mineralogy*, 3rd Edition — Chapters 9 & 10

Contents

1. What Is Crystal Symmetry?
2. The Four Symmetry Elements (Mirror, Rotation, Inversion, Rotoinversion)
3. How Symmetry Elements Combine — Special Angles & the Six Axis Combinations
4. Point Groups — The 32 Crystal Classes
5. The Six Crystal Systems — Diagnostic Symmetry & Axial Relations
6. Crystallographic Axes & Unit Cell Parameters
7. Flow Chart: How to Identify the Crystal System of Any Crystal
8. Worked Strategy for Exam Questions Like Q.16
9. Quick-Reference Summary Table

1. What Is Crystal Symmetry?

Symmetry describes the relationship between the parts of an object. For crystals, symmetry is not just a visual curiosity — it is a direct reflection of the ordered, repeating internal atomic structure.

"Symmetry is the order in arrangement and orientation of atoms in minerals, and the order in the consequent distribution of mineral properties."

— Zoltai & Stout (1984), quoted in Perkins, *Mineralogy*, p. 239

Because symmetry governs atomic order, it also governs **every physical property** of a crystal — not just its faces, but its hardness, optical behavior, cleavage, and more, in every direction. This is why identifying a crystal's symmetry (from its faces, or from properties like optical behavior) lets mineralogists infer its internal atomic arrangement, and vice versa.

Why this matters for exam questions

Most DMG-style symmetry questions give you a piece of symmetry information (a certain number/type of axes, mirror planes, or an inversion center) and ask you to name the crystal system. Solving them is a matter of matching the given symmetry description against the diagnostic symmetry of each of the six systems — covered in Section 5.

2. The Four Symmetry Elements

Every crystal's external symmetry can be fully described using combinations of just four types of symmetry elements.

(a) Mirror Plane (m)

Mirror plane (m): An imaginary plane that divides a crystal so that one half is the mirror image of the other. The symmetry operation is called *reflection*.

"We call such symmetry reflection, and we call the plane of the imaginary mirror the mirror plane. In shorthand notation, we use the letter m to designate a mirror of symmetry."

— Perkins, *Mineralogy*, p. 239

A face on one side of a mirror has an identical partner at an equal perpendicular distance on the other side, with opposite "handedness." A cube has the maximum number of mirror planes possible for any crystal:

"A perfect sphere has an infinite number of mirror planes, but... crystals cannot have more mirror planes than a cube. Cubes have nine mirrors."

— Perkins, *Mineralogy*, p. 240

(b) Rotation Axis (1, 2, 3, 4, 6)

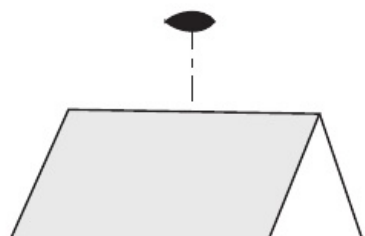
Rotation axis: A line about which a crystal can be rotated and appear unchanged one or more times during a full 360° turn. The symmetry operation is called *rotation*.

"In shorthand notation, we symbolize proper rotation axes using the numbers 1, 2, 3, 4, or 6, corresponding to rotations of 360°, 180°, 120°, 90°, and 60°, respectively."

— Perkins, *Mineralogy*, p. 241

Important restriction

Minerals can only show 1-, 2-, 3-, 4-, or 6-fold rotational symmetry. 5-fold and any symmetry greater than 6-fold (7-fold, 8-fold, etc.) is geometrically impossible for a periodic crystal lattice, even though such shapes can be drawn on paper.



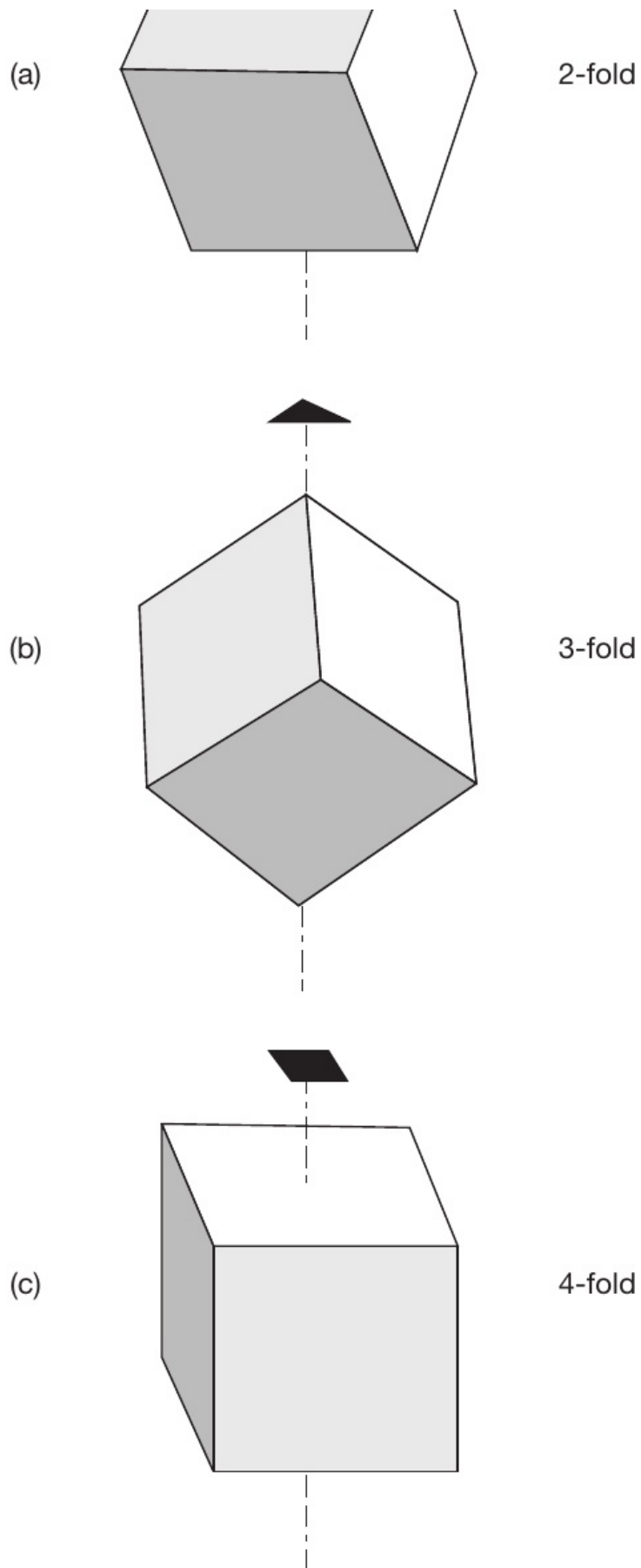


FIGURE 5 Rotation axes of a cube: (a) the edge diagonals of a cube are 2-fold rotational axes of symmetry; (b) the main diagonals of a cube are 3-fold rotational axes of

and 4-fold axes (perpendicular to faces). This is exactly why a cube is used to derive the $54^\circ 44'$ angle relevant to Q.16.

(c) Inversion Center (i, or $\bar{1}$)

Inversion center: A single point at the center of a crystal through which every face/point is repeated at an equal distance on the opposite side, with reversed orientation (like a pinhole-camera image).

"As with mirror planes, inversion centers relate two identical faces on a crystal; but while mirror planes 'reflect' faces and change their 'handedness,' inversion centers invert them."

— Perkins, *Mineralogy*, p. 241

(d) Rotoinversion Axis ($\bar{1}$, 2, 3, 4, 6)

Rotoinversion: A single, compound symmetry operation that combines rotation and inversion simultaneously — it is **not** simply "rotation, then separately inversion."

"Rotoinversion, a combination of rotation and inversion, is a symmetry operation distinct from the others. The symbols $\bar{1}$, 2, 3, 4 and 6 represent rotoinversion axes. They are articulated as 'bar-1,' 'bar-2,' etc."

— Perkins, *Mineralogy*, p. 246

Key exam fact

Perkins notes: "a 3 operation is equivalent to proper 3-fold rotation and inversion operating separately," and a 2 axis is equivalent to a mirror (m), and a 6 axis is equivalent to 3/m. But **4 is unique** — it cannot be reproduced by any combination of the other basic operations. A crystal can have a 4 axis without having either a true 4-fold axis or an inversion center.

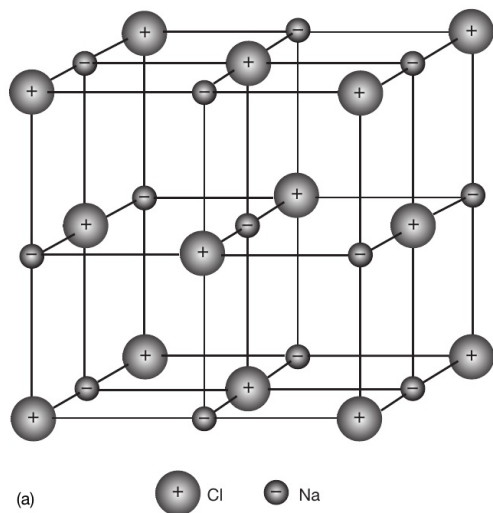


FIGURE 1 (a) The atomic structure of halite showing the cubic arrangement of Na^+ and Cl^- ions. (b) Cubic halite crystals.

the other side of the mirror. The two faces or objects are the same perpendicular distance from the mirror and have opposite “handedness”; if one points to the right, the other points to the left.

Two-dimensional drawings may have many mirror planes or no mirror planes. Circles, for example, contain an infinite number of mirror

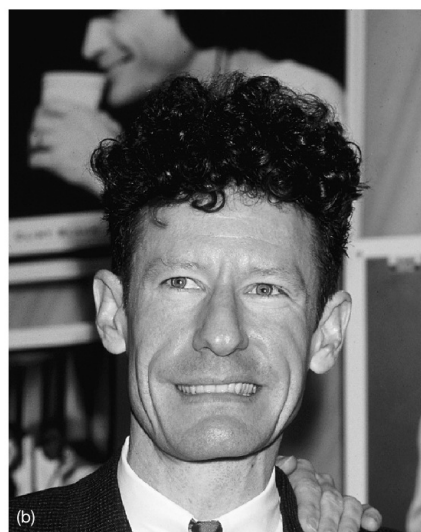
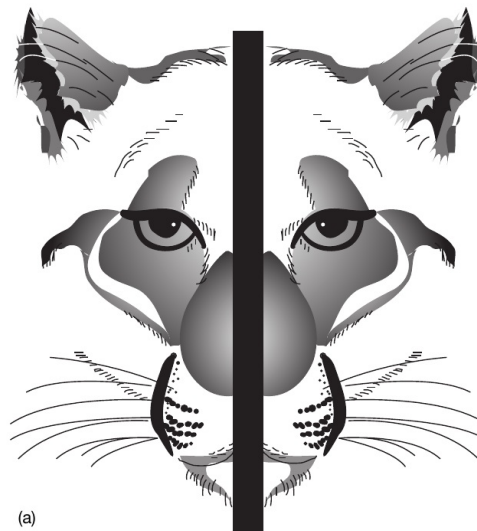


FIGURE 2 Animal, including human, faces are often quite symmetrical. (a) Because the right and left sides of the cougar's face are identical, there is a mirror plane running

Figure 12 & 13 (Perkins, Mineralogy, pp. 246–247) — A 4 rotoinversion axis operating on a point (top), and a side-by-side comparison of a true 4-fold axis, a 4 rotoinversion axis, and a 4-fold axis combined with a separate inversion center ($4/m$) — three genuinely different symmetries that can look superficially similar.

3. How Symmetry Elements Combine — Special Angles

Symmetry elements cannot combine at just any angle. Perkins shows that if two rotation axes intersect at a *non-special* angle (one that doesn't divide evenly into 360°), applying them to each other generates infinitely more axes — which is impossible in a real, finite crystal structure. So rotation axes are only ever found combined at **special angles** (30° , 45° , 60° , 90° , 120° , or the special cube-diagonal angles like $54^\circ 44'$ and $70^\circ 32'$).

This restriction means only **six** combinations of rotation axes are possible in any crystal:

Crystal Morphology and Symmetry

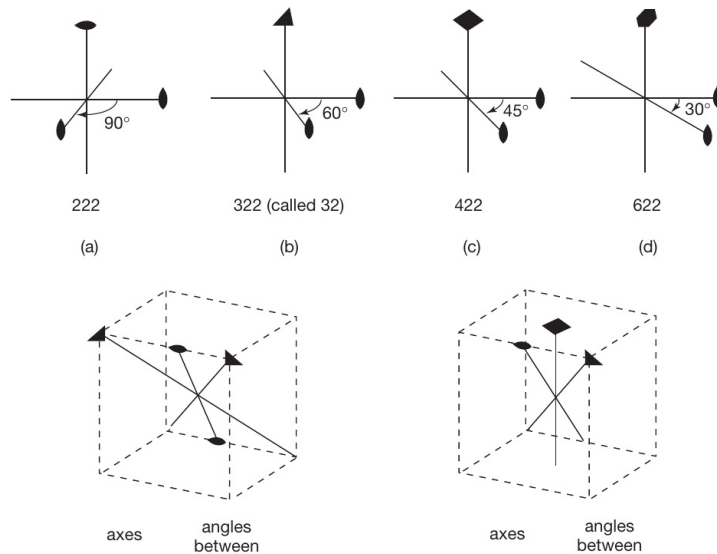


FIGURE 18 The six possible combinations of rotation axes. The angles between axes in (a), (b), (c), and (d) are shown on the drawings. For (e) and (f), based

Figure 18 (Perkins, *Mineralogy*, p. 251) — The six possible combinations of rotation axes: 222, 32, 422, 622 (simple combinations at 90°/60°/45°/30°), and two cube-based combinations (e, f) whose axes are the edges, face diagonals, and body diagonals of a cube — giving the special angles 70°32', 54°44', 45°00', and 35°16'.

Angle between a cube edge and a cube body diagonal = $\arccos(1/\sqrt{3}) = 54.7356^\circ = 54^\circ 44'$

(This is exactly the angle in Q.16, and it is only ever produced by **four** 3-fold axes — the signature of the isometric system.)

4. Point Groups — The 32 Crystal Classes

Point group: The complete set of symmetry operations (rotation axes, mirrors, inversion center, rotoinversion axes) that leave at least one point in the crystal fixed and describe its overall symmetry. There are exactly **32** possible point groups for crystals — no more, no fewer.

Because several symmetry operators are redundant with each other (e.g. a 2 axis behaves exactly like a mirror plane), crystallographers describe all crystal symmetry using just 13 independent standard operators (1, 2, 3, 4, 6, I, 2(=m), 3, 4, 6, i, plus combined symbols like 2/m, 4/m, 6/m). Point group symbols (Hermann-Mauguin notation) list the essential symmetry elements along each of a crystal's principal directions — for example, **4/m32/m** is the full symbol for the highest-symmetry cubic class (a 4-fold axis with a perpendicular mirror, four 3-fold axes, and a 2-fold axis with a perpendicular mirror).

Exam shortcut

You rarely need to identify the full point group symbol for an entrance-exam MCQ — you almost always just need the **crystal system**, which depends only on the highest-order and most numerous rotation/rotoinversion axes present (see Section 5's diagnostic rules).

5. The Six Crystal Systems

Every crystal belongs to exactly one of six crystal systems, grouped by their **diagnostic (defining) symmetry** — the minimum symmetry every member of that system must possess.

Cubic (Isometric) System

Diagnostic symmetry: Always has **four** 3-fold (or 3) axes — the cube's body diagonals. May also have three 4-fold axes, six 2-fold axes, nine mirrors, and a center.

Axial relation: $a = b = c$, all angles = 90°.

Optical character: Isotropic (same optical properties in every direction).

Rule: "All crystals with four 3-fold axes of symmetry must belong to the cubic system." (Perkins, p. 302)

Hexagonal System

Diagnostic symmetry: A **single** 3-fold, 3, or 6-fold axis (never more than one). May also have 2-fold axes and mirrors.

Axial relation: $a = b \neq c$; angles 90° or 120°.

Optical character: Uniaxial — the single optic axis is parallel to the 3-, 3-, or 6-fold axis.

Rule: "Crystals with more than one 3-fold or 3 axis belong to the cubic system." (Perkins, p. 261)

Tetragonal System

Diagnostic symmetry: One (and only one) 4-fold or 4 axis.

Axial relation: $a = b \neq c$, all angles = 90° .

Rule: "Crystals that have more than one 4-fold axis must belong to the cubic system." (Perkins, p. 261)

Orthorhombic System

Diagnostic symmetry: Three mutually perpendicular 2-fold axes (222), or one 2-fold axis with two perpendicular mirrors (mm2).

Axial relation: $a \neq b \neq c$, all angles = 90° .

Monoclinic System

Diagnostic symmetry: A single 2-fold axis and/or a single mirror plane — lower symmetry than orthorhombic.

Axial relation: $a \neq b \neq c$; $\alpha = \gamma = 90^\circ$, $\beta \neq 90^\circ$.

Triclinic System

Diagnostic symmetry: At most an inversion center (I); no rotation axes higher than 1-fold, and no mirrors. The lowest-symmetry system.

Axial relation: $a \neq b \neq c$; none of α , β , γ equal 90° .

6. Crystallographic Axes & Unit Cell Parameters

Crystallographers describe a unit cell's shape with six parameters: three edge lengths **a**, **b**, **c** and three angles **α** (between b and c), **β** (between a and c), and **γ** (between a and b).

"The symmetry of a unit cell always affects the relationships between a, b, and c. So, in the cubic system, $a = b = c$, but in the tetragonal and hexagonal systems $a = b \neq c$."

— Perkins, *Mineralogy*, p. 307



Unit Cells, Points, Lines, and Planes

To describe the details of crystals and crystal structures, it is necessary to be more quantitative. In this chapter we describe points, lines, and planes in crystal structures using a coordinate system based on unit cell geometry. The most important results are that we can give exact descriptions of crystal structures, and we can label and describe crystal faces and crystal properties in unambiguous ways.

A crystal is composed of many identical building blocks called *unit cells*. To describe a unit cell, we must give its shape, size, and composition. Symmetry axes, crystal faces, and other linear and planar features of crystals depend on the nature of the unit cell and on crystal growth. To describe them, we need some way to describe the orientation of lines and planes in crystals. Furthermore, for complete characterization of a crystal, we must specify the locations of atoms in its structure. We therefore need a quantitative way to describe points (for example, atom locations), lines, and planes in crystals and unit cells. To do this, it is convenient to have a coordinate system. The system used by crystallographers is similar to a **Cartesian coordinate system**, with the exception that the angles between the three axes vary according to the crystal being described, as do the unit lengths along each axis.

UNIT CELL PARAMETERS AND CRYSTALLOGRAPHIC AXES

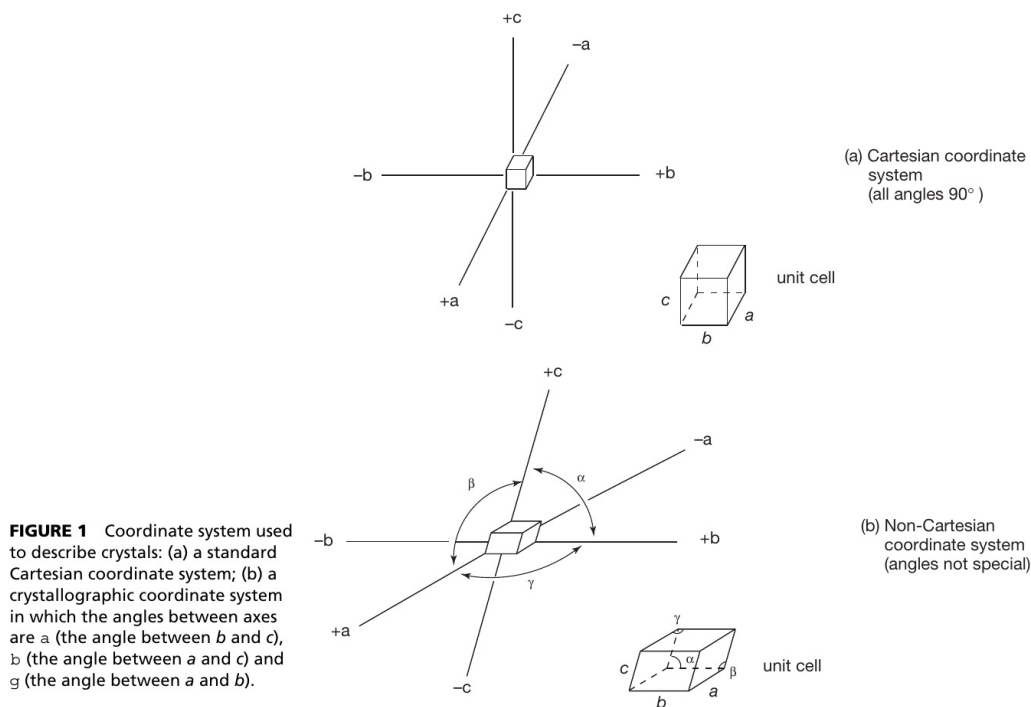
Unit cell parameters are a , b , c , α , β , and γ . a , b , and c are the lengths of unit cell edges; α , β , and γ are the angles between the edges, α being the angle between b and c , β the angle between a and c ,

and γ the angle between a and b . Mineralogists have traditionally used angstroms ($1 \text{ \AA} = 10^{-10} \text{ m}$) but some more recent literature uses nanometers ($1 \text{ nm} = 10 \text{ \AA}$) to measure a , b , and c . The angles α , β , and γ are today given as a decimal number of degrees (for example, 94.62°). Typical mineral unit cells have edges of 2 to $20 \text{ \AA}$; angles between edges vary greatly, although unit cells are often chosen so that angles are close to 90° .

Unit cell edges define the coordinate system used by crystallographers (Figure 1). Three axes, designated a , b , and c , run parallel to a , b , and c length edges of the unit cell. The angles between the axes are therefore α , β , and γ . In the cubic, tetragonal, and orthorhombic systems, α , β , and γ all equal 90° because the unit cell edges are orthogonal.

Instead of using angstroms to give distances, we may also use unit cell dimensions as a scale. For example, we might say that a certain plane intersects the axes at distances of $3a$, $2b$, and $2c$ from the origin. It is implicit that $3a$ refers to a distance equal to three unit cell edge lengths along the a axis, $2b$ a distance equal to two unit cell lengths along the b axis, and $2c$ a distance equal to two unit cell lengths along the c axis.

Figure 2, part 1 (Perkins, *Mineralogy*, pp. 307–308) — Unit cells of real minerals from the cubic, tetragonal, hexagonal, and orthorhombic systems, showing their axial lengths and angles.



The symmetry of a unit cell always affects the relationships between a , b , and c . So, in the cubic system, $a = b = c$, but in the tetragonal and hexagonal systems $a = b \neq c$. The relationships implied by crystal systems mean that, for systems other than triclinic, we need not give six values to describe unit cell shape. Table 1 gives examples of unit cell parameters for minerals from each of the crystal

systems; unnecessary information has been omitted. Figure 2 depicts the unit cells.

Consider the triclinic mineral albite, a feldspar with composition $\text{NaAlSi}_3\text{O}_8$. The last column in Table 1 gives albite's cell dimensions and Figure 2f shows a drawing of the unit cell with all edges and angles labeled. Because none of the angles are special and none of the cell edges are equal, we need six parameters to describe the unit cell shape. In

TABLE 1 Unit Cell Parameters (a , b , c , A , B , γ), Z (number of formulas per unit cell), and V (unit cell volume) for One Mineral from Each of the Crystal Systems*

Fluorite CaF_2	Rutile TiO_2	Beryl $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$	Enstatite $\text{Mg}_2\text{Si}_2\text{O}_6$	Sanidine KAlSi_3O_8	Albite $\text{NaAlSi}_3\text{O}_8$
cubic	tetragonal	hexagonal	orthorhombic	monoclinic	triclinic
$Z = 4$	$Z = 2$	$Z = 2$	$Z = 4$	$Z = 4$	$Z = 4$
$a = 5.46$	$a = 4.59$	$a = 9.23$	$a = 18.22$	$a = 8.56$	$a = 8.14$
$V = 162.77$	$c = 2.96$	$c = 9.19$	$b = 8.81$	$b = 13.03$	$b = 12.8$
	$V = 62.36$	$V = 2034.09$	$c = 5.21$	$c = 7.17$	$c = 7.16$
			$V = 836.30$	$\beta = 115.98$	$\alpha = 94.33$
				$V = 799.72$	$\beta = 116.57$
					$\gamma = 87.65$
					$V = 746.01$

* a , b , and c are in angstroms; α , β , and γ are in degrees; and V is in cubic angstroms.

Figure 2, part 2 (Perkins, *Mineralogy*, p. 308) — Monoclinic (sanidine) and triclinic (albite) unit cells — note progressively fewer axial and angular constraints as symmetry decreases.

7. Flow Chart: Identifying the Crystal System

This is the single most useful figure for solving MCQs like Q.16 — it converts a description of symmetry elements directly into a crystal system and point group.

BOX 6

Determining Crystal System and Point Group

The flow chart shown below can be used to determine a crystal's system and point group. The procedure is relatively straightforward for most well-formed crystals. However, for subhedral or anhedral crystals it is often impossible.

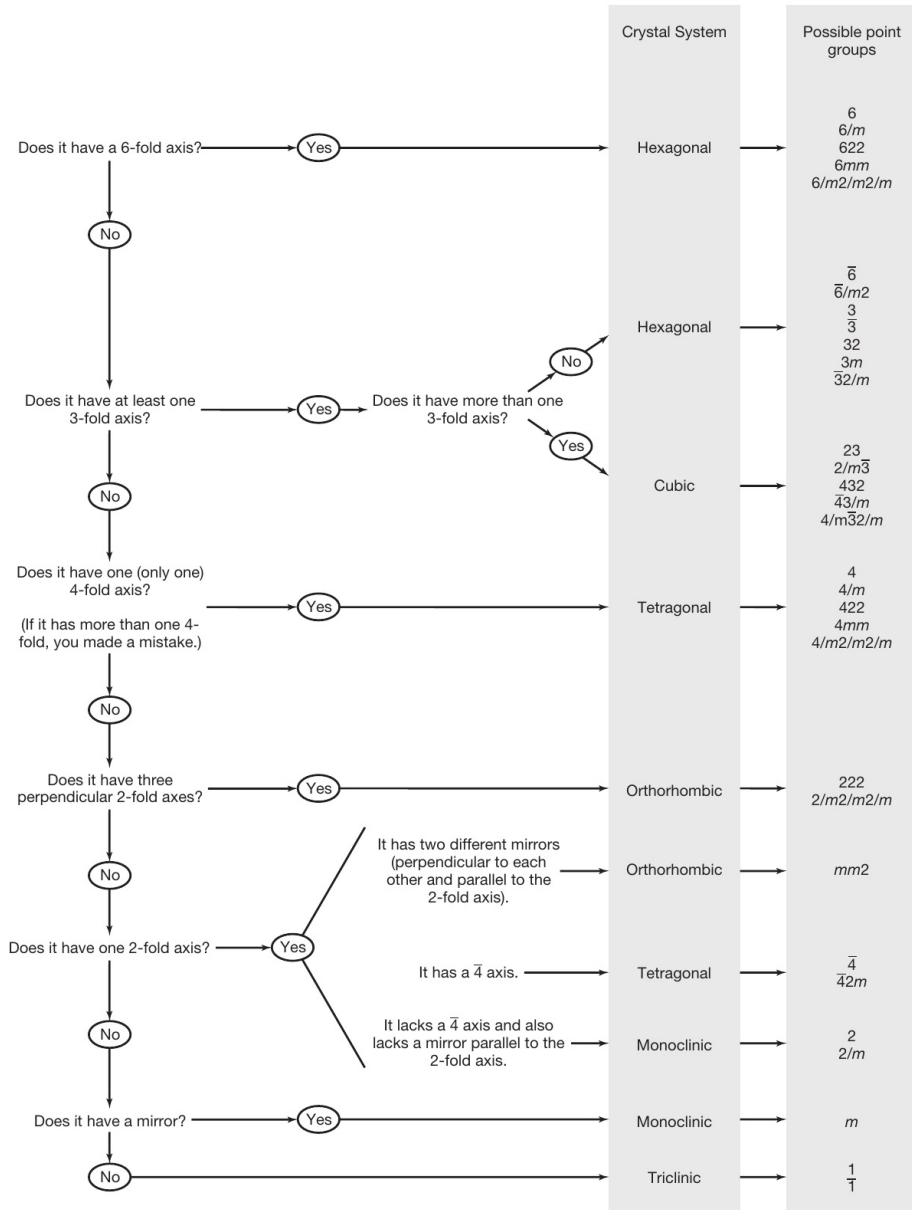


Figure 32 / Box 6 (Perkins, Mineralogy, p. 262) — Flow chart for determining crystal system and point group from observed symmetry elements.

8. Worked Strategy for Exam Questions Like Q.16

When a question describes a specific symmetry element (or combination) and asks for the crystal system, follow this sequence:

Step 1 — Count the highest-order axis and how many there are.

More than one 3-fold/3 axis → cubic, automatically. Exactly one 6-fold → hexagonal. Exactly one 4-fold → tetragonal (unless a second 4-fold appears, which forces cubic). Exactly one 3-fold/3 and no 4- or 6-fold → hexagonal.

Step 2 — If no 3-, 4-, or 6-fold axis is present, look at 2-fold axes and mirrors.

Three perpendicular 2-folds → orthorhombic. One 2-fold and/or one mirror → monoclinic. Only an inversion center or nothing → triclinic.

Step 3 — Check any stated angle against the special-angle table.

54°44' and 70°32' only arise from cube body/edge diagonals → cubic. 90°/60°/45°/30° correspond to the simple combinations

Step 4 — Eliminate distractor options using the "must belong to cubic" rules.

If an option like Hexagonal or Tetragonal is offered but the question specifies *more than one* high-order axis, it is automatically wrong — Perkins states plainly that multiple 3-fold or multiple 4-fold axes force the cubic system.

9. Quick-Reference Summary Table

System	Diagnostic Symmetry	Axial Relation	Angles
Cubic (Isometric)	Four 3-fold/3 axes	$a = b = c$	All 90°
Hexagonal	One 3-, 3-, or 6-fold axis	$a = b \neq c$	90° or 120°
Tetragonal	One 4-fold or 4 axis	$a = b \neq c$	All 90°
Orthorhombic	Three perpendicular 2-fold axes	$a \neq b \neq c$	All 90°
Monoclinic	One 2-fold axis and/or one mirror	$a \neq b \neq c$	$\alpha=\gamma=90^\circ, \beta\neq90^\circ$
Triclinic	At most an inversion center	$a \neq b \neq c$	None = 90°

Q.16 — Symmetry of Four Three-Fold Axes

DMG 2019 · Mineralogy: Crystal Morphology & Symmetry · Reference: Perkins, Mineralogy, 3rd ed.

Q.16. Four three-fold axes, each inclined at $54^{\circ}44'$ to the crystallographic axes, are characteristic symmetry of which one of the following crystal systems?

- (a) Hexagonal
- (b) Isometric
- (c) Orthorhombic
- (d) Triclinic

1. The Underlying Concept — How Rotation Axes Combine

Perkins explains that symmetry axes in crystals cannot combine in random ways — only **six** geometrically valid combinations of rotation axes exist. Four are the simple combinations 222, 32, 422, and 622. The remaining two combinations are built entirely around the geometry of a **cube**, and these are the combinations that define cubic (isometric) symmetry.

"The other two (Figures 18e and f) are best envisioned by thinking about a cube. The angles between rotation axes are all special angles (to avoid infinite symmetry). In Figures 18e and f, the angles are equal to angles between various diagonals in a cube and perpendiculars to cube faces."

— Perkins, *Mineralogy*, p. 251

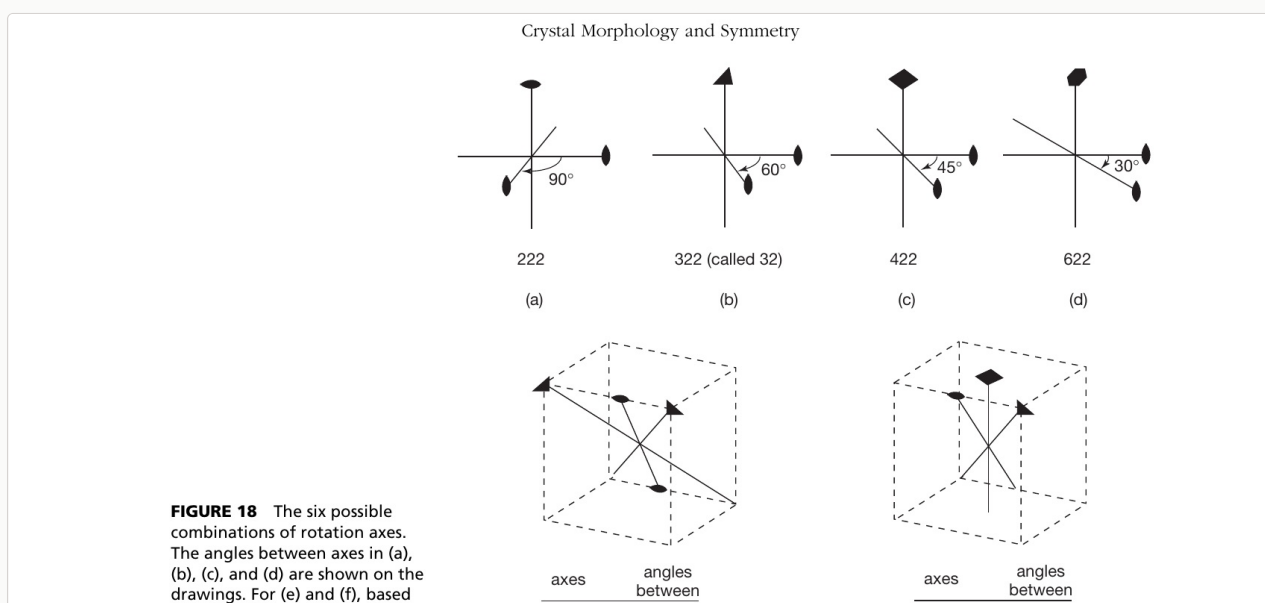


Figure 18 (Perkins, *Mineralogy*, p. 251) — The six possible combinations of rotation axes. Combinations (e) and (f) are based on a cube; their angle tables list $70^{\circ}32'$, $54^{\circ}44'$, $45^{\circ}00'$, and $35^{\circ}16'$ — the special angles between a cube's edges, face diagonals, and body diagonals.

Where $54^{\circ}44'$ comes from — the geometry

The three crystallographic axes of a cubic crystal run along the cube's **edges** (coinciding with the three 4-fold axes). The **body diagonals** of the cube are the **four 3-fold axes**. The angle between an edge and a body diagonal of a cube is fixed by simple solid geometry:

$$\theta = \arccos(1/\sqrt{3}) = 54.7356^{\circ} = 54^{\circ}44'$$

Because a cube has exactly **four** space diagonals, this construction always produces **four** 3-fold axes, each tilted $54^{\circ}44'$ from all three crystallographic axes — never more, never fewer, and only in a cubic arrangement.

2. Confirming the Diagnostic Symmetry of the Cube

"Cubes have the highest symmetry possible for crystals: three 4-fold axes, **four 3-fold axes**, six 2-fold axes, nine mirror planes, and an inversion center."

— Perkins, *Mineralogy*, p. 244

"**Cubic System:** This system is also called the **isometric system**. Crystals have high symmetry, all having **four 3-fold or 3 axes**. Some have three 4-fold axes as well, and some have 2-fold axes or mirror planes."

— Perkins, *Mineralogy*, p. 261

The book even sets this exact relationship as a self-test exercise at the end of the chapter:

"All crystals with **four 3-fold axes of symmetry must belong to the cubic system**."

— Perkins, *Mineralogy*, p. 302 (Questions for Thought, Q.3)

So the presence of exactly **four** three-fold axes — not one, not two — is the single diagnostic fingerprint of the isometric system, and their fixed 54°44' inclination to the crystallographic axes is the direct geometric fingerprint of a cube's body diagonals.

3. Evaluating Each Option

Option	Symmetry Content	Verdict
(a) Hexagonal	Only one 3-fold, 3, or 6-fold axis. Perkins: "Crystals with more than one 3-fold or 3 axis belong to the cubic system."	Eliminated
(b) Isometric	Four 3-fold axes, each inclined at 54°44' to the three crystallographic (edge) axes — exact match to the question	Correct
(c) Orthorhombic	Three mutually perpendicular 2-fold axes only; no 3-fold axes at all	Eliminated
(d) Triclinic	Lowest symmetry class — at most a center of inversion; no rotation axes higher than 1-fold	Eliminated

✓ Answer: (b) Isometric

4. Probable Follow-Up Exam Questions

Q. What is the total symmetry content of the highest-symmetry cubic point group (m3m, hexoctahedral class)?

A. Three 4-fold axes, four 3-fold axes, six 2-fold axes, nine mirror planes, and a center of symmetry.

Q. What is the general (48-face) form of the isometric system called?

A. The hexoctahedron.

Q. Why are isometric-system minerals optically isotropic?

A. The four 3-fold axes make atomic arrangement equivalent in every direction, so there is no unique optical direction — refractive index is identical in all directions.

Q. What angle does a cube's face diagonal make with its edge (relevant to the 4-2 axis combination in Figure 18f)?

A. 45°00'.

Q. Which crystal system is optically uniaxial, with its optic axis parallel to a single 3-fold, 3, or 6-fold axis?

A. Hexagonal system.

Postmagmatic Alteration Processes — Complete Concept Guide

Autometamorphism · Deuteric Alteration · Devitrification · Replacement Textures

References: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. & Perkins, *Mineralogy*, 3rd ed.

Contents

1. What Counts as "Alteration"? Autometamorphism vs. Deuteric Alteration
2. The Deuteric (Hydration) Alteration Reactions
 - a. Uralitization (pyroxene → amphibole)
 - b. Biotitization (pyroxene/amphibole → biotite)
 - c. Chloritization (any mafic → chlorite)
 - d. Sericitization (feldspar → white mica)
 - e. Saussuritization (plagioclase → epidote)
3. Olivine Alteration — Serpentine & Iddingsite
4. Symplectite & Pseudomorphs — the General Replacement Concept
5. Myrmekite — A Related (Non-Hydration) Secondary Texture
6. Devitrification of Glass — Palagonite, Felsite, Spherulites, Lithophysae
7. Quick-Reference Summary Table
8. Probable Exam Questions

1. What Counts as "Alteration"? Autometamorphism vs. Deuteric Alteration

Winter groups all of this under **secondary textures**: changes that happen to an igneous rock's minerals *after* the rock is essentially solid, but while the surrounding pluton is still hot from its own heat of crystallization — not from any later, separate metamorphic event.

"Secondary mineral reactions that occur in igneous rocks as they cool, and that are not products of a distinct later metamorphic event, are commonly called autometamorphic processes rather than metamorphic ones because they are a natural part of igneous cooling. Autometamorphic processes are more common in plutonic rocks than in volcanics because they remain at elevated temperatures for a longer time."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Important boundary case

Winter explicitly notes: "Diagenetic and weathering processes are not considered to be autometamorphic (a rather arbitrary distinction)." So if an exam option offers "diagenesis" or "weathering" as an alternative to a named deuteric process, it is always wrong — those are excluded by definition, even though the line is admittedly a judgment call.

Deuteric alteration: The specific subset of autometamorphic reactions that involve **hydration** — water either escaping from the last residual melt, or entering from outside, reacting with already-crystallized minerals at moderate temperature.

"Most, but not all, autometamorphic reactions involve minerals at moderate temperatures in an environment in which H₂O is either liberated from residual melt or externally introduced. Such alterations are a subset of autometamorphism, which involves hydration, and they are called deuteric alterations."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Note the general pattern that runs through nearly every deuteric reaction below: an anhydrous, higher-temperature igneous mineral is progressively replaced by a lower-temperature, **hydrous** mineral, and replacement typically starts at grain rims/cleavages and works inward.

"As in other deuteric alterations, hydration attacks the outer margin of a mineral, so that chlorite generally replaces the initial mafic from the rim inward. In the case of biotite, H₂O may work its way along the prominent cleavages."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

2. The Deuteric (Hydration) Alteration Reactions

(a) Uralitization — Pyroxene → Amphibole

"Pyroxene is a common primary mafic mineral in a variety of igneous rocks. If H₂O penetrates at modest temperatures, a deuteric alteration of pyroxene to amphibole results, called uralitization. Any gradation from amphibole rims on pyroxene cores to multiple patches of pyroxene in an amphibole to complete replacement is possible. The amphibole may be a single crystal of hornblende or a fibrous actinolite or hornblende aggregate. Either, when demonstrated to result from pyroxene alteration, may be called uralite, but the term is more commonly applied to the aggregates."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of pyroxene by amphibole." (p. 55)

(b) Biotitization — Pyroxene/Amphibole → Biotite

"Biotitization is a similar process of hydration/deuteric alteration that produces biotite, either directly from pyroxene, or, more commonly, from hornblende. Because biotite contains little Ca, epidote may be produced as Ca is released during the alteration of hornblende to biotite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of pyroxene or amphibole by biotite." (p. 55)

Exam trap

Biotitization is another reaction that can release Ca to form epidote as a by-product — don't confuse this incidental epidote with the epidote that defines saussuritization; the diagnostic reactant is different (hornblende here, plagioclase in saussuritization).

(c) Chloritization — Any Mafic Mineral → Chlorite

"Chloritization is the alteration of any mafic mineral to chlorite. Chlorite is a very hydrous phyllosilicate and typically replaces the less hydrous mafics at low temperature when water is available. Pyroxenes, hornblendes, and biotites are commonly observed in thin section in various stages of alteration to chlorite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

Table 1 shorthand: "Replacement of any mafic mineral by chlorite." (p. 55)

(d) Sericitization — Feldspar/Feldspathoid → White Mica

"Sericitite is a term applied to any very fine-grained white mica. Seritization is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light... K⁺ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K⁺ and may be more sericitized than associated plagioclase."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Table 1 shorthand: "Replacement of feldspar or feldspathoids by fine white micas." (p. 55)

(e) Saussuritization — Ca-Rich Plagioclase → Epidote + Albite

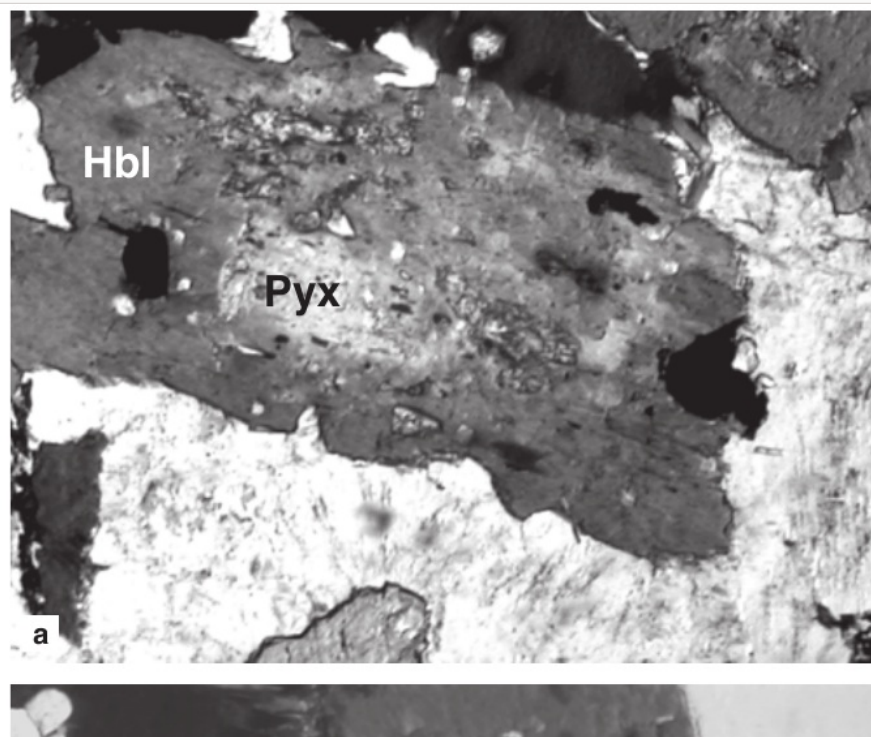
"Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite, releasing Ca and Al to form an epidote mineral ($\pm$ calcite and/or sericite). In zoned plagioclase, one can occasionally observe the products of saussuritization or seritization concentrated in the Ca-rich core or some of the more Ca-rich oscillatory bands in the original grain."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 51-52

Table 1 shorthand: "Replacement of plagioclase by epidote." (p. 55)

"Sausserite is a name given to fine-grained epidote produced by alteration of plagioclase."

— Perkins, *Mineralogy*, p. 391



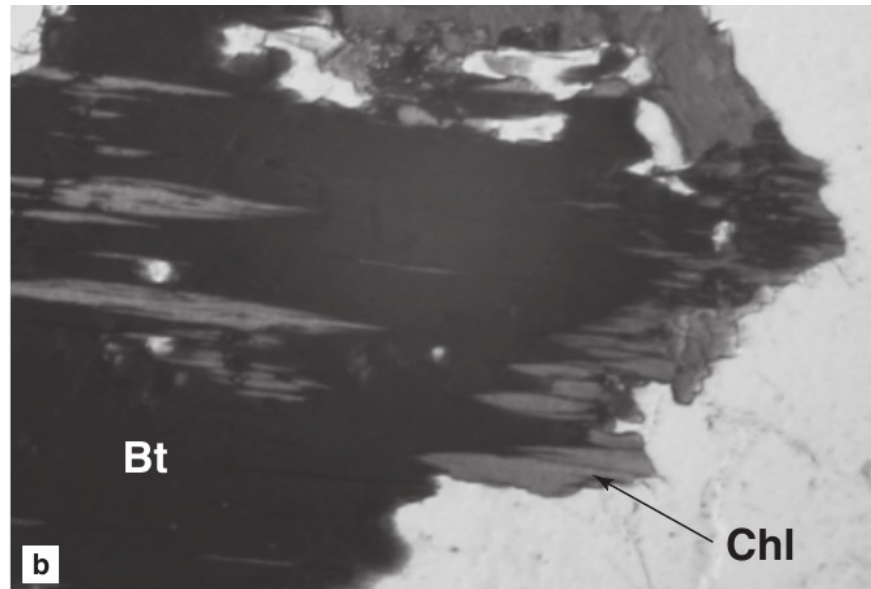


FIGURE 20 (a) Pyroxene largely replaced by hornblende. Some pyroxene remains as light areas (Pyx) in the hornblende core. Width 1 mm. (b) Chlorite (light) replaces biotite (dark) at the rim and along cleavages. Tonalite, San Diego, California. Width 0.3 mm.

generally replaces the initial mafic from the rim inward. In the case of biotite, H_2O may work its way along the prominent cleavages, and chlorite can be seen to replace biotite margins as well as along cleavage planes (Figure 20b).

Sericite is a term applied to any very fine-grained white mica. **Seritization** is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light. In more advanced stages of alteration, the feldspars appear speckled, with fine micas having yellowish birefringence and then larger clots with coarser crystals and higher birefringence. K^+ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K^+ and may be more sericitized than associated plagioclase.

Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite,

Figure 20 (Winter, Principles of Igneous and Metamorphic Petrology, p. 53) — (a) Pyroxene largely replaced by hornblende, remnant pyroxene as light patches (uralitization). (b) Chlorite (light) replacing biotite (dark) at rims and along cleavages (chloritization), tonalite, San Diego.

3. Olivine Alteration — Serpentine & Iddingsite

"Olivine is easily altered in cooling mafic rocks, even in volcanics. It is commonly rimmed or replaced by serpentine or dark brown iddingsite."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Note that olivine's alteration is distinctive enough to have its own dedicated pair of product minerals (serpentine, a hydrous Mg-phyllosilicate; iddingsite, a reddish-brown mixture of iron oxides/oxyhydroxides and clay), rather than falling under one of the five named "-ization" processes above.

4. Symplectite & Pseudomorphs — the General Replacement Concept

Pseudomorph: A replacement texture in which one or more minerals replace another, but the crystal retains the outward form of the original mineral.

Symplectite: A replacement texture in which a mineral is replaced by an intergrowth of two or more minerals growing together.

"Symplectite is a term applied to fine-grained intergrowths resulting from the combined growth of two or more minerals as they replace another mineral. As in any of the replacements described above, replacement may be partial or complete. Complete replacement and pseudomorphs are common. The fibrous actinolite-hornblende 'uralite' aggregates replacing pyroxene are an example. Biotite and epidote replacing hornblende is another."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

How this connects

Symplectite is the general textural category; uralite and biotite+epidote (from biotitization) are two named examples of symplectites that already appeared above. An exam question describing "an intergrowth of two minerals replacing an original grain" without naming a specific reaction is testing the term *symplectite*, not a specific "-ization."

5. Myrmekite — A Related (Non-Hydration) Secondary Texture

"Myrmekite is an intergrowth of dendritic quartz in a single crystal of plagioclase. The quartz appears rod-like in thin section, and numerous adjacent rods go extinct in unison, indicating that they are all parts of a single quartz crystal. Myrmekites are very common in granitic rocks and occur preferentially where plagioclase is in contact with K-feldspar. Myrmekites appear to have grown from the plagioclase-K-feldspar boundary into the K-feldspar. As the plagioclase replaces the K-feldspar, SiO₂ is released (the anorthite component of plagioclase contains less SiO₂ than the K-feldspar), thereby producing the quartz."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 52-53

Key distinction for MCQs

Myrmekite is **not** a hydration/deuteric reaction — it's driven by a silica-releasing plagioclase-K-feldspar replacement, not by water attacking a mafic or feldspathic mineral. If an option lists "myrmekite" alongside saussuritization/uralitization/chloritization as if they're all the same kind of process, that's the distractor to catch.

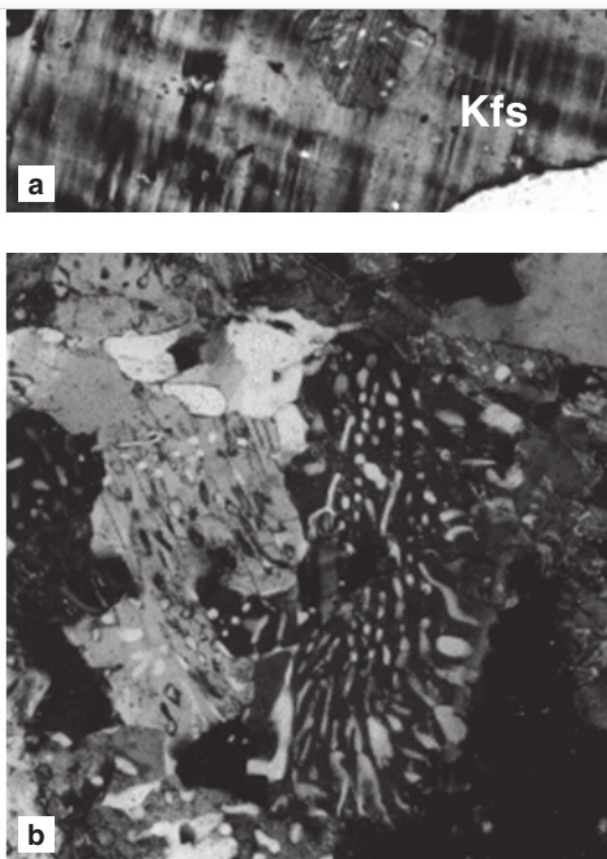


FIGURE 21 Myrmekite formed in plagioclase at the boundary with K-feldspar. Photographs courtesy L. Collins.

Figure 21 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 53) — Myrmekite formed in plagioclase at the boundary with K-feldspar (Plag, Kfs labeled). Note the wormy, dendritic quartz rods.

6. Devitrification of Glass

"Devitrification is the secondary crystallization of glass to fine-grained mineral aggregates. Glass is an inherently unstable material and is readily replaced by more stable minerals when kinetics permits. Water-quenched basaltic glass may be replaced by brown, optically isotropic oxidation-hydration products collectively known as palagonite. More silicic glassy rocks commonly devitrify to produce a microgranular mass of small, equidimensional grains of interlocking feldspar and silica minerals called felsitic texture. Crystal form is entirely suppressed in felsitic rocks, and the texture looks very much like that of chert. Devitrification of glass may also produce radial aggregates of crystals (commonly cristobalite or tridymite plus feldspar) called spherulites. Lithophysae are large cavities bordered by spherulitic growth... The minerals produced by devitrification are generally too fine to identify under the polarizing microscope, and x-ray analysis may be required."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 52-53

Devitrification Product	Description
Palagonite	Brown, optically isotropic oxidation-hydration product of water-quenched basaltic glass
Felsitic texture	Microgranular interlocking feldspar + silica minerals from silicic glass; resembles chert
Spherulites	Radial aggregates of cristobalite/tridymite + feldspar
Lithophysae	Large cavities bordered by spherulitic growth, common in rhyolites (late gas bubbles)

Why devitrification is always the wrong answer for "plagioclase → albite + epidote" questions

Devitrification only ever acts on **glass** (an amorphous, non-crystalline material). It cannot be the answer to any question describing the alteration of an already-crystalline mineral like plagioclase, pyroxene, or biotite — that's an automatic elimination clue.

7. Quick-Reference Summary Table

Process	Starting Material	Product	Category
Uralitization	Pyroxene	Amphibole (uralite)	Deuteric
Biotitization	Pyroxene / amphibole	Biotite ($\pm$ epidote as Ca by-product)	Deuteric
Chloritization	Any mafic mineral	Chlorite	Deuteric
Sericitization	Feldspar / feldspathoid	Sericite (fine white mica)	Deuteric
Saussuritization	Ca-rich plagioclase	Albite + epidote ($\pm$ calcite, sericite)	Deuteric
Olivine alteration	Olivine	Serpentine or iddingsite	Deuteric
Myrmekite formation	Plagioclase + K-feldspar boundary	Dendritic quartz + plagioclase	Non-hydration secondary texture
Devitrification	Volcanic glass	Palagonite / felsite / spherulites / lithophysae	Secondary crystallization (not deuteric hydration of a crystal)

8. Probable Exam Questions

Q. Which deuteric process converts pyroxene or amphibole to biotite, sometimes releasing Ca to also form epidote?

A. Biotitization.

Q. What potassium source allows plagioclase (rather than K-feldspar) to sericitize?

A. K⁺ released by nearby chloritization of biotite; K-feldspar itself needs no additional K⁺ and may sericitize more readily.

Q. What are the two characteristic alteration products of olivine?

A. Serpentine and dark brown iddingsite.

Q. What textural term describes a mineral being replaced by an intergrowth of two or more minerals?

A. Symplectite.

Q. Where does myrmekite preferentially form, and what drives its growth?

A. At plagioclase-K-feldspar boundaries in granitic rocks; growth is driven by excess SiO₂ released as plagioclase replaces K-feldspar (anorthite has less SiO₂ than K-feldspar).

Q. What distinguishes palagonite formation from felsitic devitrification?

A. Palagonite forms from water-quenched basaltic (mafic) glass; felsitic texture forms from more silicic glass crystallizing into interlocking feldspar + silica minerals.

Q. Are diagenesis and weathering considered autometamorphic processes?

A. No — Winter explicitly excludes both, even though he calls the distinction "rather arbitrary."

Q.17 — Deuteric Alteration of Ca-Rich Plagioclase

DMG 2019 · Igneous Petrology: Deuteric / Late-Magmatic Alteration · Reference: Perkins, Mineralogy, 3rd ed. (supplemented with standard petrology concepts)

- Q.17.** Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?
- (a) Saussuritization
 - (b) Devitrification
 - (c) Chloritization
 - (d) Uralitization

A note on sourcing: This question belongs to igneous petrology (deuteric/late-magmatic alteration processes), which sits outside the main scope of the uploaded *Mineralogy* textbook — Perkins touches it only briefly, under the mineral epidote's entry. The core textbook citation below is genuine and page-cited; the broader classification of the four alteration processes is standard igneous petrology knowledge (as covered more fully in a dedicated text such as Winter's *Igneous and Metamorphic Petrology*).

1. The Underlying Concept — What Is Deuteric Alteration?

Deuteric alteration (also called autometamorphism or autometasomatism) is the alteration of an igneous rock's primary minerals by the **magma's own residual fluids**, during the last, waning stages of crystallization — while the body is still cooling, before it becomes a fully solid, cold rock. This distinguishes it from ordinary hydrothermal alteration, which acts later, from external fluids moving through an already-solidified rock.

As a magma crystallizes, water and other volatile components that don't fit into the early-forming silicate minerals become concentrated in the last residual melt/fluid. This fluid reacts with minerals that already crystallized, converting them into new, typically hydrous, lower-temperature mineral assemblages, all while the system is still essentially self-contained ("auto-").

2. The Four Named Deuteric Alteration Processes

Process	What Alters	Product
Saussuritization	Ca-rich plagioclase (labradorite, bytownite, anorthite)	Fine-grained aggregate of albite + epidote/zoisite/clinozoisite (± sericite, calcite) — collectively "saussurite"
Devitrification	Volcanic glass	Fine-grained crystalline quartz + feldspar (spherulites, etc.) — unrelated to plagioclase alteration
Chloritization	Mafic minerals (biotite, hornblende, pyroxene)	Chlorite
Uralitization	Pyroxene (commonly augite)	Fibrous/amphibole pseudomorphs after pyroxene, called " uralite "

3. Textbook Confirmation — the Product Side of the Reaction

Perkins' *Mineralogy* directly confirms the product of this alteration, under the mineral epidote's *Varieties* section:

"Saussurite is a name given to fine-grained epidote produced by alteration of plagioclase."
— Perkins, *Mineralogy*, p. 391 (Epidote, Varieties)

The book also illustrates this exact style of alteration with a photograph:

"FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar."
— Perkins, *Mineralogy*, p. 391

octahedra. Small amounts of Ti, Fe, Mg, Na, and K may be present, but no major solid solutions are known.

Occurrences and Associations

Occurrence and Associations

Lawsonite is a metamorphic mineral typical of the blueschist facies. Associated high-pressure minerals include glaucophane, jadeite, pumpellyite, or aragonite. Other associated minerals include chlorite, plagioclase, titanite, quartz, and epidote.

Related Minerals

Hemimorphite, $\text{Zn}_4(\text{Si}_2\text{O}_7)(\text{OH})_2 \cdot \text{H}_2\text{O}$, and ilvaite, $\text{CaFe}_3\text{O}(\text{Si}_2\text{O}_7)(\text{OH})$, are other paired tetrahedral silicates in which all SiO_4 tetrahedra are paired. Other minerals usually considered in the same group are epidote, $\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$; clinozoisite, $\text{Ca}_2\text{Al}_3\text{Si}_3\text{O}_{12}(\text{OH})$; piemontite, $\text{Ca}_2(\text{Al,Mn})_3\text{Si}_3\text{O}_{12}(\text{OH})$; allanite, $(\text{Ca,Ce})_2(\text{Al,Fe,Mg})_3\text{Si}_3\text{O}_{12}(\text{OH})$; and vesuvianite, $\text{Ca}_{10}(\text{Mg,Fe})_2\text{Al}_4\text{Si}_9\text{O}_{34}(\text{OH,F})_4$.

Epidote

$\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$

Origin of Name

From the Greek word *epididonai*, meaning “increase,” referring to the base of an epidote prism, one side of which is longer than the other.

Hand Specimen Identification

Epidote is characterized by its pistachio-green color, habit, one perfect cleavage, and association. It is often fine grained and massive (Figure 19).

Physical Properties

hardness	6 to 7
specific gravity	3.4 to 3.5
cleavage/fracture	perfect (001), poor (100)/uneven
luster/transparency	vitreous/transparent to translucent
color	pistachio-green, yellow-green to black
streak	white



FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar.

The Epidote entry (Perkins, *Mineralogy*, p. 391), including Figure 19 — a vein of epidote in granite formed by feldspar alteration — and the Varieties note defining saussurite as fine-grained epidote produced by alteration of plagioclase.

Why the reaction yields albite + epidote specifically

Plagioclase is a solid solution between albite ($\text{NaAlSi}_3\text{O}_8$) and anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$). During saussuritization:

Anorthite component + H_2O ($\pm \text{CO}_2$) $\rightarrow$ Epidote/Zoisite + (Al, Ca redistributed) | Albite component largely unaltered

The **Ca-Al-rich (anorthite) component** breaks down, releasing Ca and Al that recombine with the residual fluid to form epidote-group minerals, while the **Na-rich (albite) component** survives essentially untouched. This is exactly why the reaction is described as producing "almost pure albite **and** epidote" — both phases persist side by side as breakdown products.

4. Evaluating Each Option

Option	Reasoning	Verdict
(a) Saussuritization	Defining reaction is exactly $\text{Ca-plagioclase} \rightarrow \text{albite} + \text{epidote/zoisite}$; confirmed by Perkins' saussurite entry	Correct
(b) Devitrification	Applies to volcanic glass turning crystalline — no plagioclase or albite/epidote relationship at all	Eliminated
(c) Chloritization	Produces chlorite from mafic (Fe-Mg) minerals, not albite + epidote from feldspar	Eliminated
(d) Uralitization	Affects pyroxene, producing amphibole (uralite) — unrelated to feldspar breakdown	Eliminated

✓ Answer: (a) Saussuritization

5. Probable Follow-Up Exam Questions

Q. What is "saussurite"?

A. A fine-grained, greenish, massive aggregate of albite, epidote/zoisite, sericite, and calcite that replaces Ca-rich plagioclase; classically seen in "saussuritized gabbro."

Q. What does uralitization produce, and from what mineral?

A. It converts pyroxene (typically augite) into a fibrous amphibole pseudomorph called uralite.

Q. Is devitrification an alteration of a crystalline mineral?

A. No — it specifically refers to crystallization of previously glassy (amorphous) volcanic material, not alteration of an existing crystal.

Q. Which rock type most classically shows saussuritized plagioclase?

A. Gabbro and diorite — "saussuritized gabbro" is the standard textbook example.

Q. How does deuteric alteration differ from hydrothermal alteration?

A. Deuteric alteration is caused by the magma's own residual fluid during the final stages of crystallization (autometamorphic); hydrothermal alteration is caused by external fluids acting on an already-solidified rock, often much later.

Q.17 — Deuteric Alteration of Ca-Rich Plagioclase

DMG 2019 · Igneous Petrology: Secondary/Deuteric Alteration · References: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. & Perkins, *Mineralogy*, 3rd ed.

Q.17. Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?

- (a) Saussuritization
- (b) Devitrification
- (c) Chloritization
- (d) Uralitization

1. The Underlying Concept — What Is Deuteric Alteration?

Winter classifies this whole family of processes under **secondary textures** — changes that occur in an igneous rock after most of it has crystallized, driven by heat and fluid still left over from the magma itself:

"Secondary mineral reactions that occur in igneous rocks as they cool, and that are not products of a distinct later metamorphic event, are commonly called autometamorphic processes rather than metamorphic ones because they are a natural part of igneous cooling."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

"Most, but not all, autometamorphic reactions involve minerals at moderate temperatures in an environment in which H₂O is either liberated from residual melt or externally introduced. Such alterations are a subset of autometamorphism, which involves hydration, and they are called deuteric alterations."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

In short: **deuteric alteration = hydration reactions driven by the magma's own residual water**, acting on minerals that have already crystallized, while the pluton is still cooling. It is a subset of the broader autometamorphic category, and (per Winter) is distinct from diagenesis and weathering, which are explicitly excluded from this category.

2. The Named Deuteric/Secondary Alteration Processes

Winter's Table 1 (Specific Mineral Replacements) gives the definitive one-line summary of each:

Process	Winter's Definition
Saussuritization	"Replacement of plagioclase by epidote."
Uralitization	"Replacement of pyroxene by amphibole."
Biotitization	"Replacement of pyroxene or amphibole by biotite."
Chloritization	"Replacement of any mafic mineral by chlorite."
Sericitization	"Replacement of feldspar or feldspathoids by fine white micas."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 55, Table 1

Devitrification is treated separately, since it doesn't replace one crystalline mineral with another — it involves glass:

"Devitrification is the secondary crystallization of glass to fine-grained mineral aggregates. Glass is an inherently unstable material and is readily replaced by more stable minerals when kinetics permits."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

3. Saussuritization in Full Detail

"Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more

nearly pure albite, releasing Ca and Al to form an epidote mineral ($\pm$ calcite and/or sericite)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 51-52

This is precisely the reaction the question describes: Ca-rich plagioclase $\rightarrow$ nearly pure albite + epidote. Winter even notes where to look for it texturally:

"In zoned plagioclase, one can occasionally observe the products of saussuritization or seritization concentrated in the Ca-rich core or some of the more Ca-rich oscillatory bands in the original grain."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 52

Perkins' *Mineralogy* corroborates this from the mineralogical (product) side, under the epidote entry:

"Saussurite is a name given to fine-grained epidote produced by alteration of plagioclase."

— Perkins, *Mineralogy*, p. 391 (Epidote, Varieties)

"FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar."

— Perkins, *Mineralogy*, p. 391

Ca-rich plagioclase (anorthite component) + $H_2O \rightarrow$ nearly pure albite + epidote/zoisite ($\pm$ calcite, sericite)
The Na-rich (albite) component survives largely unaltered — hence "almost pure albite **and** epidote" as products side by side.

4. Contrast with Uralitization & Chloritization (the Other Deuteric Options)

"Pyroxene is a common primary mafic mineral in a variety of igneous rocks. If H_2O penetrates at modest temperatures, a deuteric alteration of pyroxene to amphibole results, called uralitization. Any gradation from amphibole rims on pyroxene cores to multiple patches of pyroxene in an amphibole to complete replacement is possible."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

"Chloritization is the alteration of any mafic mineral to chlorite. Chlorite is a very hydrous phyllosilicate and typically replaces the less hydrous mafics at low temperature when water is available."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 51

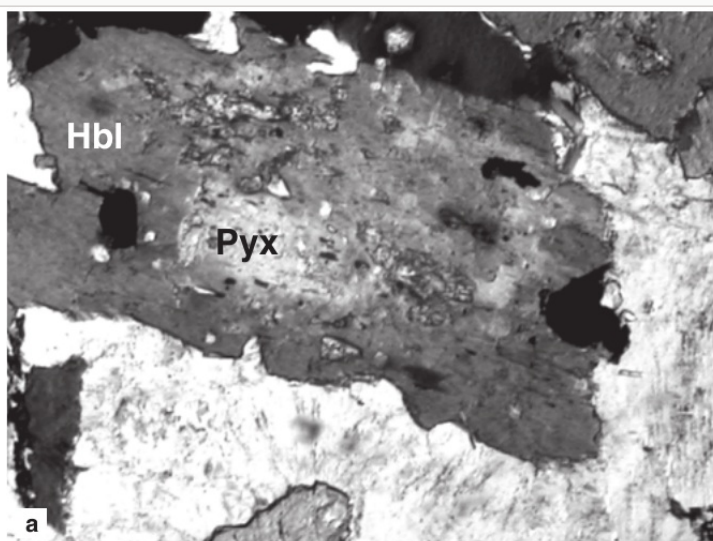




FIGURE 20 (a) Pyroxene largely replaced by hornblende. Some pyroxene remains as light areas (Pyx) in the hornblende core. Width 1 mm. (b) Chlorite (light) replaces biotite (dark) at the rim and along cleavages. Tonalite, San Diego, California. Width 0.3 mm.

generally replaces the initial mafic from the rim inward. In the case of biotite, H_2O may work its way along the prominent cleavages, and chlorite can be seen to replace biotite margins as well as along cleavage planes (Figure 20b).

Sericite is a term applied to any very fine-grained white mica. **Seritization** is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light. In more advanced stages of alteration, the feldspars appear speckled, with fine micas having yellowish birefringence and then larger clots with coarser crystals and higher birefringence. K^+ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K^+ and may be more sericitized than associated plagioclase.

Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite,

Figure 20 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 53) — (a) Pyroxene largely replaced by hornblende (uralitization); remnant pyroxene shown as light areas in the hornblende core. (b) Chlorite (light) replacing biotite (dark) at grain rims and along cleavages (chloritization/biotite-to-chlorite alteration). Neither process involves plagioclase or produces albite + epidote.

octahedra. Small amounts of Ti, Fe, Mg, Na, and K may be present, but no major solid solutions are known.

Occurrence and Associations

Lawsonite is a metamorphic mineral typical of the blueschist facies. Associated high-pressure minerals include glaucophane, jadeite, pumpellyite, or aragonite. Other associated minerals include chlorite, plagioclase, titanite, quartz, and epidote.

Related Minerals

Hemimorphite, $Zn_4(Si_2O_7)(OH)_2 \cdot H_2O$, and ilvaite, $CaFe_3O(Si_2O_7)(OH)$, are other paired tetrahedral silicates in which all SiO_4 tetrahedra are paired. Other minerals usually considered in the same group are epidote, $Ca_2(Al,Fe)_3Si_3O_{12}(OH)$; clinozoisite, $Ca_2Al_3Si_3O_{12}(OH)$; piemontite, $Ca_2(Al,Mn)_3Si_3O_{12}(OH)$; allanite, $(Ca,Ce)_2(Al,Fe,Mg)_3Si_3O_{12}(OH)$; and vesuvianite,



Epidote

$\text{Ca}_2(\text{Al,Fe})_3\text{Si}_3\text{O}_{12}(\text{OH})$

Origin of Name
From the Greek word *epididonai*, meaning “increase,” referring to the base of an epidote prism, one side of which is longer than the other.

Hand Specimen Identification
Epidote is characterized by its pistachio-green color, habit, one perfect cleavage, and association. It is often fine grained and massive (Figure 19).

Physical Properties

hardness	6 to 7
specific gravity	3.4 to 3.5
cleavage/fracture	perfect (001), poor (100)/uneven
luster/transparency	vitreous/transparent to translucent
color	pistachio-green, yellow-green to black
streak	white



FIGURE 19 A vein of epidote in a granite, probably formed by alteration of feldspar.

The Epidote entry (Perkins, Mineralogy, p. 391), including Figure 19 (a vein of epidote in granite from feldspar alteration) and the Varieties note defining saussurite.

5. Evaluating Each Option

Option	What Actually Alters → Product	Verdict
(a) Saussuritization	Plagioclase → albite + epidote (Winter, p. 55, Table 1) — exact match	Correct
(b) Devitrification	Glass → fine crystalline aggregate (felsite/spherulites); no plagioclase involved	Eliminated
(c) Chloritization	Any mafic mineral → chlorite; not feldspar-related	Eliminated
(d) Uralitization	Pyroxene → amphibole; not feldspar-related	Eliminated

✓ Answer: (a) Saussuritization

6. Probable Follow-Up Exam Questions

Q. What single-line definition would Winter's Table 1 give for saussuritization?

A. "Replacement of plagioclase by epidote."

Q. Why does higher-temperature (Ca-rich) plagioclase break down preferentially over its sodic counterpart?

A. Because Ca-rich plagioclase is less stable than the sodic (albite) component at low temperatures, so on cooling it decomposes, releasing Ca and Al to form epidote while albite persists.

Q. What textural evidence in a zoned plagioclase crystal points to saussuritization?

A. Alteration products concentrated in the Ca-rich core or in Ca-rich oscillatory zoning bands of the original grain (Winter, p. 52).

Q. What is uralite, and how does it form?

A. An amphibole (single hornblende crystal or fibrous actinolite/hornblende aggregate) formed by deuteric alteration of pyroxene; the term is more commonly applied to the fibrous aggregates.

Q. Is saussuritization considered autometamorphic, diagenetic, or metamorphic?

A. Autometamorphic (specifically deuteric) — Winter explicitly excludes diagenesis and weathering from the autometamorphic category, even though the distinction is "rather arbitrary."

Q. What mineral commonly forms alongside epidote when biotite replaces hornblende (a related deuteric process)?

A. Epidote — because biotite contains little Ca, the Ca released during hornblende-to-biotite alteration forms epidote (biotitization, Winter p. 51).

Compatible & Incompatible Trace Elements — Complete Concept Guide

Partition Coefficients · Fractional Crystallization · Partial Melting · HFS & LIL Elements

Reference: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. — Chemical Petrology II

Contents

1. What Makes an Element "Compatible" or "Incompatible"?
2. The Distribution Coefficient (K_D) & Bulk Distribution Coefficient
3. HFS vs. LIL Incompatible Elements
4. Winter's Partition Coefficient Table — Reading It Correctly
5. How Fractional Crystallization Concentrates Incompatible Elements
6. How Partial Melting Concentrates Incompatible Elements (Batch Melting)
7. Applying This to Rock Types — Why Granite Wins
8. Q.18 Worked Solution
9. Quick-Reference Summary & Probable Exam Questions

1. What Makes an Element "Compatible" or "Incompatible"?

Every element distributes itself unevenly between a solid mineral and a coexisting liquid when the two are in equilibrium — this is called **chemical fractionation**. Whether an element ends up preferring the crystal or the melt depends on how well its ionic size and charge fit into the crystal lattice sites available.

"Practically all elements distribute themselves unevenly between any two phases at equilibrium. This effect is known as chemical fractionation."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 167

Compatible element: Fits comfortably into common rock-forming mineral lattices (especially mafic minerals); strongly partitions into the **solid**.

Incompatible element: Too large, too small, or too highly charged to fit comfortably into common mineral lattices; strongly partitions into the **liquid**.

"By convention, incompatible trace elements are concentrated in the melt more than the solid, and K_D (or D)... is considerably less than 1. Compatible trace elements concentrate in the solid and K_D (or D) $\gg 1$."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 167

"Of the major elements, then, Mg and Fe are generally considered compatible, whereas K and Na are incompatible."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 167

2. The Distribution Coefficient (K_D) & Bulk Distribution Coefficient

For a simple exchange reaction between a solid and liquid, Winter defines a distribution constant:

$$K_D = X_i^{\text{solid}} / X_i^{\text{liquid}}$$

where X_i is the mole fraction of element i in each phase. When a rock contains several minerals, the **bulk distribution coefficient** (D_i) accounts for all of them, weighted by how much of each mineral is present:

$$D_i = \sum (W^A \times D_i^A)$$

where W^A is the weight fraction of mineral A in the rock, and D_i^A is that mineral's own partition coefficient for element i . Winter works a garnet lherzolite example (60% olivine, 25% orthopyroxene, 10% clinopyroxene, 5% garnet) for erbium and gets $D_{\text{Er}} = 0.459$ — and notes that even a small amount of a mineral with a very different D (here, garnet) can shift the whole rock's bulk value substantially.

Why trace elements are more diagnostic than major elements

"Major element K_D values between common phases are usually within an order of magnitude of 1.0. K_D values for trace elements, on the other hand, can range over several orders of magnitude... As a result, trace elements can be very sensitive to distribution and fractionation processes." (Winter, p. 167)

3. HFS vs. LIL Incompatible Elements

Incompatible elements are further split into two families based on ionic charge and radius:

Family	Characteristics	Example Elements
High Field Strength (HFS)	Smaller, more highly charged ions	REEs, Th, U, Ce, Pb ⁴⁺ , Zr, Hf, Ti, Nb, Ta
Large Ion Lithophile (LIL)	Low field strength, larger/lower-charge ions; more mobile, especially if a fluid phase is present	K, Rb, Cs, Ba, Pb ²⁺ , Sr, Eu ²⁺

"Small, low-valence elements are usually compatible, including the trace elements Ni, Cr, Cu, W, Ru, Rh, Pd, Os, Ir, Pt, and Au."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 167

4. Winter's Partition Coefficient Table — Reading It Correctly

This table (mineral/melt partition coefficients for common trace elements in basaltic-andesitic systems) is the single most useful reference for these questions — scan down any mineral's column to see which elements it accepts ($D > 1$) and which it rejects ($D < 1$).

TABLE 1 Partition Coefficients (C_S/C_L) for Some Commonly Used Trace Elements in Basaltic and Andesitic Rocks							
	Olivine	Opx	Cpx	Garnet	Plag	Amph	Magnetite
Rb	0.010	0.022	0.031	0.042	0.071	0.29	
Sr	0.014	0.040	0.060	0.012	1.830	0.46	
Ba	0.010	0.013	0.026	0.023	0.23	0.42	
Ni	14	5	7	0.955	0.01	6.8	29
Cr	0.70	10	34	1.345	0.01	2.00	7.4
La	0.007	0.03	0.056	0.001	0.148	0.544	2
Ce	0.006	0.02	0.092	0.007	0.082	0.843	2
Nd	0.006	0.03	0.230	0.026	0.055	1.340	2
Sm	0.007	0.05	0.445	0.102	0.039	1.804	1
Eu	0.007	0.05	0.474	0.243	0.1/1.5*	1.557	1
Dy	0.013	0.15	0.582	3.17	0.023	2.024	1
Er	0.026	0.23	0.583	6.56	0.020	1.740	1.5
Yb	0.049	0.34	0.542	11.5	0.023	1.642	1.4
Lu	0.045	0.42	0.506	11.9	0.019	1.563	

Data from Rollinson (1993) and <http://earthref.org>

Table 1 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 168) — Partition coefficients (C_{solid}/C_{liquid}) for Rb, Sr, Ba, Ni, Cr, and the REEs in olivine, orthopyroxene, clinopyroxene, garnet, plagioclase, amphibole, and magnetite.

Key pattern to notice

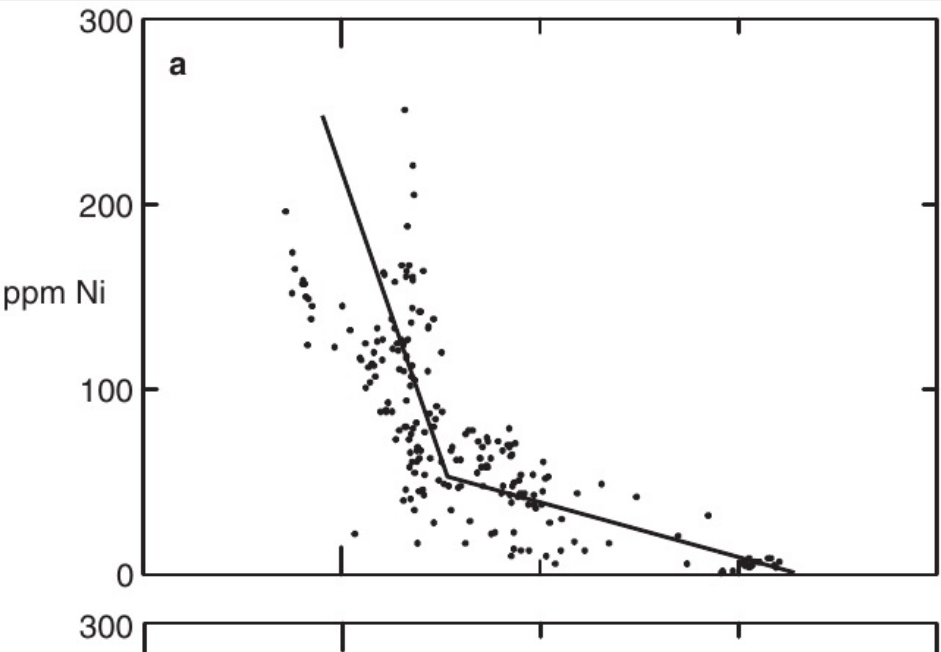
Olivine's D-values for Rb (0.010), Sr (0.014), Ba (0.010), and the light REEs (La = 0.007, Ce = 0.006) are all tiny — olivine **rejects** these incompatible elements almost entirely. But olivine's D for Ni is 14 and for Cr is 0.70 — it happily **accepts** these compatible elements. This single contrast explains why olivine-rich rocks (dunite, peridotite) are incompatible-element-poor and compatible-element-rich.

5. How Fractional Crystallization Concentrates Incompatible Elements

"Crystal fractionation of a basaltic magma increases the enrichment of Rb and the other incompatibles in the late liquids, whereas Ni and the compatibles will be selectively removed into the early crystals of olivine or pyroxene."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 168

As magma cools, minerals crystallize roughly in Bowen's Reaction Series order (olivine and Ca-rich plagioclase first, then pyroxenes, amphibole, biotite, K-feldspar, quartz last). Because early mafic minerals accept compatible elements and reject incompatible ones, every increment of crystallization concentrates incompatible elements further into the remaining liquid:



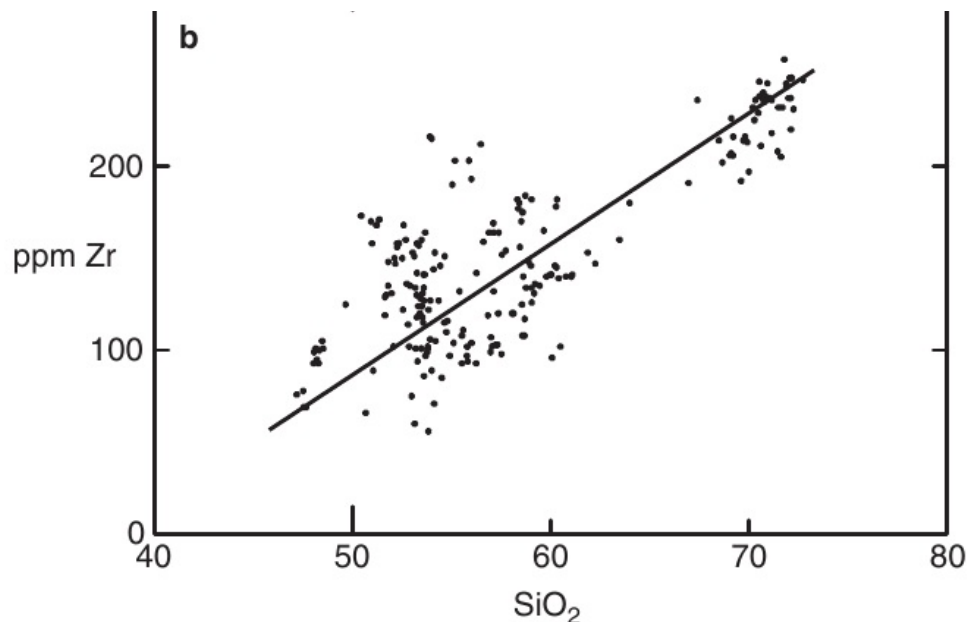


FIGURE 1 Harker diagrams of (a) Ni versus SiO_2 and (b) Zr versus SiO_2 for the Crater Lake suite of volcanic rocks.

Conversely, a trace element with a very small $\bar{D}_i$ is preferentially concentrated in the liquid and will reflect the proportion of liquid at a given state of crystallization or melting because its concentration varies inversely with the amount of liquid diluting it. The plot of Zr versus SiO_2 in Figure 1b is an illustration. Some relatively simple but useful quantitative models have been developed that can treat multiple fractionating phases and various crystallization or melting schemes using a variety of trace elements. We shall discuss a few of these next.

2 MODELS FOR SOLID–MELT PROCESSES

2.1 Batch Melting

Figure 1 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 169) — Harker diagrams of Ni and Zr versus SiO_2 for the Crater Lake volcanic suite. Ni (compatible) collapses sharply once olivine begins fractionating; Zr (incompatible) instead climbs steadily as SiO_2 (a proxy for the degree of fractionation) increases.

"The process of fractional crystallization tends to remove only a few elements from the liquid in significant quantities, and a number of incompatible, LIL, and non-lithophile elements become concentrated in the latest liquid fraction."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 219

Taken to the extreme, this is why **pegmatites** (essentially very coarse-grained granites formed from the last residual, volatile-rich liquid) show "a tremendous concentration of incompatible elements" (Li, Be, REEs, W, Zr) and are mined specifically for them (Winter, p. 219).

6. How Partial Melting Concentrates Incompatible Elements (Batch Melting)

The same logic runs in reverse during partial melting of a mantle source: the first small increment of melt strips out the incompatible elements disproportionately, because they don't want to stay in the solid residue. Winter's batch melting equation:

$$C_L / C_0 = 1 / [D_i(1 - F) + F]$$

where C_0 is the original concentration before melting, C_L is the concentration in the melt, D_i is the bulk distribution coefficient, and F is the weight fraction of melt produced.

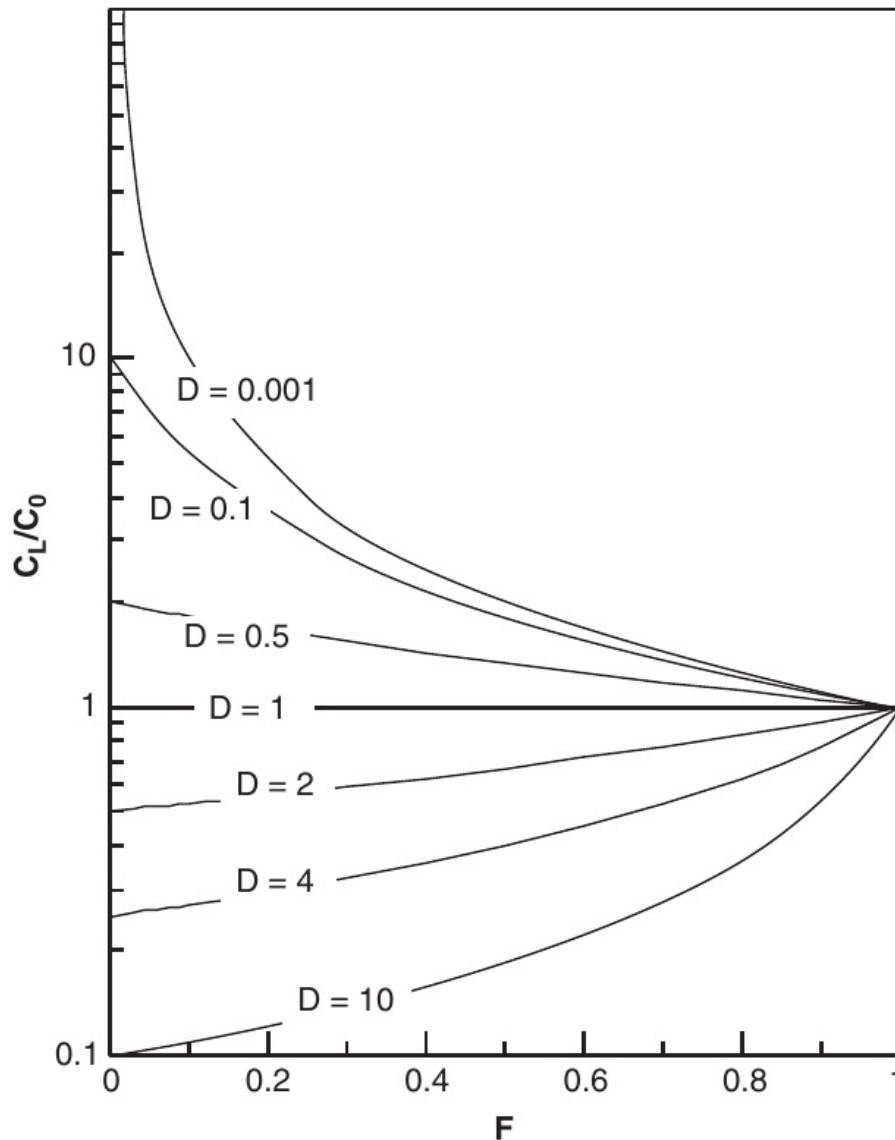


FIGURE 2 Variation in the relative concentration of a trace element in a liquid versus source rock as a function of $\bar{D}$ and the fraction melted, using Equation (5) for equilibrium batch melting.

Figure 2 shows the variation in C_L/C_0 with F for various values of $\bar{D}_i$, using Equation (5). Many petrologists consider values of F greater than 0.4 to be unlikely for batch melting in the mantle because such large amounts of melt should separate and rise before higher values are reached. The left side of Figure 2 is thus of most interest.

"For very incompatible elements, as D_i approaches 0... if we know the concentration of a very incompatible element in both a magma and the source rock, we can determine the fraction of partial melt produced. This is another useful way trace elements can be used to evaluate melting processes at depth."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 168

7. Applying This to Rock Types — Why Granite Wins

Rock	Petrogenetic Role	Incompatible Element Level
Dunite	Nearly monomineralic olivine cumulate or highly depleted mantle residue — olivine's D for incompatibles is tiny, so they were never retained	Very low ("the most depleted")
Peridotite	Mantle source rock (olivine + pyroxenes); low bulk D for incompatibles keeps them in whatever melt has been extracted, not the residue	Very low to low (fertile varieties slightly higher, but still far below granite)
Basalt	Relatively primitive partial melt of mantle peridotite; incompatibles are concentrated relative to the source, but this is only the first step of differentiation	Low-moderate
Granite	The most fractionated common igneous rock — the residual liquid left after extensive removal of mafic, compatible-element-rich minerals (or a crustal partial melt enriched the same way)	Highest

"Fertile xenoliths thus have more incompatible elements, which can be correlated with their potential to yield magma before becoming more refractory like the depleted samples... dunites are the most depleted."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 203

8. Q.18 Worked Solution

Q.18. Which one of the following rocks is normally expected to have the highest abundance of incompatible trace elements?

- (a) Basalt
- (b) Dunite
- (c) Peridotite
- (d) Granite

Option	Reasoning	Verdict
(a) Basalt	Primitive-to-moderately evolved mantle melt; some incompatible enrichment, but far less than a fully fractionated felsic rock	Eliminated
(b) Dunite	Nearly pure olivine; olivine's D-values for Rb/Sr/Ba/REE are all $\ll 1$ — incompatibles were never retained here	Eliminated
(c) Peridotite	Mantle source dominated by olivine + pyroxene; same low-D argument applies, even for "fertile" varieties	Eliminated
(d) Granite	Represents the latest, most fractionated residual liquid — incompatible elements have been progressively concentrated into it at every prior crystallization step	Correct

✓ Answer: (d) Granite

9. Quick-Reference Summary & Probable Exam Questions

Concept	Key Fact
Compatible elements	$D \gg 1$; concentrate in solid; e.g., Ni, Cr, Mg, Fe
Incompatible elements	$D \ll 1$; concentrate in liquid; e.g., K, Rb, Ba, Sr, REE, Zr, Nb, Th, U
HFS elements	Small, highly charged: REE, Th, U, Zr, Hf, Ti, Nb, Ta
LIL elements	Larger, lower charge, fluid-mobile: K, Rb, Cs, Ba, Sr
Fractional crystallization effect	Concentrates incompatibles in residual liquid → granite/pegmatite

Q. What does a bulk distribution coefficient close to 1 imply about an element during crystallization or melting?

A. No significant fractionation occurs — its concentration stays essentially the same in solid and liquid.

Q. Why can even a small modal percentage of garnet drastically change a rock's bulk D for the heavy REEs?

A. Garnet's own D-values for heavy REEs (e.g., Yb = 11.5, Lu = 11.9) are extremely high compared to other minerals, so even 5% garnet can dominate the weighted bulk average.

Q. Which is more incompatible in olivine: Sr or Rb?

A. Both are strongly incompatible ($D \ll 1$), but per Winter's table, Rb ($D=0.010$) and Sr ($D=0.014$) are very close; Ba is essentially the same as Rb ($D=0.010$).

Q. What happens to CL/C_0 for a highly incompatible element as the degree of partial melting (F) approaches 1?

A. It approaches 1 — because essentially all the source rock has melted, the melt's composition converges on the original source composition.

Q. Why are pegmatites economically important?

A. Because late-stage fractional crystallization concentrates incompatible and rare elements (Li, Be, REEs, W, Zr) into the residual fluid that forms them, making them valuable ore sources and gem localities.

Magma Series & Plate Tectonic Settings — Complete Concept Guide

Alkaline · Tholeiitic · Calc-Alkaline · Classification Diagrams & Tectonic Distribution

Reference: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. — Chemical Petrology I

Contents

1. What Is a Magma Series?
2. The First Split — Alkaline vs. Subalkaline
3. The Second Split — Tholeiitic vs. Calc-Alkaline
4. Classification Diagrams: TAS, AFM, and Al_2O_3 -Plagioclase
5. Magma Series and Plate Tectonic Setting — Table 7
6. Komatiite — A Time-Restricted, Not Tectonically-Restricted, Rock Type
7. Q.19 Worked Solution
8. Quick-Reference Summary & Probable Exam Questions

1. What Is a Magma Series?

Magma series: A group of igneous rocks that share chemical (and often mineralogical) characteristics and show a consistent evolutionary trend on a variation diagram, implying they descended from a common parental magma via a shared evolutionary process.

"A group of rocks that share some chemical (and perhaps mineralogical) characteristics and shows a consistent pattern on a variation diagram, suggesting a genetic relationship, can be referred to as a magma series. Synonymous terms include association, lineage, magma type, and clan."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 158

The concept traces back to Iddings (1892), who first proposed that igneous rocks fall into either an **alkaline** or a **subalkaline** series, each evolving from a distinct type of parental magma toward more silicic derivatives. Harker (1909) later observed the same pattern geographically, dividing Cenozoic circum-oceanic volcanics into an "Atlantic" (alkaline) branch and a "Pacific" (subalkaline) branch.

2. The First Split — Alkaline vs. Subalkaline

"Alkaline rocks are silica undersaturated, whereas subalkaline rocks are silica saturated to oversaturated."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 158

This split is visualized on a **Total Alkali-Silica (TAS) diagram** — weight % (Na₂O + K₂O) plotted against weight % SiO₂:

"The alkaline and subalkaline series are distinguished in a total alkali versus silica diagram... the alkaline rocks plot distinctly above and the subalkaline rocks below the dividing lines chosen by MacDonald (1968) and Irvine and Baragar (1971)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159

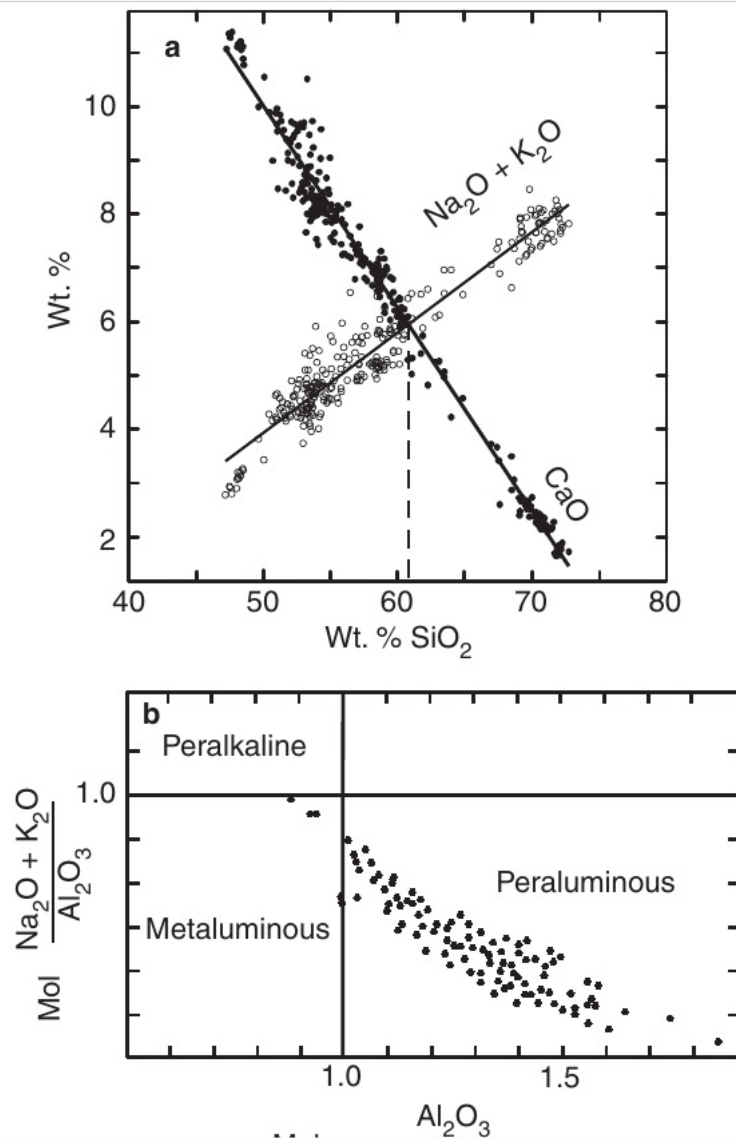


Figure 11 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159) — Total alkalis versus silica diagram for the alkaline (open circles) and subalkaline rocks of Hawaii. The alkaline field sits clearly above the subalkaline field at any given SiO_2 content.

Evolutionary pathway difference

"The common evolutionary sequence in the alkaline series begins with an alkali olivine basalt and proceeds through trachybasalts and trachyandesites, to trachytes or phonolites." Subalkaline magmas instead evolve toward either quartz-normative (silica-saturated) or olivine-normative compositions across the "thermal divide" at the albite composition, which prevents the two series from crossing into one another as they cool (p. 159-160).

3. The Second Split — Tholeiitic vs. Calc-Alkaline

"Most investigators continue to recognize Iddings's (1892a) original two series, alkaline and subalkaline, and accept that the subalkaline can be further divided into the tholeiitic series and calc-alkaline series."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 159

"The subalkaline series was further subdivided into a tholeiitic and a calc-alkaline series by Tilley (1950)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160

Interestingly, these two series **cannot** be told apart on the alkali-silica or Ne-OI-Q diagrams used to separate alkaline from subalkaline — they need different, dedicated diagrams (Section 4).

4. Classification Diagrams: TAS, AFM, and Al_2O_3 -Plagioclase

Because tholeiitic and calc-alkaline trends overlap on a TAS diagram, petrologists instead use an **AFM diagram** (Alkalis-FeO(total)-MgO ternary) to separate them:

"If we compare on AFM diagrams... we see that the Skaergård trend is clearly tholeiitic, and the Crater Lake trend is calc-alkaline. Both series progress along the basalt-andesite-dacite-rhyolite trend, but there are distinctive mineralogical and chemical differences between the two series that are most evident for intermediate compositions."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160

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both CaO and alkalis (as the original Peacock criteria do).

Irvine and Baragar (1971) further divided the alkali series into a **sodic series** and **potassic series**, and Middlemost (1975) recommended adding a **high-K series** as well (Figure 16). The remaining terms, such as *peraluminous*, *metaluminous*, etc., although they may yet be shown to characterize some distinctive magma series, are more useful as descriptive terms when some chemical characteristic is emphasized for a particular igneous rock or province.

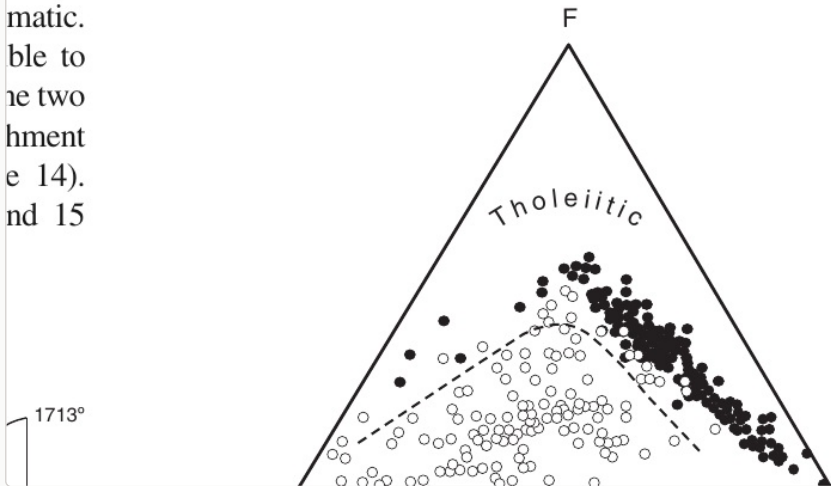


Figure 14 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 160) — AFM diagram distinguishing tholeiitic rocks (Iceland, Mid-Atlantic Ridge, Columbia River Basalts, Hawaii — solid circles) from calc-alkaline rocks of the Cascade volcanics (open circles). Tholeiitic liquids show strong iron enrichment before differentiating toward silica; calc-alkaline liquids differentiate toward silica-rich compositions without significant Fe-enrichment.

A second diagnostic diagram plots wt.% Al₂O₃ against the anorthite content of normative plagioclase (Winter's Figure 15, Irvine & Baragar 1971) — tholeiitic and calc-alkaline rocks again plot as distinct fields, useful when AFM data isn't available.

5. Magma Series and Plate Tectonic Setting

This is the section that directly answers Q.19. Winter cross-tabulates each series against every major tectonic setting:

cons of magmas at mid-ocean ridges. Magma series are important to our understanding of igneous petrogenesis. Earlier, as we first regarded the spectrum of igneous rocks that occur on Earth, I asked a basic	were several parents, each with its own range of differentiation products? It would appear that the latter is the case and that there are at least three principal magma series, and perhaps several other minor ones.
Summary Petrologists use the chemical composition of rocks and minerals to characterize and categorize individual types, to identify series of related rocks, and to model processes that relate the rocks in a consanguineous series. Analyses were initially performed by using tedious gravimetric methods but are now quickly and routinely accomplished by using instrumental spectroscopic techniques based on emission or absorption of energy when directed at prepared samples. Analyses are typically reported in wt. % oxide for the major	used, particularly ones in which various components are plotted versus some chemical parameter that is considered a measure of chemical fractionation (e.g., SiO₂, Zr, Mg#). Ternary diagrams, such as the AFM diagram, are also common. By using bivariate diagrams, such as Harker or Pearce element ratio diagrams, we can evaluate specific differentiation mechanisms (e.g., fractional crystallization of certain minerals) deemed responsible for the chemical evolution within a suite of supposedly consanguineous rocks. Quanti-

Table 7 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 162, after Wilson 1989) — Occurrence of the alkaline, tholeiitic, and calc-alkaline series

Calc-Alkaline — Convergent Margins ONLY

"First, calc-alkaline magmas are essentially restricted to subduction-related plate tectonic processes. This must carry some genetic significance and will certainly be a factor in any model for subduction zone magma genesis. Recognition of calc-alkaline characteristics in the composition of ancient volcanic rocks may help determine their tectonic framework."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 161

Repeated later: "Because the calc-alkaline series is restricted mostly to convergent plate boundaries in complex settings, we shall set it aside [when modeling oceanic basalt generation]." (p. 191)

Tholeiitic — Everywhere, Dominant at Divergent Margins

"Second, tholeiitic magmas are practically the exclusive magma type associated with divergent boundaries. Although some alkaline rocks are found, they play a subordinate role, usually in the initial stages of continental rifting."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 161

Tholeiitic basalts (as MORB) are generated at mid-ocean ridges and are also found at oceanic islands, convergent margins, and continental settings — the most volumetrically dominant and tectonically widespread series of the three.

Alkaline — Widespread Except at Divergent Margins

Per Table 7: found at convergent margins, oceanic within-plate settings (classic ocean-island alkaline basalts), and continental within-plate settings (continental rifting, intraplate volcanism) — but essentially absent as a dominant type at divergent (mid-ocean ridge) margins, where tholeiitic dominates instead.

6. Komatiite — A Time-Restricted, Not Tectonically-Restricted, Rock Type

Komatiite (glossary definition): "Ultramafic volcanic (mostly Archean): wt.% MgO > 18%, TiO₂ < 1%." (Winter, p. 34)

"Komatiites, ultramafic volcanic rocks, are almost completely restricted to the Archean, when the geothermal gradient was much higher, and a larger fraction of partial melting (F) was apparently easier to generate."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 198

Why this matters for Q.19

Komatiite is defined by **composition and geologic age** (an ultramafic lava essentially unique to the hotter Archean mantle), not by plate tectonic setting. It is not one of the three principal magma series (alkaline/tholeiitic/calc-alkaline) at all, so it cannot be "restricted to subduction zones" in the sense the question asks.

7. Q.19 Worked Solution

Q.19. Which one of the following magma series is found only in subduction zones?

- (a) Calc-alkaline
- (b) Tholeiitic
- (c) Alkaline
- (d) Komatiite

Option	Tectonic Distribution (per Table 7)	Verdict
(a) Calc-alkaline	Convergent only — the single "yes" in the whole table	Correct
(b) Tholeiitic	Convergent, divergent, oceanic, and continental — the <i>most</i> widespread series	Eliminated
(c) Alkaline	Convergent, oceanic, and continental (absent only at divergent margins)	Eliminated
(d) Komatiite	Not a tectonic-setting category; an Archean-age, MgO-rich ultramafic lava type	Eliminated

✓ Answer: (a) Calc-alkaline

8. Quick-Reference Summary & Probable Exam Questions

Series	Silica Saturation	Key Diagram(s)	Tectonic Setting
Alkaline	Undersaturated	TAS (Fig. 11)	Convergent, oceanic & continental within-plate (not divergent)
Tholeiitic	Saturated-oversaturated	AFM (Fig. 14), Al ₂ O ₃ -plag (Fig. 15)	All settings; dominant at divergent margins
Calc-alkaline	Saturated-oversaturated	AFM (Fig. 14), Al ₂ O ₃ -plag (Fig. 15)	Convergent (subduction) only

Q. Which two series cannot be distinguished on a TAS or Ne-Ol-Q diagram, and what diagram must be used instead?

A. Tholeiitic and calc-alkaline; distinguished instead via the AFM diagram or an Al₂O₃ vs. normative-plagioclase plot.

Q. What geochemical behavior distinguishes tholeiitic differentiation from calc-alkaline differentiation on an AFM diagram?

A. Tholeiitic liquids show strong early iron enrichment before evolving toward silica-rich compositions; calc-alkaline liquids evolve toward silica-rich compositions without significant iron enrichment.

Q. What is the "thermal divide," and what does it prevent?

A. A compositional barrier at the albite composition on the Ne-Q system that prevents alkaline and subalkaline liquids from crossing into each other's fields as they cool — keeping the two series distinct.

Q. Who first proposed the alkaline/subalkaline distinction, and who first subdivided subalkaline into tholeiitic and calc-alkaline?

A. Iddings (1892) proposed alkaline/subalkaline; Tilley (1950) subdivided subalkaline into tholeiitic and calc-alkaline.

Q. Why can't komatiite be considered a "magma series" in the same sense as alkaline/tholeiitic/calc-alkaline?

A. It's defined by a compositional threshold (MgO > 18%, TiO₂ < 1%) and near-total restriction to the Archean, not by a distinct evolutionary lineage tied to a tectonic setting.

Q. At which tectonic setting is tholeiitic magma overwhelmingly dominant, with alkaline playing only a subordinate role?

A. Divergent boundaries (mid-ocean ridges); alkaline rocks appear there only in minor amounts, usually during early continental rifting.

The Rb-Sr Isotope System — Complete Concept Guide

Decay Scheme · Isochron Method · Crust-Mantle Evolution · Reservoir Rb/Sr Values

References: Misra, *Introduction to Geochemistry: Principles and Applications* & White, *Geochemistry* (2005)

Contents

1. The Rb-Sr Decay Scheme
2. The Isochron Equation & Isochron Diagram
3. Why Rb/Sr Differs Between Crust and Mantle
4. Reference Reservoirs: BABI, UR, and CHUR
5. The Crust-Mantle Sr Evolution Diagram (Figure 10.14)
6. Rb-Sr as a Petrogenetic Indicator
7. Comparison with the Sm-Nd System
8. Q.20 Worked Solution
9. Quick-Reference Summary & Probable Exam Questions

1. The Rb-Sr Decay Scheme

"Rubidium has two naturally occurring isotopes - radioactive ^{87}Rb (27.835 atom %) and stable ^{85}Rb (72.165 atom %) - whereas strontium has four - all of which are stable. The Rb-Sr dating method is based on the β^- decay of ^{87}Rb to ^{87}Sr in Rb-bearing minerals and rocks ($\lambda_{\text{Rb}} = 1.42 \times 10^{-11} \text{ yr}^{-1}$; $T_{1/2}(\text{Rb}) = 48.8 \times 10^9 \text{ yr}$)."

— Misra, *Introduction to Geochemistry*, p. 230

Why this pair works for geochronology & petrogenesis: ^{87}Rb decays by beta emission to ^{87}Sr with a half-life of ~48.8 billion years — long enough that Rb/Sr ratios accumulated over geologic time still preserve a measurable, meaningful signal today.

2. The Isochron Equation & Isochron Diagram

Using ^{86}Sr (non-radiogenic) for normalization, the growth equation for ^{87}Sr is:

$$^{87}\text{Sr}/^{86}\text{Sr} = (^{87}\text{Sr}/^{86}\text{Sr})_0 + (^{87}\text{Rb}/^{86}\text{Sr})(e^{\lambda_{\text{Rb}}t} - 1)$$

— Misra, *Introduction to Geochemistry*, p. 230, Eq. 10.16

For a suite of comagmatic rocks/minerals that crystallized at the same time with the same initial ratio, this equation plots as a straight line (an **isochron**) on a plot of $^{87}\text{Sr}/^{86}\text{Sr}$ versus $^{87}\text{Rb}/^{86}\text{Sr}$:

$$\text{intercept} = (^{87}\text{Sr}/^{86}\text{Sr})_0 \quad | \quad \text{slope (m)} = e^{\lambda_{\text{Rb}}t} - 1$$

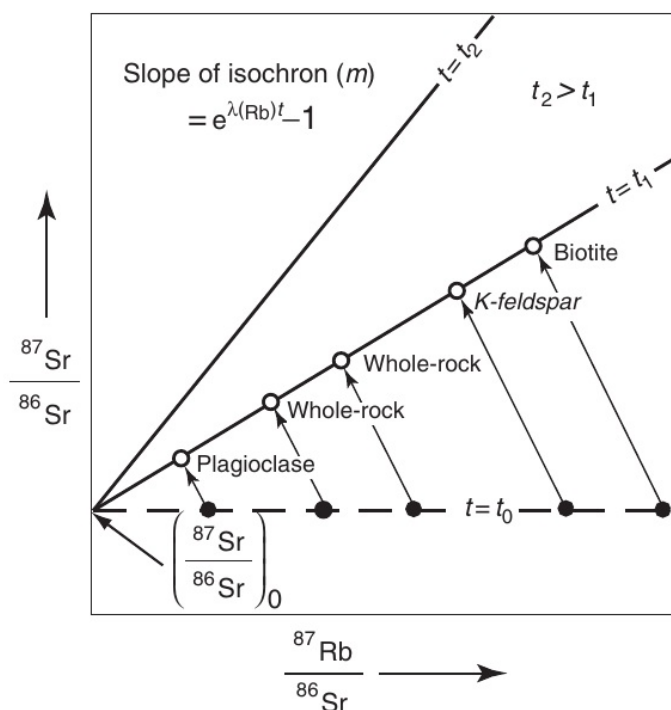


Fig. 10.4 Schematic Rb–Sr isochron diagram for two different suites of comagmatic igneous minerals and whole-rocks. The two suites are of different age (t_1 and t_2) but formed from homogeneous magmas having the same initial strontium ratio. The horizontal line is also an isochron (t_0), representing the time when the minerals and rocks had the same initial ratio and no radiogenic ^{87}Sr . The arrows indicate the generation of radiogenic ^{87}Sr by the decay of ^{87}Rb over the time period t_1 . The radiogenic ^{87}Sr is proportional to the ^{87}Rb originally present in the sample. Note that the $(^{87}\text{Sr}/^{86}\text{Sr})_0$ of a comagmatic suite can be approximated by the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of a sample with a very low concentration of ^{87}Rb .

Figure 10.4 (Misra, *Introduction to Geochemistry*, p. 232) — Schematic Rb–Sr isochron diagram for two comagmatic suites of different ages (t_1 , t_2) that formed from magmas sharing the same initial ratio. The steeper the isochron slope, the older the suite (more time for radiogenic ^{114}Sr to accumulate from ^{114}Rb decay).

Key insight from the diagram

The $(^{114}\text{Sr}/^{113}\text{Sr})_0$ of a comagmatic suite can be approximated directly by the $^{114}\text{Sr}/^{113}\text{Sr}$ ratio of any sample in the suite with a very low Rb/Sr — since almost no radiogenic ^{114}Sr has accumulated in it, its present ratio is essentially still the original, initial ratio.

3. Why Rb/Sr Differs Between Crust and Mantle

"The Earth developed an inhomogeneous continental crust that was dominated by granitic rocks enriched in Rb (which substitutes easily for K in silicate minerals), whereas Sr (which substitutes easily for Ca in silicate and carbonate minerals) was preferentially retained in the residual, depleted mantle. Consequently, crustal rocks have evolved with elevated Rb/Sr ratios and have developed much higher present-day $^{114}\text{Sr}/^{113}\text{Sr}$ ratios, whereas the depleted mantle... has evolved with much lower Rb/Sr ratios."

— Misra, *Introduction to Geochemistry*, p. 246

"The new crust has a higher Rb/Sr ratio than the mantle because Rb is more incompatible than Sr."

— White, *Geochemistry* (2005), p. 325

This is a straightforward consequence of incompatible-element behavior (see the earlier trace-elements guide): during partial melting, Rb (highly incompatible, substitutes for K) partitions strongly into the melt that eventually crystallizes as continental crust, while Sr (less incompatible, substitutes for Ca) is more evenly retained. The result is a crust that is Rb-enriched relative to Sr, and a mantle progressively depleted in Rb relative to Sr as more crust is extracted from it.

4. Reference Reservoirs: BABI, UR, and CHUR

BABI (Basaltic Achondrite Best Initial): The initial ¹¹⁴Sr/¹¹³Sr ratio of the solar system, determined from basaltic achondrite meteorites.

"Papanastassiou and Wasserburg (1969) analyzed seven basaltic achondrites and established an isochron that gave a model age of 4.39 ± 0.26 Ga... and an initial ¹¹⁴Sr/¹¹³Sr ratio of 0.69899 ± 0.00005 , which they referred to as Basaltic Achondrite Best Initial (BABI)... Thus, we can conclude that the Earth was born 4.5 ± 0.1 Ga ago with a primordial ¹¹⁴Sr/¹¹³Sr ratio of approximately 0.699; this is now accepted as the ¹¹⁴Sr/¹¹³Sr ratio for a hypothetical 'uniform reservoir' (UR) of planetary material."

— Misra, *Introduction to Geochemistry*, p. 246

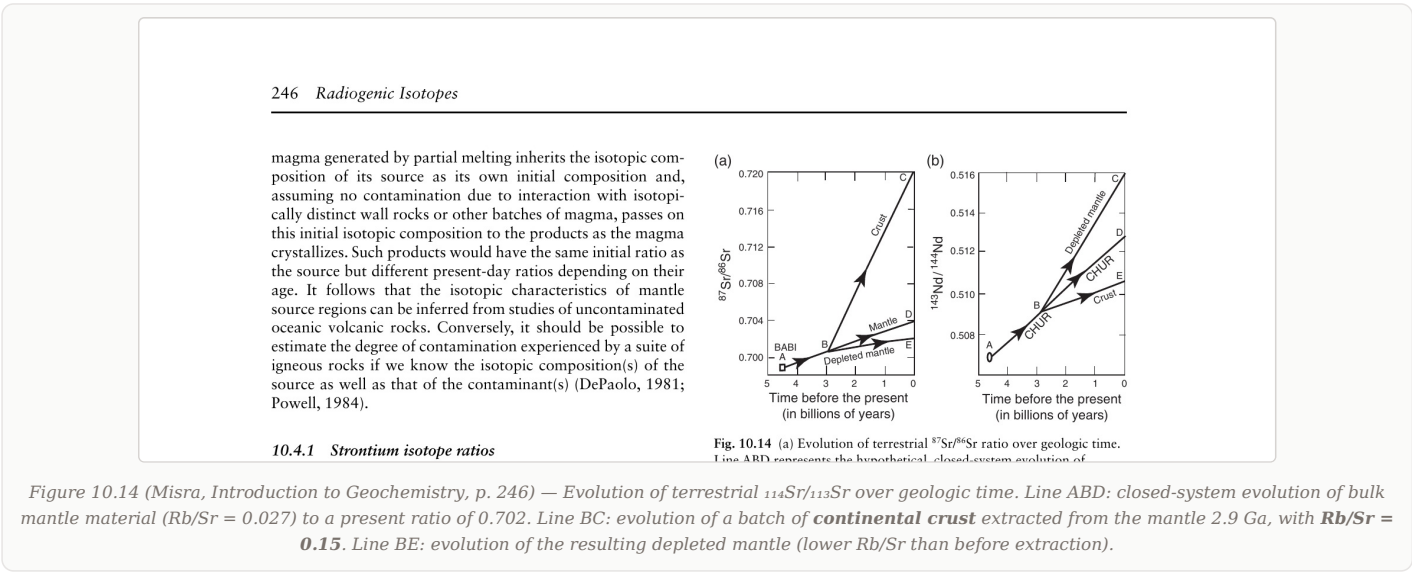
CHUR (Chondritic Uniform Reservoir): The equivalent reference reservoir concept for the Sm-Nd system, based on the assumption that the bulk Earth's Sm/Nd ratio matches that of chondritic meteorites.

"It can be calculated that to produce a ratio of 0.704 from the BABI value of 0.699 over 4.55 billion years (the Earth's age) would require an average Rb/Sr ratio of 0.027. The Rb/Sr ratio of the mantle must have decreased as a function of time because of preferential enrichment of Rb in the partial melts extracted from the mantle."

— Misra, *Introduction to Geochemistry*, p. 246

5. The Crust-Mantle Sr Evolution Diagram

Misra's Figure 10.14a is the single most important diagram for this topic — it models the divergent Sr-isotope evolution paths of the bulk mantle, a segment of extracted continental crust, and the resulting depleted mantle, all starting from the common BABI/UR point.



Reservoir	Representative Rb/Sr (Fig. 10.14a)	Present-day ¹¹⁴ Sr/ ¹¹³ Sr
Bulk mantle (undifferentiated)	0.027	~0.702
Continental crust (Line BC)	0.15	Higher, steeper slope from extraction point
Depleted mantle (post-extraction)	Lower than 0.027	Lower than undepleted mantle

A note on precision: Misra's text states the continental crust value as "0.15" (to two decimal places), rather than to three. Among the exam's four options (0.124, 0.158, 0.182, 0.242), **0.158** is by a clear margin the closest match — 0.008 away, versus 0.026, 0.032, and 0.092 away for the other three. This is presented as the best-supported answer from the uploaded reference, while being transparent that the source text itself rounds to two decimal places.

6. Rb-Sr as a Petrogenetic Indicator

"This difference allows us to use initial $^{114}\text{Sr}/^{113}\text{Sr}$ ratios to discriminate between mantle and crustal components in igneous rocks... rocks having initial $^{114}\text{Sr}/^{113}\text{Sr}$ ratios greater than 0.706 indicate a crustal component."

— Misra, *Introduction to Geochemistry*, p. 246

This is why Sr isotopes are routinely used to test whether a magma has been contaminated by continental crust during ascent: a mantle-derived magma with no crustal contamination should show initial $^{114}\text{Sr}/^{113}\text{Sr}$ in the range $\sim 0.702\text{--}0.706$ (present-day mantle-derived basalts), while assimilation of old, Rb-enriched crust pushes the ratio measurably higher.

7. Comparison with the Sm-Nd System

The Sm-Nd system is often used alongside Rb-Sr because it behaves in the **opposite** sense with respect to crust/mantle partitioning:

"The Sm-Nd system is in many ways opposite of the Rb-Sr system. First, the Sm/Nd ratio of the mantle is higher than that of the crust and hence the $^{143}\text{Nd}/^{144}\text{Nd}$ ratio is higher in the mantle than in the crust... Third, neither Sm nor Nd is particularly mobile, so ages and initial ratios are relatively insensitive to weathering and metamorphism."

— White, *Geochemistry* (2005), p. 325

Feature	Rb-Sr	Sm-Nd
Higher parent/daughter ratio in	Crust	Mantle
Element mobility (weathering/metamorphism)	Rb and Sr relatively mobile	Sm and Nd relatively immobile
Best suited for dating	Rb-rich rocks (granites, pegmatites, shales)	Mafic/ultramafic rocks (crustal rocks too uniform in Sm/Nd)

8. Q.20 Worked Solution

Q.20. Average continental crust reservoir is represented by which one of the following Rb/Sr values of radiogenic parent-daughter pairs?

(a) 0.124
(b) 0.158
(c) 0.182
(d) 0.242

Option	Distance from textbook value (0.15)	Verdict
(a) 0.124	0.026	Eliminated
(b) 0.158	0.008 — closest	Correct
(c) 0.182	0.032	Eliminated
(d) 0.242	0.092	Eliminated

✓ Answer: (b) 0.158

9. Quick-Reference Summary & Probable Exam Questions

Concept	Key Value / Fact
^{114}Rb decay constant (λ_{Rb})	$1.42 \times 10^{-11} \text{ yr}^{-1}$
^{114}Rb half-life	48.8×10^9 years
BABI initial $^{114}\text{Sr}/^{113}\text{Sr}$	~ 0.699
Bulk mantle Rb/Sr (Fig. 10.14a)	0.027
Continental crust Rb/Sr (Fig. 10.14a)	~ 0.15 (closest option: 0.158)
Crustal-contamination threshold (initial $^{114}\text{Sr}/^{113}\text{Sr}$)	> 0.706 indicates crustal component

Q. What is the significance of a sample with very low Rb/Sr in an isochron plot?

A. Its present $^{87}\text{Sr}/^{86}\text{Sr}$ closely approximates the suite's initial ratio, since it has generated almost no radiogenic ^{87}Sr since crystallization.

Q. Why does the mantle's Rb/Sr ratio decrease over geologic time?

A. Because each episode of partial melting preferentially extracts Rb (more incompatible) into the crust-forming melt, progressively lowering the residual mantle's Rb/Sr.

Q. What initial $^{87}\text{Sr}/^{86}\text{Sr}$ threshold indicates a crustal contribution to a magma?

A. Greater than about 0.706.

Q. Why is Rb-Sr less reliable than Sm-Nd for dating rocks that have experienced weathering or metamorphism?

A. Rb and Sr are relatively mobile elements, so their ratios can be disturbed by fluid-related alteration; Sm and Nd are comparatively immobile.

Q. What does the slope of an Rb-Sr isochron represent mathematically?

A. $e^{(\lambda_{\text{Rb}}t)} - 1$, from which the age t of the comagmatic suite can be calculated.

REE Diagrams & the Europium Anomaly — Complete Concept Guide

Chondrite Normalization · Lanthanide Contraction · LREE/HREE · Eu & Ce Anomalies

Reference: Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed. — Chemical Petrology II

Contents

1. What Are the Rare Earth Elements?
2. Why REE Diagrams Are Normalized to Chondrite
3. How to Read the Diagram — Slope, LREE & HREE
4. The Europium Anomaly — Mechanism
5. Locating the Anomaly — Why Sm-Gd
6. Positive vs. Negative Eu Anomalies
7. The Cerium Anomaly — A Contrasting Case
8. Q.21 Worked Solution
9. Quick-Reference Summary & Probable Exam Questions

1. What Are the Rare Earth Elements?

"The rare earth elements (REE), the series from lanthanum to lutetium (atomic numbers 57–71), are members of Group IIIA of the periodic table... They all have similar chemical and physical properties, causing them to behave as a coherent series (the 'lanthanide series'). They all have a 3+ oxidation state, as a rule, and their ionic radius decreases steadily with increasing atomic number (called the lanthanide contraction)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 171–172

La → Ce → Pr → Nd → (Pm) → Sm → **Eu** → Gd → Tb → Dy → Ho → Er → Tm → Yb → Lu
Increasing atomic number (57–71) | Decreasing ionic radius | Increasing compatibility

"The decrease in ionic radius causes the heavier REEs to be progressively less compatible... Note that plagioclase, for example, is insensitive to the effect of ionic radius, whereas garnet strongly favors the heavy rare earth elements (HREE)."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 172

The two valence exceptions

"There are two exceptions to the 3+ valence norm for the REE. At low values of oxygen fugacity (f_{O_2}), Eu can have a 2+ valence... Eu^{2+} substitutes for Ca in plagioclase (but is too large to do so in clinopyroxene or most other Ca-bearing phases)... Under oxidizing conditions, Ce can also have a valence of 4+." (Winter, p. 172) These two exceptions are the entire reason REE diagrams show anomalies at all — every other REE behaves as a smoothly-varying 3+ ion.

2. Why REE Diagrams Are Normalized to Chondrite

"A direct plot of concentration versus atomic number suffers from the jagged, or 'sawtooth,' Oddo-Harkins effect, whereby atoms with an even atomic number are more abundant than their immediate neighbors with odd atomic numbers. The effect is eliminated by normalizing, or dividing the concentration of each REE by the concentration of the same REE in a standard... the most commonly used standards are estimates of primordial mantle or chondrite meteorite concentrations."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 172

REE diagram construction: Concentration (normalized to chondrite, plotted on the y-axis, log scale) versus each REE arranged by **increasing atomic number** on the x-axis (La on the left, Lu on the right). The subscript "N" denotes a normalized value, e.g., La_N , Eu_N .

3. How to Read the Diagram — Slope, LREE & HREE

"The increasing D_i values as atomic number (Z) increases... reflect the lanthanide contraction and the progressively greater proportion of the heavy REE (HREE) in the solid and the light REE (LREE) in the liquid, resulting in a negative slope for the curves in an REE diagram... a positive slope is impossible, whatever you may try, because the LREE will always favor the liquid compared to the HREE."

— Winter, *Principles of Igneous and Metamorphic Petrology*, pp. 172–173

TABLE 5 Spreadsheet to Calculate an REE Diagram for th

	A	B	C	D	E
1	Rare Earth Element Diagram				
2	Element	Atom #	chondr	sample	samp/ch
3	La	57	0.237	1.75	7.40
4	Ce	58	0.612	4.86	7.94
5	Nd	60	0.467	3.37	7.22
6	Sm	62	0.153	0.98	6.43
7	Eu	63	0.058	0.15	2.62
8	Dy	66	0.254	1.00	3.94
9	Er	68	0.1655	0.44	2.65
10	Yb	70	0.17	0.37	2.17
11	Lu	71	0.0254	0.05	2.00
12			in ppm	in ppm	
13					
14					
15			Distribution Coefficients		
16		Olivine	Opx	Cpx	Garnet
17	La	0.007	0.03	0.056	0.00
18	Ce	0.006	0.02	0.092	0.00
19	Nd	0.006	0.03	0.230	0.02
20	Sm	0.007	0.05	0.445	0.10
21	Eu	0.007	0.05	0.474	0.24
22	Dy	0.013	0.15	0.582	3.17
23	Er	0.026	0.23	0.583	6.56
24	Yb	0.049	0.34	0.542	11.5
25	Lu	0.045	0.42	0.506	11.9

Figure 4 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 173) — Chondrite-normalized REE concentrations for melts produced at various values of F (fraction melted) from a garnet lherzolite. Note the x-axis runs by atomic number, La(57) through Lu(71), and every curve has a negative slope — steeper for smaller F .

"The slope on an REE diagram can easily be approximated mathematically by the ratio of the normalized concentration of an element on the left side, such as La or Ce, divided by an element on the right, such as Yb or Lu... Similarly $(La/Sm)_N$ or $(La/Eu)_N$ can be used to measure enrichment within the LREE, whereas $(Tb/Yb)_N$ will do the same for the HREE."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 173

4. The Europium Anomaly — Mechanism

"The distribution coefficient in the spreadsheet table for europium in plagioclase is for a high-Eu²⁺ component. If you assign 20% plagioclase to the source rock, there will be a pronounced dip in the REE pattern at Eu... This is referred to as a europium anomaly, reflecting the Eu²⁺ substitution for Ca in plagioclase."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 174

time. As I have suggested, this is a common problem for most of our models. Using them, we might be able to identify

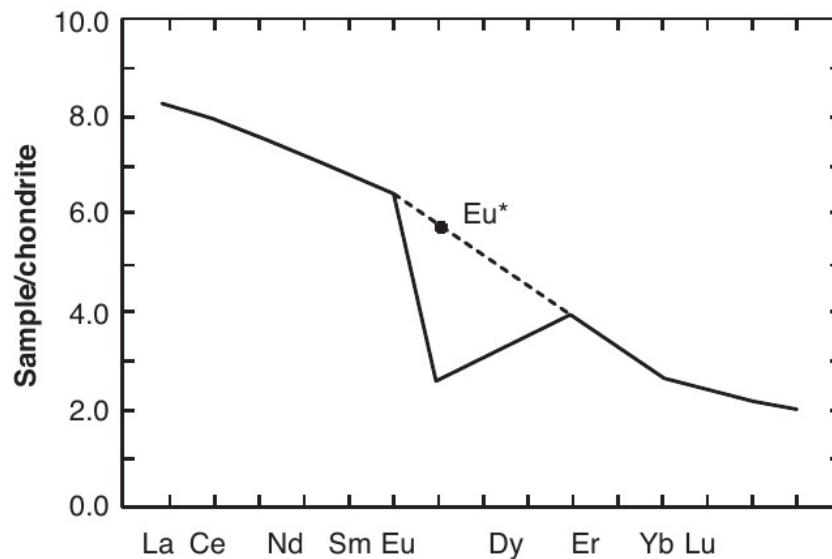


FIGURE 5 REE diagram for 10% batch melting of a hypothetical lherzolite with 20% plagioclase. Because Eu^{2+} is retained in the plagioclase of the source, the extracted melt is depleted in Eu, resulting in a pronounced negative europium anomaly.

Figure 5 (Winter, *Principles of Igneous and Metamorphic Petrology*, p. 174) — REE diagram for 10% batch melting of a lherzolite source with 20% plagioclase, showing the characteristic dip (negative Eu anomaly) at europium relative to the smooth trend defined by its REE neighbors.

Why plagioclase and only plagioclase

Eu^{2+} is similar in ionic radius/charge to Ca^{2+} , which plagioclase readily accepts in its crystal structure. None of the neighboring REE (all still REE^{3+}) substitute into plagioclase nearly as well, so plagioclase selectively "grabs" Eu while leaving Sm and Gd (its immediate neighbors) largely unaffected — creating a visible dip or spike isolated at exactly one point on the curve.

5. Locating the Anomaly — Why Sm-Gd

"The magnitude of the Eu anomaly is commonly expressed as Eu/Eu^ , where Eu^* is the hypothetical value of Eu if no Eu^{2+} was captured by plagioclase (the value on a straight line between the nearest neighboring elements)."*

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 174

$$\text{Eu}^* = \text{interpolated value on the straight line between } \text{Sm}_N \text{ and } \text{Gd}_N$$

$$\text{Eu anomaly} = \text{Eu}_N / \text{Eu}^*$$

Because Eu sits directly between Sm ($Z=62$) and Gd ($Z=64$) in the REE sequence, "the nearest neighboring elements" used to define Eu^* are unambiguously **Sm and Gd**. This is the direct textbook basis for locating the Eu anomaly between the Sm-Gd pair on any REE diagram.

6. Positive vs. Negative Eu Anomalies

"The anomaly may be either negative...or positive, depending on whether plagioclase was removed or accumulated, respectively. A negative europium anomaly is a good indicator that the liquid was at one time in equilibrium with now-absent plagioclase, but it cannot easily determine whether the reason for this is a plagioclase-bearing melt source or the removal of plagioclase phenocrysts from the melt at a later time."

— Winter, *Principles of Igneous and Metamorphic Petrology*, p. 174

Anomaly Type	What It Indicates
Negative Eu anomaly (dip)	Melt was in equilibrium with plagioclase that is now absent — either a plagioclase-retaining melting residue, or plagioclase phenocrysts fractionated out of the melt later
Positive Eu anomaly (spike)	Plagioclase was accumulated (a plagioclase cumulate rock)

White's *Geochemistry* gives the classic crustal-scale example of this principle:

"[Continental crustal] residues of those melts were rich in plagioclase, hence retaining somewhat more of the Eu in the lower crust, and creating a complimentary Eu-depleted upper crust."
— White, *Geochemistry* (2005), p. 267

7. The Cerium Anomaly — A Contrasting Case

Cerium is the other REE with a valence exception (Ce^{4+} under oxidizing conditions), and it produces an anomaly through an entirely different mechanism — not mineral partitioning, but redox chemistry in surface/seawater environments:

"Both the pelagic sediment and seawater display a negative Ce anomaly, a consequence of Ce being in the +4 oxidation state."
— White, *Geochemistry* (2005), p. 267

Exam distinction

Eu anomalies are diagnostic of **igneous plagioclase fractionation**; Ce anomalies are diagnostic of **oxidizing surface/marine conditions** (Ce^{4+} is far less soluble than the other trivalent REE and gets preferentially removed from seawater). Don't confuse the two mechanisms if a question asks about anomaly causes.

8. Q.21 Worked Solution

Q.21. In a rare earth element (REE) diagram, europium anomaly is seen between which one of the following REE pairs?

- (a) La-Nd
- (b) Nd-Sm
- (c) Sm-Gd
- (d) Gd-Dy

Option	Reasoning	Verdict
(a) La-Nd	Light REEs at the start of the sequence; nowhere near Eu (Z=63)	Eliminated
(b) Nd-Sm	Close to Eu's position, but Eu comes after Sm, not between Nd and Sm	Eliminated
(c) Sm-Gd	Sm (Z=62) and Gd (Z=64) are exactly the "nearest neighboring elements" used to define Eu* — Eu (Z=63) sits directly between them	Correct
(d) Gd-Dy	This pair lies past Gd, on the far side of Eu	Eliminated

✓ Answer: (c) Sm-Gd

9. Quick-Reference Summary & Probable Exam Questions

Concept	Key Fact
REE atomic number range	La (57) to Lu (71)
Normal REE valence	3+ (all REE, as a rule)
Eu's special valence	2+ at low fO_2 ; substitutes for Ca in plagioclase
Ce's special valence	4+ under oxidizing conditions; drives seawater/sediment Ce anomaly
Eu* definition	Interpolated value between Sm_N and Gd_N

Negative Eu anomaly	Plagioclase removed (residue or fractionated out)
Positive Eu anomaly	Plagioclase accumulated (cumulate)
REE diagram slope sign	Always negative or flat; never positive (LREE always favor liquid over HREE)

Q. Why is a positive slope on an REE diagram considered impossible?

A. Because the LREE are always less compatible than the HREE due to the lanthanide contraction, so the LREE will always be enriched in the liquid relative to HREE, giving a negative (or at most flat) slope.

Q. Which mineral has the strongest effect on producing steep negative REE slopes when present in the source?

A. Garnet, due to its very high distribution coefficients for the HREE.

Q. What ratio is commonly used to measure LREE enrichment, and what ratio measures HREE enrichment?

A. $(La/Sm)_N$ or $(La/Eu)_N$ for LREE enrichment; $(Tb/Yb)_N$ for HREE enrichment.

Q. Why can't Eu^{2+} substitute into clinopyroxene the way it does into plagioclase?

A. Eu^{2+} is too large for the Ca site in clinopyroxene and most other Ca-bearing phases, unlike the larger, more accommodating plagioclase structure.

Q. What does the Oddo-Harkins effect refer to, and how is it eliminated on REE diagrams?

A. The "sawtooth" pattern where even-atomic-number elements are naturally more abundant than odd-numbered neighbors; it's eliminated by normalizing each REE to its chondrite (or primordial mantle) concentration.

Q. If a granite shows a strong negative Eu anomaly, what does this suggest about its petrogenesis?

A. That plagioclase fractionated out of (or was retained as a residual phase from) the parental magma, removing Eu preferentially relative to its REE neighbors.

Epsilon Notation & the Chondritic Uniform Reservoir — Complete Concept Guide

CHUR · ϵNd & ϵSr · The Mantle Array · Depleted Mantle vs. Crustal Contamination

References: Misra, *Introduction to Geochemistry: Principles and Applications* & White, *Geochemistry* (2005)

Contents

1. What Is CHUR?
2. Why Isotope Ratios Are Expressed as Deviations — the ϵ Notation
3. The ϵNd Formula
4. Reading the Sign of ϵNd
5. The Companion Parameter — ϵSr
6. The Mantle Array — Plotting ϵNd vs. ϵSr
7. Why Not δ , ω , or ν ?
8. Q.22 Worked Solution
9. Quick-Reference Summary & Probable Exam Questions

1. What Is CHUR?

CHUR (Chondritic Uniform Reservoir): A hypothetical reference reservoir whose Sm/Nd ratio and Nd isotopic composition are assumed to equal the bulk Earth's, based on the assumption that chondritic meteorites represent unfractionated, primordial solar-system material.

"The terrestrial evolution of Nd isotopes is described in terms of an initial 'chondritic uniform reservoir' (CHUR; DePaolo and Wasserburg, 1976) whose Sm/Nd ratio = 0.31 and whose Nd isotope ratios were uniform and the same as those of chondrites. Because the chondrites have not been affected by magmatic processes, they provide an estimate of the bulk-earth values of Sm/Nd and $^{143}\text{Nd}/^{144}\text{Nd}$."

— Misra, *Introduction to Geochemistry*, p. 247

"From analysis of several chondrites and an achondrite, Jacobsen and Wasserburg (1980) determined an isochron age of 4.6 Ga for the meteorites and an initial $^{143}\text{Nd}/^{144}\text{Nd}$ ratio of 0.505828... The present-day values of the $^{143}\text{Nd}/^{144}\text{Nd}$ ratio... and the $^{145}\text{Sm}/^{144}\text{Nd}$ ratio of CHUR are 0.512638 and... 0.1967."

— Misra, *Introduction to Geochemistry*, pp. 247–248

White adds the mechanistic picture of why crust and mantle diverge from this common starting point:

"The Earth as a whole has a $^{145}\text{Sm}/^{144}\text{Nd}$ ratio identical to that of chondrites (0.1967), and the initial $^{143}\text{Nd}/^{144}\text{Nd}$ of the Earth is also the same as that of chondrites. Thus the bulk Earth evolves along the same line as chondrites. Because Nd is more incompatible than Sm and will therefore be more enriched in a partial melt than Sm, the crust, formed by partial melting of the mantle, has a low Sm/Nd ratio and evolves along a line of lower slope than the bulk Earth. Partial melting depletes the mantle more in Nd than in Sm, so the mantle is left with a high Sm/Nd ratio and evolves along a steeper slope."

— White, *Geochemistry* (2005), p. 326

2. Why Isotope Ratios Are Expressed as Deviations — the ϵ Notation

"In general, differences in the Nd isotope ratios between the rocks of interest and CHUR are quite small, and it is convenient to represent the comparison by the 'epsilon parameter' introduced by DePaolo and Wasserburg (1976)."

— Misra, *Introduction to Geochemistry*, p. 248

"It is often more meaningful to consider variations in $^{143}\text{Nd}/^{144}\text{Nd}$ relative to the bulk Earth, or chondritic, value. This is the purpose of ϵ notation. ϵNd is simply the relative deviation of the $^{143}\text{Nd}/^{144}\text{Nd}$ ratio from the chondritic ratio in parts in 10^4 ."

— White, *Geochemistry* (2005), p. 326

Why bother with a special notation at all?

White explains the practical benefit directly: "Most rocks will have ϵNd values in the range of -20 to +10, and it is certainly easier to work with a one or two digit number than a five or six digit one" (p. 326). Raw $^{143}\text{Nd}/^{144}\text{Nd}$ ratios only differ from CHUR in the 4th or 5th decimal place — ϵ notation rescales that tiny difference into an easy, comparable number.

3. The ϵNd Formula

$$\epsilon\text{Nd} = [(^{143}\text{Nd}/^{144}\text{Nd})_{\text{meas}} / (^{143}\text{Nd}/^{144}\text{Nd})_{\text{p(CHUR)}} - 1] \times 10^4$$

— Misra, *Introduction to Geochemistry*, Eq. 10.55, p. 248

where $(^{143}\text{Nd}/^{144}\text{Nd})_{\text{meas}}$ is the measured ratio in the sample, and $(^{143}\text{Nd}/^{144}\text{Nd})_{\text{p(CHUR)}}$ is CHUR's present-day value (0.512638, normalized to $^{146}\text{Nd}/^{144}\text{Nd} = 0.7219$).

A note on notation across sources

Some references write the ratio inside the brackets differently ("measured minus CHUR, divided by CHUR") — algebraically equivalent to Misra's form. Either way, the essential idea is the same: ϵNd measures how far a rock's $^{143}\text{Nd}/^{144}\text{Nd}$ sits from the chondritic baseline, scaled by 10,000.

4. Reading the Sign of ϵNd

"A value of zero for ϵ_{Nd} ... would indicate that the corresponding measured isotopic ratio of the igneous rock is same as that of the CHUR... at present. A positive ϵ_{Nd} value implies that the magma was derived from a depleted mantle source; a negative ϵ_{Nd} value implies that the magma was derived from, or contaminated by, a source with a lower Sm/Nd ratio, such as old crustal material that had separated from CHUR."

— Misra, *Introduction to Geochemistry*, p. 248

ϵ_{Nd} value	Interpretation
$\epsilon_{Nd} = 0$	Rock's $^{143}Nd/^{144}Nd$ matches CHUR exactly (chondritic composition)
$\epsilon_{Nd} > 0$	Magma derived from a depleted mantle source (higher Sm/Nd than chondritic, having lost Nd to earlier crust extraction)
$\epsilon_{Nd} < 0$	Magma derived from, or contaminated by, old crustal material (lower Sm/Nd than chondritic)

"A negative value of ϵ_{Nd} implies that on average over the history of the Earth the Sm/Nd ratio of that rock or its precursors has been lower than chondritic. This in turn implies the rare earth pattern of the rock or its precursors was light rare earth-enriched. We can draw the opposite inference from a positive ϵ_{Nd} value."

— White, *Geochemistry* (2005), p. 326

5. The Companion Parameter — ϵ_{Sr}

$$\epsilon_{Sr} = [(^{87}Sr/^{86}Sr)_{meas} / (^{87}Sr/^{86}Sr)_{p(UR)} - 1] \times 10^4$$

— Misra, *Introduction to Geochemistry*, Eq. 10.56, p. 248

"We often consider ϵ_{Nd} values together with ϵ_{Sr} values... where UR represents 'uniform reservoir', equivalent to bulk earth, and $(^{87}Sr/^{86}Sr)_{p(UR)}$ is the present value of this ratio in the uniform reservoir (= 0.7045)."

— Misra, *Introduction to Geochemistry*, p. 248

The sign convention flips for ϵ_{Sr}

"The implications of positive and negative ϵ_{Sr} values are just the opposite of each other; for example, a crustal source of magma or its contamination by a crustal source is indicated by **positive** ϵ_{Sr} values." (Misra, p. 248) Compare this to ϵ_{Nd} , where crustal contamination gives a **negative** value — the two systems are mirror images because Rb/Sr is elevated in the crust while Sm/Nd is elevated in the depleted mantle.

6. The Mantle Array — Plotting ϵ_{Nd} vs. ϵ_{Sr}

"Figure 10.15 is a plot of measured $^{87}Sr/^{86}Sr$ versus $^{143}Nd/^{144}Nd$ ratios of selected young oceanic, subduction-related, and continental basalts... (OIBs), often referred to as the **Mantle Array**."

— Misra, *Introduction to Geochemistry*, pp. 246, 249 (paraphrased from surrounding text)

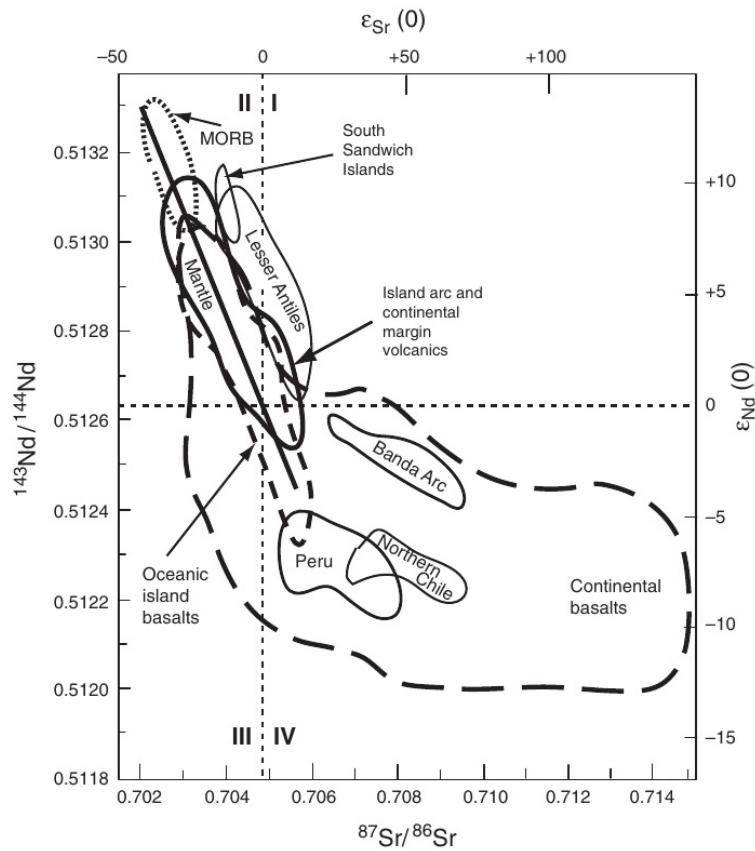


Fig. 10.15 Nd and Sr isotope compositional fields of young oceanic basalts (MORB and OIB), subduction-related basalts (island arcs formed

Figure 10.15 (Misra, *Introduction to Geochemistry*, p. 249) — Nd and Sr isotope compositional fields (the "Mantle Array") for young oceanic, subduction-related, and continental basalts, plotted as $^{87}\text{Sr}/^{86}\text{Sr}$ versus $^{143}\text{Nd}/^{144}\text{Nd}$ (equivalently, ϵSr vs. ϵNd).

Quadrant I: $\epsilon\text{Nd} > 0$, $\epsilon\text{Sr} > 0$

Rare — simultaneous Nd and Sr enrichment is geochemically inconsistent, since they normally move in opposite directions.

Quadrant II: $\epsilon\text{Nd} > 0$, $\epsilon\text{Sr} < 0$

The dominant mantle field — MORBs plot here, with the **lowest $^{87}\text{Sr}/^{86}\text{Sr}$ and highest ϵNd** , reflecting a long-term depleted mantle source. Island-arc basalts also largely plot here.

Quadrant III: $\epsilon\text{Nd} < 0$, $\epsilon\text{Sr} < 0$

Rare, like Quadrant I; some continental basalts plot here due to contamination from old basement granulites.

Quadrant IV: $\epsilon\text{Nd} < 0$, $\epsilon\text{Sr} > 0$

Continental margin volcanics (e.g., South America) plot here — reflecting subduction of oceanic crust under continental crust and contamination by Nd/Sr from felsic crustal rocks.

7. Why Not δ , ω , or ν ?

Symbol	What It's Actually Used For
--------	-----------------------------

	Reserved for stable isotope ratios (e.g., $\delta^{18}\text{O}$, $\delta^{13}\text{C}$) — expressed as per mil (‰) deviation from a defined standard (e.g., SMOW, PDB).
--	--

δ (Delta)	A completely different notation system for a different (non-radiogenic) class of isotope variation.
ω (Omega)	Not a standard symbol for isotope-deviation notation in radiogenic or stable isotope geochemistry.
υ (Upsilon)	Likewise not used for any standard isotope-deviation parameter.
ϵ (Epsilon)	The defined, standard symbol for deviation of a radiogenic isotope ratio (Nd, Sr, Hf) from its respective chondritic/uniform reference reservoir, in parts per 10⁴.

Exam trap to remember

δ -notation and ϵ -notation look similar (both express a "deviation from a reference") but apply to fundamentally different isotope systems: δ for stable isotopes (mass-dependent fractionation, per mil), ϵ for radiogenic isotopes (time-dependent ingrowth relative to CHUR/UR, parts per 10⁴). Don't let the superficial similarity cause you to pick δ for a radiogenic-isotope question.

8. Q.22 Worked Solution

Q.22. Which one of the following symbols denotes the deviation of the Nd-isotopic composition of a rock from that of the chondritic uniform reservoir?

- (a) ω (Omega)
- (b) ϵ (Epsilon)
- (c) δ (Delta)
- (d) υ (Upsilon)

Option	Reasoning	Verdict
(a) Omega	Not used for isotopic deviation notation	Eliminated
(b) Epsilon	Exactly the DePaolo & Wasserburg (1976) parameter both textbooks define for CHUR-relative deviation	Correct
(c) Delta	Used for stable isotope per mil notation, not radiogenic CHUR-relative deviation	Eliminated
(d) Upsilon	Not used for any standard isotope-deviation notation	Eliminated

✓ Answer: (b) ϵ (Epsilon)

9. Quick-Reference Summary & Probable Exam Questions

Concept	Key Value / Fact
CHUR present-day $^{143}\text{Nd}/^{144}\text{Nd}$	0.512638
CHUR present-day $^{145}\text{Sm}/^{144}\text{Nd}$	0.1967 (Misra) / matches chondrite
UR (bulk Earth) present-day $^{87}\text{Sr}/^{86}\text{Sr}$	0.7045
$\epsilon\text{Nd} > 0$ meaning	Depleted mantle source
$\epsilon\text{Nd} < 0$ meaning	Crustal source or contamination
$\epsilon\text{Sr} > 0$ meaning	Crustal source or contamination (opposite sign convention to ϵNd)
Who introduced ϵ notation	DePaolo and Wasserburg (1976)

Q. Why do ϵNd and ϵSr have opposite sign conventions for indicating crustal contamination?

A. Because Rb/Sr is elevated in the crust (raising $^{87}\text{Sr}/^{86}\text{Sr}$, giving positive ϵSr), while Sm/Nd is elevated in the depleted mantle, not the crust (so crustal contamination lowers $^{143}\text{Nd}/^{144}\text{Nd}$, giving negative ϵNd).

Q. Which basalt type plots with the lowest $^{87}\text{Sr}/^{86}\text{Sr}$ and highest ϵNd on the Mantle Array?

A. MORBs (mid-ocean ridge basalts), reflecting their long-term depleted mantle source.

Q. What does $\epsilon\text{Nd} = 0$ for a rock indicate?

A. That its measured $^{143}\text{Nd}/^{144}\text{Nd}$ ratio equals CHUR's present-day value — a chondritic composition.

Q. Why are Quadrants I and III of the Mantle Array sparsely populated?

A. Because simultaneous enrichment or depletion of both Nd and Sr relative to their reservoirs is inconsistent with their opposing geochemical behavior during mantle melting.

Q. What Sm/Nd ratio does Misra assign to CHUR?

A. 0.31 (present-day), corresponding to a $^{145}\text{Sm}/^{144}\text{Nd}$ of 0.1967.

Q. Continental margin volcanics from subduction of oceanic crust under continental crust typically plot in which Mantle Array quadrant?

A. Quadrant IV ($\epsilon\text{Nd} < 0$, $\epsilon\text{Sr} > 0$), reflecting crustal contamination.

The U-Pb Concordia-Discordia System — Complete Concept Guide

Dual Decay Schemes · The Concordia Curve · Discordia Chords · Pb Loss & Zircon Geochronology

Reference: Misra, *Introduction to Geochemistry: Principles and Applications*

Contents

1. Why U-Pb Dating Uses Two Independent Clocks
2. Why Zircon Is the Preferred Mineral
3. The Concordia Curve — Definition & Construction
4. The Discordia Line — What Causes Discordance
5. Reading the Concordia-Discordia Diagram (Figure 10.7)
6. The Related Pb-Pb Isochron
7. Q.23 Worked Solution
8. Quick-Reference Summary & Probable Exam Questions

1. Why U-Pb Dating Uses Two Independent Clocks

Uranium has two isotopes that each decay to a different lead isotope, via independent decay constants:

Parent	Daughter	Half-life
^{238}U	^{206}Pb	~ 4.47 billion years (λ^{238})
^{235}U	^{207}Pb	~ 0.70 billion years (λ^{235} , faster decay)

Because both schemes operate simultaneously within the very same mineral grain, a mineral that has remained a closed system since crystallization should yield the **same age** from both:

"If a uranium-bearing mineral behaves as a closed system, equations (10.24) and (10.26) should, in principle, yield concordant ages. In reality, however, even zircon crystals often yield discordant ages because of some Pb loss from the system of interest."
— Misra, *Introduction to Geochemistry*, p. 234

Why this dual-clock design is so powerful
Having two independent geochronometers running in the same mineral means the system can self-check: if both clocks agree, the age is trustworthy (concordant). If they disagree (discordant), that disagreement itself carries information — typically revealing that the mineral lost lead (or gained uranium) at some point after crystallization.

2. Why Zircon Is the Preferred Mineral

"Zircon is by far the most preferred mineral as it has a relatively high concentration of U (because U^{4+} substitutes quite readily for Zr^{4+} due to identical charge and similar ionic radius), is widely distributed (present in most felsic to intermediate igneous rocks), and contains very little initial Pb (because Pb^{2+} does not substitute easily for U^{4+} in the zircon crystal structure). Zircon is extremely resistant to hydrothermal alteration, and can survive diagenesis, metamorphism, or weathering that may modify or destroy its host rock."
— Misra, *Introduction to Geochemistry*, p. 233

A useful bonus feature
"Another advantage of using zircon crystals is that they may be zoned, the different zones representing separate events such as magmatic crystallization or metamorphism. In such cases, the zones... may provide a chronology of successive events experienced by the host rock." (Misra, p. 233)

3. The Concordia Curve — Definition & Construction

Concordia curve: The locus of points on a $^{206}\text{Pb}^*/^{238}\text{U}$ vs. $^{207}\text{Pb}^*/^{235}\text{U}$ diagram where the age calculated from each decay scheme is identical (concordant).

"It seems that it may still be possible to interpret the isotopic data by means of the concordia diagram devised by Wetherill (1956a). The concordia curve is constructed by plotting compatible values of $^{206}\text{Pb}^/^{238}\text{U}$ (y axis) against $^{207}\text{Pb}^*/^{235}\text{U}$ (x axis) calculated for chosen values of t , so that every point on the concordia corresponds to a particular value of t , if the system can be presumed to have been closed to U, Pb, and the intermediate daughters."*
— Misra, *Introduction to Geochemistry*, p. 234

"For a zircon plotting on the concordia (e.g., point Q), an identical age (t) can be calculated from either of the equations. Any U-Pb system that plots on the concordia yields concordant dates, which can be calculated from the coordinates of the point on the concordia."
— Misra, *Introduction to Geochemistry*, p. 235

4. The Discordia Line — What Causes Discordance

Discordia: A straight-line chord connecting a point on the concordia to another point (often the origin), representing a population of cogenetic zircons that have all experienced varying degrees of Pb loss (or U gain).

"With variable loss of Pb, the changed isotopic compositions of the zircon crystals would lie on a straight line joining point X and the origin of the concordia diagram (the coordinates of any zircon crystal from which all the radiogenic Pb has been lost). Such a chord

is called a discordia because U–Pb systems lying on it yield discordant dates."

— Misra, *Introduction to Geochemistry*, p. 235

Recently weathered zircons plot differently

"Discordant zircons that have lost varying degrees of U as a result of chemical weathering recently may also lie on a discordia that passes through the origin but they should plot **above** the concordia." (Misra, p. 235) — recent U loss pushes points the opposite direction from ancient Pb loss.

5. Reading the Concordia-Discordia Diagram

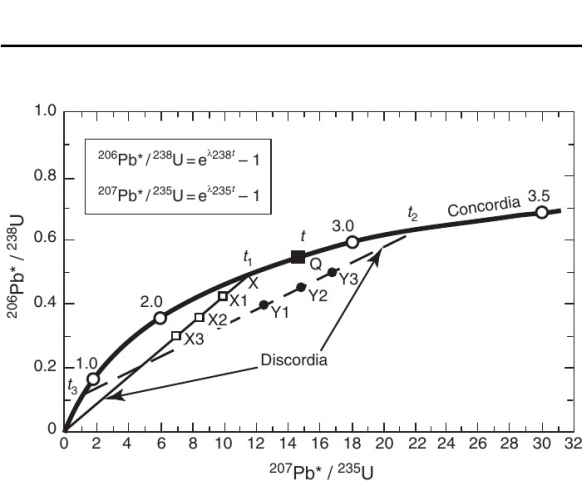


Fig. 10.7 U–Pb concordia diagram illustrating interpretations of lead isotopic data for uranium-bearing minerals such as zircon. The concordia is generated by plotting $^{206}\text{Pb}^*/^{238}\text{U}$ against $^{207}\text{Pb}^*/^{235}\text{U}$, which are calculated for selected values of t (a few of which are marked on the concordia), using equations (10.22) and (10.24). For a zircon plotting on the concordia (e.g.,

the crystallization age of zircon (t_2). Assuming was instantaneous, the lower intercept is inte (t_3) of the event that disturbed the system (Y the intercept has no age significance if the Pb tinuous diffusion over a long period of time logical information would be required to dis episodic and continuous Pb loss models for a ples. The computer code *Isoplot* (version (2003) can be used for constructing the cor fitting of discordias to analytical data, an coordinates of the points of intersection wit

Isotopic composition of lead in U- and Th-bearing minerals and whole rocks

An equation relating the isotopic compo U- and Th-bearing minerals to model age ca combining equations (10.21) and (10.22) an the present-day $^{238}\text{U}/^{235}\text{U}$ ratio, which is equ

Figure 10.7 (Misra, *Introduction to Geochemistry*, p. 235) — U–Pb concordia diagram. The curved concordia line is marked with selected t values. Point Q on the concordia gives a single concordant age. Points X_1 , X_2 , X_3 lie on a discordia through the origin, representing zircons of age t_1 that experienced varying Pb loss. Points Y_1 , Y_2 , Y_3 lie on a discordia whose upper intercept with the concordia (t_2) gives the original crystallization age, and whose lower intercept (t_3) gives the age of a later, episodic Pb-loss event.

"In some cases, zircon crystals separated from a single rock (e.g., a granite) plot on a discordia that intersects the concordia at two points. The upper intercept represents... the crystallization age (t_2)... The lower intersection (t_3) is the age of the lead-loss event if the lead loss was episodic; it has no age significance if the Pb loss was by continuous diffusion over a long period of time."

— Misra, *Introduction to Geochemistry*, p. 235

Feature	Meaning
Point on the concordia (e.g., Q)	A single, trustworthy concordant age — both decay schemes agree
Discordia through the origin (X_1 , X_2 , X_3)	Zircons of one true age t_1 with varying degrees of Pb loss down to total loss (origin)
Discordia with two intercepts (Y_1 , Y_2 , Y_3)	Upper intercept t_2 = crystallization age; lower intercept t_3 = age of episodic Pb-loss event

6. The Related Pb-Pb Isochron

A separate but related diagram uses $^{207}\text{Pb}/^{204}\text{Pb}$ versus $^{206}\text{Pb}/^{204}\text{Pb}$ (both normalized to non-radiogenic ^{204}Pb) to construct an ordinary isochron:

"A suite of closed U–Pb systems having the same age and the same initial lead isotopic composition should form a straight line array in a plot of $^{207}\text{Pb}/^{204}\text{Pb}$ (y axis) versus $^{206}\text{Pb}/^{204}\text{Pb}$ (x axis), the measured present-day ratios in the samples. This straight line, an isochron, will pass through a point whose coordinates are $(^{207}\text{Pb}/^{204}\text{Pb})_0$ and $(^{206}\text{Pb}/^{204}\text{Pb})_0$."

— Misra, *Introduction to Geochemistry*, p. 236

$$\text{slope (m)} = (1/137.88) \times [(e^{\lambda_{235}t} - 1) / (e^{\lambda_{238}t} - 1)]$$

Don't mix up the two diagram types

The Concordia-Discordia diagram plots Pb^*/U ratios (radiogenic lead relative to remaining parent) against each other for a single sample or population, testing internal consistency between two decay schemes. The Pb-Pb isochron plots Pb/Pb ratios (normalized to ^{204}Pb) across multiple cogenetic samples, exactly like an Rb-Sr or Sm-Nd isochron, and doesn't require knowing U or Pb concentrations at all — only isotope ratios.

7. Q.23 Worked Solution

Q.23. In a $^{238}\text{U}/^{207}\text{Pb}$ versus $^{235}\text{U}/^{207}\text{Pb}$ ratio diagram, the locus of points having the same $^{238}\text{U}/^{207}\text{Pb}$ and $^{235}\text{U}/^{207}\text{Pb}$ ages is known as

- (a) Mixing line
- (b) Isochron line
- (c) Concordia curve
- (d) Discordia curve

Option	Reasoning	Verdict
(a) Mixing line	Describes isotopic mixing between two source end-members — unrelated concept	Eliminated
(b) Isochron line	A different diagram (Pb/Pb ratios across a cogenetic suite), not the Pb^*/U vs. Pb^*/U concordant-age locus	Eliminated
(c) Concordia curve	Exactly the locus where the $^{238}\text{U}/^{207}\text{Pb}$ age equals the $^{235}\text{U}/^{207}\text{Pb}$ age — direct textbook match	Correct
(d) Discordia curve	A chord connecting points with <i>disagreeing</i> ages due to Pb loss/U gain — the opposite of what's asked	Eliminated

✓ Answer: (c) Concordia curve

8. Quick-Reference Summary & Probable Exam Questions

Concept	Key Fact
Concordia deviser	Wetherill (1956)
Concordia axes	$^{207}\text{Pb}^*/^{238}\text{U}$ (y) vs. $^{207}\text{Pb}^*/^{235}\text{U}$ (x)
Discordia through origin	Represents one true age with varying Pb loss
Discordia upper intercept	Crystallization age
Discordia lower intercept	Age of episodic Pb-loss event (if episodic; meaningless if continuous diffusion)
Present-day $^{238}\text{U}/^{235}\text{U}$	137.88
Preferred mineral	Zircon (high U, low initial Pb, highly resistant to alteration)

Q. What happens to recently U-depleted (weathered) zircons on the concordia diagram?

A. They plot on a discordia through the origin but lie above the concordia, distinguishing them from ancient Pb-loss discordias.

Q. Why does zircon retain so little initial (common) lead?

A. Because Pb^{2+} does not substitute easily for U^{4+} in the zircon crystal structure, so almost all Pb present is radiogenic.

Q. What advantage does zonation in zircon crystals provide geochronologists?

A. Different growth zones can represent separate events (e.g., magmatic crystallization vs. later metamorphism), letting a single grain record a chronology of multiple events.

Q. What information is needed to construct a Pb-Pb isochron that is NOT needed for the concordia diagram?

A. Actually the reverse — the Pb-Pb isochron does not require U or Pb concentrations, only isotope ratios; the concordia diagram requires Pb^*/U ratios, which do depend on knowing radiogenic Pb relative to remaining U.

Q. Under what condition does the lower intercept of a discordia have no age significance?

A. When the Pb loss occurred by continuous diffusion over a long period, rather than as a discrete episodic event.

Silicate Melt Architecture: Network Formers, Modifiers & Magma Rheology

Comprehensive Textbook Study Guide & Complete Solution for DGMS 2019 Exam (Question 24)

Standard References: William M. White, *Geochemistry* (2005) | John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010) | Dexter Perkins, *Mineralogy*

PART I

Theoretical Foundation & Direct Textbook Analysis

Detailed conceptual walkthrough of silicate liquid structure, bridging vs. non-bridging oxygens, cation classification, and melt viscosity

1. THE ATOMIC ARCHITECTURE OF SILICATE LIQUIDS & GLASSES

A foundational insight in modern igneous petrology is that silicate magmas are **not random, unstructured ionic fluids**. Spectroscopic studies (Raman, IR, NMR, and EXAFS) of silicate glasses—which represent supercooled liquids that preserve the structural state of the melt—reveal that liquids retain the short-range atomic architecture of the crystalline minerals that crystallize from them.

"Most, though not all, of our knowledge of the structure has come from studies of glasses rather than melts. While the thermodynamic properties of silicate liquids and their respective glasses differ, other studies have confirmed the general structural similarities of glasses and liquids. Spectral studies of glasses, which in some respects can be viewed as supercooled liquids, have revealed that silicate liquids have structures rather similar to those of silicate solids. In fact, the principal difference between silicate liquids and solids is the absence of long-range ordering in the former; short range ordering is similar. As in silicate minerals, the primary structural element of silicate liquids is the silicon tetrahedron, consisting of a silicon atom surrounded by four oxygens."

— William M. White, *Geochemistry* (2005), Chapter 4: Applications of Thermodynamics, p. 136

The primary structural unit in all silicate melts is the $[\text{SiO}_4]^{4-}$ **tetrahedron**, with a Si–O distance of $\sim 1.62 \text{ \AA}$ and O–Si–O bond angles of $\sim 109.5^\circ$. Tetrahedra link together by sharing oxygen atoms:

- **Bridging Oxygen (BO, O^0):** An oxygen shared between two tetrahedral centers ($\equiv\text{Si}-\text{O}-\text{Si}\equiv$ or $\equiv\text{Si}-\text{O}-\text{Al}\equiv$). It has a formal charge of zero with respect to the network, linking tetrahedra into continuous polymer chains, sheets, and 3D frameworks.
- **Non-Bridging Oxygen (NBO, O^-):** An oxygen bonded covalently to only one tetrahedral silicon atom ($\equiv\text{Si}-\text{O}^-$). Its extra unit of negative charge is balanced by electrostatic attraction to nearby modifier metal cations (Ca^{2+} , Mg^{2+} , Fe^{2+} , Na^+ , K^+).

- **Free Oxygen (FO, O^{2-}):** An oxygen anion not bonded to any silicon atom, present only in highly depolymerized, ultrabasic, or carbonatitic melts and coordinated entirely by metal cations.

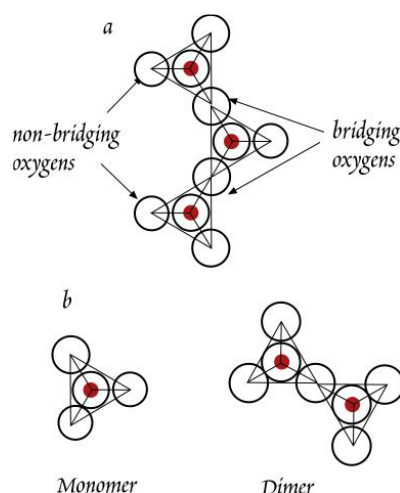


Figure 4.21 (White, *Geochemistry*, 2005, p. 136): Silicate structures in melts. (a) Short-range silicate structures in melts resemble those in solids. Individual tetrahedra may be linked by bridging oxygens shared between two silicon atoms, or terminated by non-bridging oxygens. (b) Structural units in silicate melts include isolated monomers ($[SiO_4]^{4-}$, with zero bridging oxygens) and dimers ($[Si_2O_7]^{6-}$, where only 1 of 4 oxygens in each tetrahedron is bridging).

2. CATION CATEGORIZATION: NETWORK FORMERS VS. NETWORK MODIFIERS

How a dissolved cation behaves in a silicate melt depends fundamentally on its **ionic potential (Z/r , valence divided by ionic radius)** and its ability to fit into tetrahedral 4-fold coordination with oxygen, as governed by **Pauling's First Rule (Radius Ratio Rule)**.

*"The degree to which the silica tetrahedra are linked, or polymerized, in silicate liquids affects chemical and physical properties. The degree of polymerization in turn depends on other cations present. **These may be divided into two groups, network formers and network modifiers.** Relatively small, highly charged cations such as Al^{3+} and Fe^{3+} (more rarely, Ti^{4+} , P^{3+} , B^{3+} as well) often substitute for silicon in tetrahedral sites and, along with Si, are termed network formers. **The other common cations of natural silicate liquids, Ca^{2+} , Mg^{2+} , K^+ , Na^+ , and H^+ , are network modifiers. These ions cannot substitute for silicon in tetrahedra and their positive charges can only be balanced by non-bridging oxygens. Addition of these ions disrupts the linkages between silica tetrahedra. Thus as silicate melts become richer in these network modifiers they become progressively depolymerized.**"*

— William M. White, *Geochemistry* (2005), Chapter 4: Applications of Thermodynamics, p. 136

Crystallographic Reason: Why Ca^{2+} and Mg^{2+} Cannot Form Networks

According to Pauling's Rules, the coordination number (CN) of a cation surrounded by oxygen anions ($r_{\text{O}^{2-}} \approx 1.40 \text{ \AA}$) is dictated by the radius ratio $r_{\text{cation}} / r_{\text{oxygen}}$:

- **Tetrahedral (CN = 4) stability window:** $0.225 \leq r/r_{\text{O}} < 0.414$.

Si^{4+} ($r = 0.26 \text{ \AA}$, $r/r_{\text{O}} = 0.19$) and Al^{3+} ($r = 0.39 \text{ \AA}$, $r/r_{\text{O}} = 0.28$) fit easily into tetrahedral sites.

- **Octahedral (CN = 6) stability window:** $0.414 \leq r/r_{\text{O}} < 0.732$.

Mg^{2+} ($r = 0.72 \text{ \AA}$, $r/r_{\text{O}} = 0.514$) is far too large for a tetrahedron and requires 6-fold octahedral coordination.

- **Cubic / Antiprismatic (CN = 8) stability window:** $0.732 \leq r/r_{\text{O}} < 1.000$.

Ca^{2+} ($r = 1.00 \text{ \AA}$, $r/r_{\text{O}} = 0.714$ to 0.75) requires 6- to 8-fold coordination.

Because Ca^{2+} and Mg^{2+} cannot occupy tetrahedral sites, they sit in the open interstitial spaces between tetrahedra. To achieve local charge balance, they pull electrons from adjacent oxygens, rupturing the $\equiv\text{Si}-\text{O}-\text{Si}\equiv$ bridging bonds:

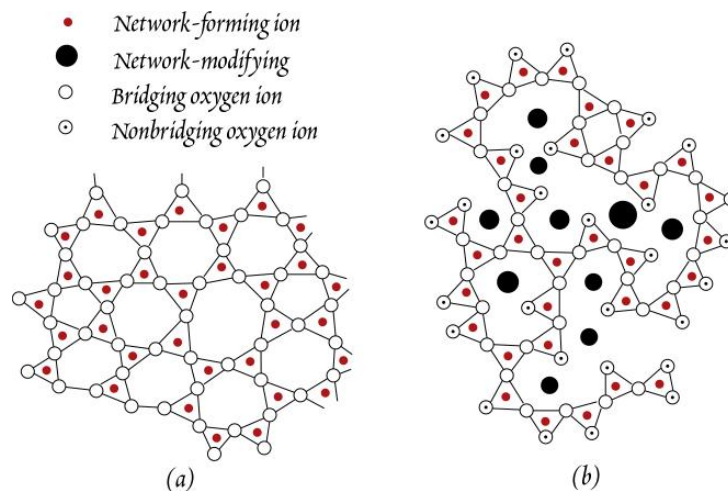


Figure 4.22 (White, *Geochemistry*, 2005, p. 136): (a) Structure of pure silica glass (liquid), consisting of a fully polymerized, continuous 3D network where all oxygens are bridging (BO). (b) A silica-rich glass with additional component ions: network-modifying cations (solid black circles, e.g., Ca^{2+} , Mg^{2+}) enter the structure, severing bridging bonds and creating non-bridging oxygens (open circles, NBO), thus disrupting the continuous polymer network.

3. QUANTIFYING DEPOLYMERIZATION: THE NBO/T PARAMETER & TRACE ELEMENTS

To parameterize the structural state of magmas in quantitative petrology, scientists use the **NBO/T ratio (Non-Bridging Oxygens per Tetrahedral Cation)**:

$$\text{NBO/T} = (2\text{O} - 4\text{T}) / \text{T}$$

where **O** is total oxygen atoms per formula unit and **T** is total tetrahedral cations ($\text{Si}^{4+} + \text{Al}^{3+} + \text{Fe}^{3+} + \text{P}^{5+}$).

- **Quartz / Pure SiO_2 Melt:** $\text{O} = 2, \text{T} = 1 \Rightarrow \text{NBO/T} = (4 - 4)/1 = \mathbf{0.0}$ (Completely polymerized framework).
- **Feldspar Melt ($\text{NaAlSi}_3\text{O}_8$):** $\text{O} = 8, \text{T} = 4 \Rightarrow \text{NBO/T} = (16 - 16)/4 = \mathbf{0.0}$ (Fully polymerized; Na^+ charge-balances Al^{3+}).
- **Pyroxene Melt ($\text{CaMgSi}_2\text{O}_6$):** $\text{O} = 6, \text{T} = 2 \Rightarrow \text{NBO/T} = (12 - 8)/2 = \mathbf{2.0}$ (Infinite single chains).
- **Olivine Melt (Mg_2SiO_4):** $\text{O} = 4, \text{T} = 1 \Rightarrow \text{NBO/T} = (8 - 4)/1 = \mathbf{4.0}$ (Completely depolymerized isolated monomers).

"The ratio of non-bridging oxygens to tetrahedral cations (Si^{4+} and Al^{3+}), written NBO/T, provides a measure of the degree of melt depolymerization. Kohn and Schofield (1994) carried out a series of experiments demonstrating the effect of melt polymerization on olivine/melt partitioning of Zn and Mn... the Zn partition coefficient decreases from about 4.5 in highly polymerized melts (low NBO/T) to about 0.8 in depolymerized melts (high NBO/T)."

— William M. White, *Geochemistry* (2005), Chapter 7: Trace Elements, p. 278

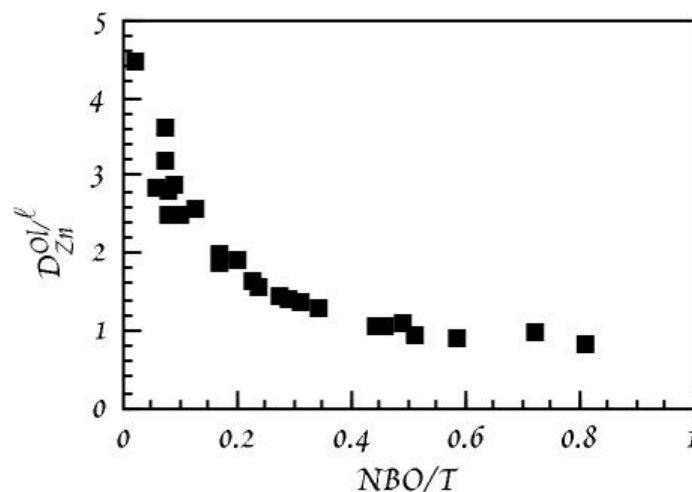


Figure 7.13 (White, *Geochemistry*, 2005, p. 278): Variation of the zinc olivine/liquid partition coefficient ($D_{\text{Zn}}^{\text{ol/l}}$) as a function of the number of non-bridging oxygens per tetrahedral cation (NBO/T) in experiments by Kohn and Schofield (1994). In depolymerized melts (high NBO/T, rich in Ca^{2+} and Mg^{2+}), abundant octahedral sites exist in the liquid, lowering the partition coefficient.

Confirmation in Thermodynamic Activity Models

In the two-lattice thermodynamic melt model of Nielsen & Dungan (1983) presented in White (2005, p. 278, Eqs. 7.35–7.36), network modifier oxides (ΣNM) and network former oxides (ΣNF) are separated mathematically:

$$\begin{aligned}\Sigma\text{NM} &= X_{\text{AlO}1.5} + X_{\text{CaO}} + X_{\text{MgO}} + X_{\text{FeO}} + X_{\text{P}_2\text{O}_5} + X_{\text{TiO}_2} + X_{\text{MnO}} + X_{\text{CrO}1.5} - \\ &\quad X_{\text{NaO}0.5} - X_{\text{KO}0.5} \\ \Sigma\text{NF} &= X_{\text{SiO}_2} + X_{\text{NaO}0.5} + X_{\text{KO}0.5}\end{aligned}$$

Note that CaO and MgO are explicitly summed as network modifiers (ΣNM), providing mathematical textbook confirmation.

4. MELT RHEOLOGY: VISCOSITY, DENSITY & THE ROLE OF VOLATILES (H_2O VS. CO_2)

Melt structure is the single most important factor controlling the physical behavior of erupting magmas. High polymerization creates an interconnected network of strong Si–O–Si and Si–O–Al bonds that strongly resists viscous shear flow.

"Because melts approximate the structures of the corresponding minerals, basaltic melts tend to have more isolated Si-O complexes, with no bridging oxygens (as in olivine) and partially bridged -Si-O-Si-O- chains (as in pyroxene), than fully bridged three-dimensional -Si-O-Al-O- networks (as in plagioclase). Because they yield much more feldspar and quartz, rhyolitic melts tend to be dominated by three-dimensional -Si-O-Al-O- networks. The more polymerized -Si-O-Al-O- networks create strong interconnected bonds throughout the rhyolitic melt, and thus higher viscosity. Greater viscosity thus generally correlates with higher silica content."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010), Chapter 3, p. 61

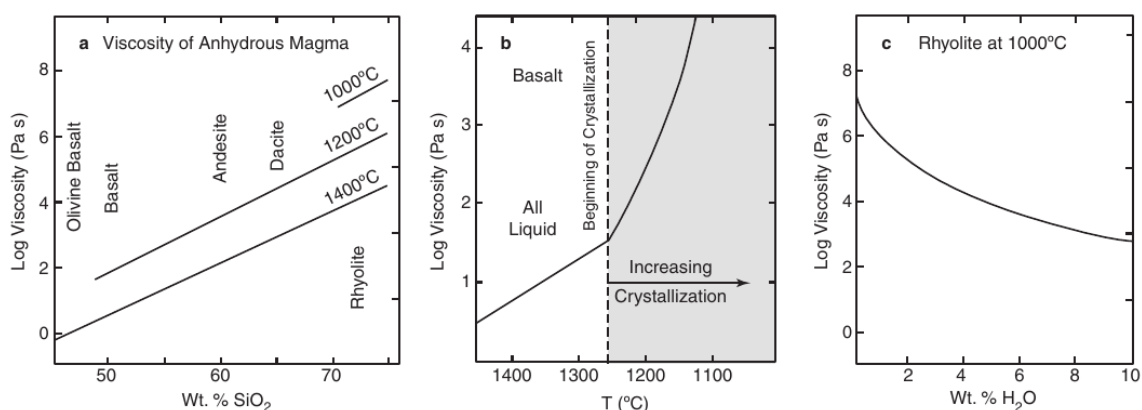


Figure 1 (Winter, *Principles of Igneous and Metamorphic Petrology*, 2010, p. 62): (a) Calculated viscosities of anhydrous silicate liquids at 1 atmosphere pressure (Bottinga & Weill, 1972; Hess, 1989): Rhyolite viscosity is $\sim 10^4$ Pa·s at 1400°C and skyrockets to 10^7 Pa·s at 1000°C, whereas depolymerized

basalt is ~ 1 Pa·s. (b) Abrupt increase in viscosity of basalt as it crystallizes. (c) Dramatic reduction in rhyolite viscosity at 1000°C as dissolved H_2O depolymerizes the melt network (Shaw, 1965).

Contrasting Volatiles: H_2O (Depolymerizer) vs. CO_2 (Polymerizer)

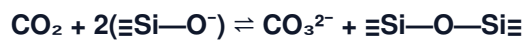
Winter (2010, Chapter 5, pp. 145–146) details a profound contrast in how dissolved volatile gases interact with silicate melt structures:

- **Water (H_2O):** Dissolves by attacking bridging oxygens:



H_2O behaves as a powerful **network modifier**. Adding just 2 wt% H_2O to a rhyolitic melt drops its viscosity by two full orders of magnitude (from 10^7 to 10^5 Pa·s; Figure 1c above).

- **Carbon Dioxide (CO_2):** In marked contrast, CO_2 does not attack bridging bonds. Instead, in mafic melts rich in Ca^{2+} and Mg^{2+} , CO_2 reacts with two non-bridging oxygens to form carbonate complexes:



CO_2 actually consumes non-bridging oxygens and produces bridging oxygens, increasing the degree of polymerization of the melt! Therefore, CO_2 raises melt viscosity and dissolves preferentially in depolymerized, Ca-Mg-rich mafic melts.

PART II

DGMS 2019 Examination: Question & Worked Solution

Rigorous option-by-option critical analysis and definitive textbook-grounded answer

ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

Q.24. In the atomic structure of silicate melts, Ca^{2+} and Mg^{2+} ions are categorized as

- (a) Network modifiers
- (b) Network formers
- (c) Bridging atoms
- (d) Non-bridging atoms

Correct Answer: (a) Network modifiers



Ca^{2+} and Mg^{2+} are large, divalent cations that cannot enter 4-fold tetrahedral coordination with oxygen. They occupy interstitial positions and disrupt the polymerized framework by converting bridging oxygens into non-bridging oxygens.

OPTION-BY-OPTION CRITICAL ANALYSIS

✓ (a) Network modifiers

CORRECT

Why it is correct: As explicitly established in White (2005, p. 136) and Winter (2010, p. 61), cations in silicate liquids are classified based on whether they build or break the tetrahedral polymer network. Because of their large ionic radii ($\text{Mg}^{2+} \approx 0.72 \text{ \AA}$, $\text{Ca}^{2+} \approx 1.00 \text{ \AA}$) and low charge (+2), Ca^{2+} and Mg^{2+} cannot enter tetrahedral sites. Instead, they occupy interstitial 6- to 8-fold coordinated sites where their charges are satisfied by attracting non-bridging oxygens. **This disrupts the $\equiv\text{Si}-\text{O}-\text{Si}\equiv$ linkages, depolymerizing the melt, lowering its viscosity, and increasing its density.** Hence, they are universally classified as network modifiers.

X (b) Network formers

INCORRECT

Why it is incorrect: Network formers are small, highly charged cations (Si^{4+} , Al^{3+} , Fe^{3+} , and minor Ti^{4+} , P^{5+}) with ionic potentials $Z/r \geq 6$ that can enter 4-fold tetrahedral coordination ($[\text{TO}_4]$). Pauling's radius ratio rule requires $0.225 \leq r/r_{\text{O}} < 0.414$ for tetrahedral coordination. With r/r_{O} values of 0.514 (Mg^{2+}) and 0.714 (Ca^{2+}), these cations are crystallographically excluded from tetrahedral network sites.

X (c) Bridging atoms

INCORRECT

Why it is incorrect: In silicate melt nomenclature, the term "bridging" applies exclusively to **oxygen anions (O^0)** that are shared between two adjacent $[\text{SiO}_4]$ tetrahedra. Ca^{2+} and Mg^{2+} are positively charged metal counterions, not bridging atoms.

X (d) Non-bridging atoms

INCORRECT

Why it is incorrect: "Non-bridging" also strictly describes an **oxygen species (O^-)** bonded to a single silicon atom ($\equiv\text{Si}-\text{O}^-$). While Ca^{2+} and Mg^{2+} associate electrostatically with non-bridging oxygens, the cations themselves are categorized functionally as **network modifiers**.

PART III

Synthesis Reference Tables & High-Yield Practice Questions

Quick-reference diagnostic tables and 8 curated exam questions for DGMS, UPSC GSI, and IIT-JAM

SYNTHESIS TABLES: CATION CLASSIFICATION & MAGMA PROPERTIES

Table 1: Comprehensive Classification of Cations in Silicate Melts

Category	Cations	Coordination (CN)	Structural Action	Effect on Polymerization
Network Formers	Si ⁴⁺ , Al ³⁺ , Fe ³⁺ (Ti ⁴⁺ , P ⁵⁺ , B ³⁺)	4 (Tetrahedral)	Build 3D -Si-O-Si- / -Si-O-Al- framework	Increases polymerization & viscosity
Network Modifiers	Ca ²⁺ , Mg ²⁺ , Fe ²⁺ , Mn ²⁺ , Na ⁺ , K ⁺ , H ⁺	6 to 8 (Octahedral / Interstitial)	Rupture bridging bonds; create NBOs	Depolymerizes melt; lowers viscosity
Charge Balancers	Na ⁺ , K ⁺ (coupled with Al ³⁺)	Near [AlO ₄] ⁻ tetrahedra	Maintains neutrality: Na ⁺ + Al ³⁺ ↔ Si ⁴⁺	Allows Al ³⁺ to act as network former
Amphoteric / Intermediate	Fe ³⁺ , Al ³⁺ , Ti ⁴⁺	4 or 6 (Depends on alkalinity/P/T)	Former in alkaline; modifier if excess	Variable depending on peralkalinity

Table 2: Comparison Across the Natural Magma Spectrum

Magma Type	SiO ₂ (wt%)	Key Modifiers	NBO/T	Dominant Units	Viscosity (Pa·s)	Density (g/cm ³)
Rhyolite / Granite	70–76%	Very Low (Alkalis balance Al)	0.02–0.15	3D Framework (Tectosilicate-like)	10 ⁵ – 10 ⁸	~2.20 – 2.35
Andesite / Diorite	57–63%	Moderate	0.30–0.50	Framework + Sheets/Chains	10 ² – 10 ⁴	~2.45 – 2.60
Basalt / Gabbro	45–52%	High (Ca ²⁺ , Mg ²⁺ , Fe ²⁺ rich)	0.60–0.95	Short Chains + Monomers	1 – 10 ²	~2.65 – 2.85
Komatiite / Ultramafic	40–45%	Extreme (MgO up to 30%)	1.20–1.80	Monomers ([SiO ₄] ⁴⁻) dominant	0.01 – 0.1	~2.85 – 3.05

Table 3: The Three Oxygen Species in Silicate Melts

Oxygen Species	Symbol	Coordination & Bonding	Formal Charge	Dominant In
Bridging Oxygen	O ⁰ (BO)	Bonded to 2 tetrahedral cations (Si–O–Si, Si–O–Al)	0 (Neutral in network)	Rhyolites, Dacites, Granites (Polymerized)
Non-Bridging Oxygen	O ⁻ (NBO)	Bonded to 1 tetrahedral cation + 1 modifier (Si–O ⁻ ... M ²⁺)	-1 (Local ionic bond)	Basalts, Andesites, Alkaline melts
Free Oxygen	O ²⁻ (FO)	Unbonded to Si; bonded only to modifier cations (M ²⁺ ... O ²⁻)	-2 (Purely ionic)	Ultra-depolymerized melts (Carbonatite, Komatiite)

PROBABLE HIGH-YIELD EXAM QUESTIONS (UPSC GSI, DGMS, IIT-JAM & GATE)

Q1. In silicate melts, Al³⁺ acts as a network former only when:

- (a) Water content is greater than 5 wt%
- (b) It is charge-balanced by an equimolar proportion of alkali (Na⁺, K⁺) or alkaline earth (Ca²⁺) cations
- (c) Melt temperature exceeds 1200°C
- (d) Fe³⁺ is absent from the melt

→ **Correct Answer: (b)**

Explanation: Al³⁺ substitutes for Si⁴⁺ in tetrahedral sites ([AlO₄]⁵⁻). Because it has a charge of +3 instead of +4, an adjacent cation (such as Na⁺, K⁺, or ½Ca²⁺) is required to maintain local electrical neutrality. Without charge balancing, excess Al³⁺ enters 6-fold coordination as a network modifier.

Q2. For a hypothetical melt having pure forsterite stoichiometry (Mg₂SiO₄), the value of NBO/T is:

- (a) 0.0 (b) 1.0 (c) 2.0 (d) 4.0

→ **Correct Answer: (d) 4.0**

Explanation: Total oxygens O = 4, tetrahedral cations T = 1 (Si). NBO/T = (2O – 4T)/T = (2 × 4 – 4 × 1)/1 = (8 – 4)/1 = 4.0. In forsterite melt, all tetrahedra are isolated monomers [SiO₄]⁴⁻ with zero bridging oxygens.

Q3. Which of the following cations has the highest tendency to act as a network modifier in silicate melts?

- (a) Si⁴⁺ (b) Ti⁴⁺ (c) P⁵⁺ (d) Ca²⁺

→ **Correct Answer: (d) Ca²⁺**

Explanation: Ca²⁺ has low charge (+2) and large radius (1.00 Å), giving it a very low ionic potential (Z/r = 2.0) and 8-fold coordination preference. Si⁴⁺, Ti⁴⁺, and P⁵⁺ are all high field-strength cations that act as network formers.

Q4. The addition of MgO or CaO to an anhydrous rhyolitic melt causes:

- (a) Increase in viscosity and decrease in density
- (b) Decrease in viscosity and increase in density
- (c) Increase in both viscosity and density
- (d) Decrease in both viscosity and density

→ **Correct Answer: (b) Decrease in viscosity and increase in density**

Explanation: As stated in White (2005, p. 136): Network modifiers rupture bridging bonds, depolymerizing the melt into shorter segments (lowering viscosity) and collapsing the open tetrahedral framework into a closer-packed structure (increasing density).

Q5. How does dissolved CO₂ affect the polymerization of mafic silicate melts compared to H₂O?

- (a) Both CO₂ and H₂O depolymerize the melt equally
- (b) H₂O depolymerizes the melt, whereas CO₂ reacts with non-bridging oxygens to increase polymerization
- (c) CO₂ depolymerizes the melt, whereas H₂O polymerizes it
- (d) Neither CO₂ nor H₂O has any effect on melt structure

→ **Correct Answer: (b)**

Explanation: Winter (2010, p. 146) highlights that CO₂ reacts with two non-bridging oxygens to produce carbonate (CO₃²⁻) and a bridging oxygen (Si–O–Si): $\text{CO}_2 + 2(\text{Si}-\text{O}^-) \rightleftharpoons \text{CO}_3^{2-} + \text{Si}-\text{O}-\text{Si}$. This creates bridging bonds and increases polymerization, directly contrasting with H₂O which breaks bridging bonds.

Q6. Divalent transition metals (e.g., Zn²⁺, Mn²⁺) have olivine/melt partition coefficients (D) that:

- (a) Increase dramatically as the melt becomes more depolymerized (higher NBO/T)
- (b) Decrease significantly as the melt becomes more depolymerized (higher NBO/T)
- (c) Are entirely independent of melt polymerization
- (d) Drop to zero in basaltic liquids

→ **Correct Answer: (b) Decrease significantly as the melt becomes more depolymerized**

Explanation: Kohn & Schofield (1994) and White (2005, p. 278) showed that D_{Zn} drops from ~4.5 in highly polymerized melts (low NBO/T) to ~0.8 in depolymerized melts (high NBO/T), because depolymerized melts provide abundant energetically favorable 6-fold octahedral sites that retain Zn²⁺ in the liquid.

Q7. Which mineral group has an atomic structure most analogous to a silicate melt dominated by dimer units [Si₂O₇]⁶⁻?

- (a) Nesosilicates
- (b) Sorosilicates
- (c) Inosilicates
- (d) Phyllosilicates

→ **Correct Answer: (b) Sorosilicates**

Explanation: Dimers consist of two [SiO₄] tetrahedra sharing exactly one bridging oxygen ([Si₂O₇]⁶⁻, NBO/T = 3.0), which is the defining structural hallmark of sorosilicates (e.g., epidote, hemimorphite).

Q8. In the 2-lattice melt solution model of Nielsen & Dungan (1983) and White (2005, p. 278), the sum of network modifiers (ΣNM) explicitly includes:

- (a) SiO₂ and NaO_{0.5}
- (b) CaO and MgO
- (c) AlO_{1.5} and SiO₂
- (d) Only H₂O

→ **Correct Answer: (b) CaO and MgO**

Explanation: In thermodynamic melt activity modeling, mole fractions of CaO, MgO, FeO, and MnO are grouped in the network-modifier component (ΣNM), whereas SiO_2 is the primary component in the network-former term (ΣNF).

Tapaswi Saikia · Geology Student · DGMS & Competitive Geological Examination Preparation

Source Exam: DGMS 2019 (Junior Scientific Officer), Question 24 · Subject: Petrology & Geochemistry

Textbook Authorities: William M. White, *Geochemistry* (2005), pp. 136–138, 278–280 | John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2010), pp. 61–62, 145–146 | Dexter Perkins, *Mineralogy*